

# External Reviewer Packet

Adam Okulicz-Kozaryn

Tuesday 28<sup>th</sup> April, 2026 14:50

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1 statements

2 cv

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<http://scholar.google.com/citations?hl=en&user=pz9RYIoAAAAJ>

## EMPLOYMENT

Associate Professor of Public Policy, Rutgers University, the State University of New Jersey, Camden, 2018-??

Assistant Professor of Public Policy, Rutgers University, the State University of New Jersey, Camden, 2012-18

Research Scientist at Texas Schools Project (2010-11) and Clinical Assistant Professor, UT-Dallas, 2010-12

## EDUCATION

Post-doctoral Fellow, Institute for Quantitative Social Science, Harvard University, 2008-10

PhD Public Policy and Political Economy, The University of Texas at Dallas, 2008

MS Corporate and Economic Policy, Poznan University of Economics (Poland), 2003

BS Banking and Finance, Poznan School of Banking (Poland), 2001

## MAJOR PUBLICATIONS (all on Google Scholar; publications with students listed under service; AOK=Adam Okulicz-Kozaryn; click text for PDF)

2026 Golden L, Valente RR, AOK, Mikhaeil E, Do Unpredictable and Unstable Work Hours Reduce Subjective Wellbeing? Social Indicators Research

Urbanization and Happiness. A Research Agenda for Happiness. Why we want to know, what we know, and what we want to know additionally. Editors: Jan Ott and Ruut Veenhoven. Edward Elgar Publishing

2025 AOK, Martinez L Colombia: Unlivable but Happy. Fool's Paradise? (No, a Real Paradise, Better than the US) Applied Research in Quality of Life

2024 D'Italia M, AOK, Constructing General Human Agency Indicators (GHAIs) and a General Personal Agency Scale (GPAS) Social Indicators Research

Urban Regrets (Unhappy Metros: Satisfaction With Life Scale (SWLS)) Heliyon

2023 Magdy Zaki E, AOK, Valente RR, Subjective Well-being and Urbanization in Egypt Cities

AOK, Everett B, Magdy Zaki E, Happiness is In The Air if It Grows (Growing Places are Happier than Shrinking ones) Urban Affairs Review

2022 Unhappy Metros: Panel Evidence Applied Research in Quality of Life

AOK, Valente R, Misanthropolis: Do Cities Promote Misanthropy? Cities

2021 AOK, Valente R, Urban unhappiness is common Cities

AOK, Morawski L, A similar effect of volunteering and pensions on subjective wellbeing of elderly Economics and Sociology

2020 Valente R, AOK, Religiosity and Trust: Evidence from the United States Review of Religious Research

AOK, Valente R, The Perennial Dissatisfaction of Urban Upbringing Cities

AOK, Morawski L, Effect of volunteering and pensions on subjective wellbeing of elderly—are there cross-country differences?  
Applied Research in Quality of Life

2019 Are we happier among our own race? Economics and Sociology

AOK, Altman M, The Energy Paradox: Energy Use And Happiness Applied Research in Quality of Life

2018 AOK, Valente R, No Urban Malaise for Millennials Regional Studies

2017 AOK, Valente R, Livability and Subjective Wellbeing Across European Cities Applied Research in Quality of Life

Strzelecka M, AOK, Is tourism conducive to residents' social trust? Evidence from large-scale surveys. Tourism Review

AOK, Valente R, Life Satisfaction of Career Women and Housewives Applied Research in Quality of Life

AOK, Valente R, City Life: Glorification, Desire, and the Unconscious Size Fetish, ed. by I Kapoor, in Psychoanalysis and the Global, University of Nebraska Press

AOK, Golden L, Happiness is flextime Applied Research in Quality of Life

2016 AOK, Strzelecka M, Happy Tourists, Unhappy Locals Social Indicators Research

Happiness Research for Public Policy and Administration Transforming Government: People, Process and Policy

Unhappy Metropolis (When American City Is Too Big) Cities

AOK, Mazelis J M, More Unequal in Income, More Unequal in WellBeing Social Indicators Research

AOK, Mazelis J M, Urbanism and Happiness: A Test of Wirth's Theory of Urban Life Urban Studies

2015 Freedom and Life Satisfaction in Transition Society and Economy

AOK, Nash T, Tursi N O, Luxury Car Owners Are Not Happier Than Frugal Car Owners International Review of Economics

The More Religiosity, the Less Creativity Across US Counties Business Creativity & the Creative Economy

Income Inequality and Wellbeing Applied Research in Quality of Life

! !: this book is excluded from the promotion packet (this is a popular science book) Happiness and Place: Cities v Nature. Why Life is Better Outside of the City? Palgrave Macmillan

2014 AOK, Holmes IV O, Avery D R, The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals  
Journal of Applied Psychology

Winners and Losers in Transition: Preferences For Redistribution and Nostalgia For Communism In Eastern Europe  
Kyklos

'Freedom From' and 'Freedom To' across countries Social Indicators Research

Natural Sprawl Administration & Society (Disputatio Sine Fine section) (section of the journal reviewed by editor only)

2013 Cluttered Writing: Adjectives and Adverbs in Academia Scientometrics

## City Life: Rankings (Livability) vs Perceptions (Satisfaction) Social Indicators Research

2012 Does Religious Diversity Make Us Unhappy? Mental Health, Religion & Culture

Income and Well-being across European Provinces Social Indicators Research

2011 Berry B J L, AOK, An Urban-Rural Happiness Gradient Urban Geography

Geography of European Life Satisfaction Social Indicators Research

Europeans Work To Live and Americans Live To Work (Who is Happy to Work More: Americans or Europeans?)

Journal of Happiness Studies

2010 Religiosity and Life Satisfaction Across Nations Mental Health, Religion & Culture

2009 Berry B J L, AOK, Dissatisfaction with City Life: A New Look at Some Old Questions Cities

## EXTERNAL GRANTS

190k PLN (~\$50k) PI, "The effect of social transfers and social capital on happiness of elderly." Vistula University, Warsaw Poland, Polish National Science Foundation (Narodowe Centrum Nauki NCN) 2016/21/B/HS4/03058 .. 2017-2019

\$1.5k Consultant (summer salary), "Health information inequity in South Jersey-Vineland" South Jersey Institute for Population Health (SJIPH) Catalyst Grant <https://sjiph.org/projects/health-information-inequity-in-south-jersey-vineland> 2026

## INTERNAL GRANTS (AWARDS and FELLOWSHIPS)

\$12k faculty advisor, Rutgers Camden chancellor's dissertation completion award for Marlo Rossi .....2026

\$30k faculty advisor, Rutgers Camden urban innovation fund grant to Ojobo Agbo Eje .....2024-2026

\$15k PI, Rutgers Camden chancellor's interdisciplinary research collaboration grant ..... 2023-2024

\$1k Rutgers-Camden Research Council Grant Award ..... 2016-2017

\$1k Rutgers-Camden Digital Teaching Fellow ..... 2016-2017

\$4k Rutgers-Camden Rand Institute Faculty Fellow ..... 2013-2014

\$1k Rutgers-Camden Civic Engagement Faculty Fellow ..... 2013-2014

## TEACHING (note curriculum development: [developed/designed] or [re-designed])

2025 recipient of department's Jay A Sigler award for "outstanding service and dedication to students" (awarded by students); the plaque says "this award is given to the faculty member who exhibits a commitment to student's academic development and personal growth."

GIS for data science <https://theaok.github.io/gisPy/> [developed/designed] ..... 2023-present

Data Processing (data management for data science) <https://theaok.github.io/datManPy> [developed/designed] 2023-present

Data Visualization <https://theaok.github.io/vis> [developed/designed] ..... 2023-present

|                                                                                                                                             |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Coping with Extremes: Ecological resilience & human well-being [innovative class co-taught with biology dept] [co-developed/designed] ..... | 2019            |
| Community & WellBeing (Happiness and Place) [developed/designed] .....                                                                      | 2018-present    |
| Data Management [developed/designed] .....                                                                                                  | 2016-present    |
| GIS for Public Sector/Urban mapping [developed/designed] .....                                                                              | 2012-present    |
| Quantitative Methods 2 [re-designed] .....                                                                                                  | 2013-2017, 2024 |
| Research Methods [re-designed] .....                                                                                                        | 2012-present    |
| Introduction to Regional and Economic Development [re-designed] .....                                                                       | 2012-2014       |
| Quantitative Methods 1 [re-designed] .....                                                                                                  | 2012-2013       |

## SERVICE

### SERVICE TO THE FIELD/PROFESSION:

Review papers (approx 3-4/mo or 36-48/year as of Jan 2026; “top journals” underlined; starting in 2022: last 2 digits of year of review shown):

- <informal/for a colleague> [24]
- Acta Sociologica
- Applied Economics [23,23]
- Applied Research in Quality of Life [23,23,23,24,24,24,24,26]
- Basic and Applied Social Psychology [22]
- BMC Psychology [26]
- BMJ Open
- British Journal of Political Science [25]
- Cities [22,23,23,23,23,23,24,24,24,24,24,25,25,25,25,25,25,25,25]
- City, Culture and Society
- City, Territory and Architecture [24]
- Cogent Arts & Humanities [23]
- Cognition and Emotion
- Communications Psychology [23]
- Comparative Economic Studies [26]
- Discover Public Health [25,26]
- Ecological Economics [24]
- Environment and Planning A
- Economics and Politics
- European Journal of Social Psychology
- European Journal of Political Economy [23,23,24]
- European Societies
- European Urban and Regional Studies [22,22]
- Frontiers in Psychology [23]
- Global Transitions [24]

- Humanities and Social Sciences Communications [23,23,23,24,24,25,25]
- Heliyon [23,24]
- International Journal of Comparative Sociology [23,24,24,25]
- International Journal of Environmental Research and Public Health
- International Journal of Psychology
- International Journal of Social Economics [22]
- International Journal of Wellbeing [25]
- International Perspectives in Psychology: Research, Practice, Consultation
- International Sociology
- Journal of Comparative Economics
- Journal of Contemporary Religion
- Journal of Economic Growth
- Journal of Economic Psychology
- Journal of Economic and Social Geography
- Journal of Personality and Social Psychology: Personality Processes and Individual Differences[16]
- Journal of Personality and Social Psychology: Attitudes and Social Cognition[26]
- Journal of Religion and Health
- Journal of Regional Science
- Journal of Happiness Studies [22,23,24,26,26,26]
- Journal of Hospitality and Tourism Management
- Journal of Public and Nonprofit Affairs
- Journal of Public Policy [23,23]
- Journal of the American Planning Association
- Journal of Urbanism
- Land Use Policy [23,24,24]
- Local Development and Society [25]
- Mental Health, Religion and Culture
- Millah: Journal of Religious Studies [25]
- Nature Communications [24]
- Open Psychology Journal
- Quality of Life Research
- Papers in Regional Science [22,22]
- PLOS ONE [22]
- Political Behavior [22]
- PNAS Nexus [24]
- Problems of Post-Communism [22]
- Psychology of Religion and Spirituality
- Public Administration Review [24,25]
- Regional Studies [22,22]
- Review of Political Economy
- Routledge (book publisher) [22]
- Rural Sociology [22]
- Urban Geography
- Urban Studies [25]

- Urban Affairs Review
- Utilities Policy [25,25]
- Science of the Total Environment
- Scientific Reports [24]
- Scienze Regionali Italian Journal of Regional Science
- Scientometrics [22,22]
- Semestre Economico (Universidad de Medellín) [22]
- Social Forces
- Social Psychological and Personality Science
- Social Indicators Research [as reviewer] [22,22,23,23,23,23,24,24,24,24,25,25,25,25,25]
- Social Indicators Research [as co-editor]<sup>1</sup> [ 25, 25, 25, 25, 25 ]
- Social Science And Medicine [24,24,25,25,25,25,25]
- Social Science Research [23]
- Social Sciences & Humanities Open [23]
- Sociological Spectrum
- Sociology of Religion
- Sustainability
- Sustainable Cities and Society
- Sustainable Futures [25]
- The Lancet Planetary Health (IF >10)
- Third World Quarterly [22,23]
- Wellbeing, Space & Society [22,25]

#### Reviewed books chapters:

- 3 chapters in “A Research Agenda for Happiness. Why we want to know, what we know, and what we want to know additionally.” Editors: Jan Ott and Ruut Veenhoven. Edward Elgar Publishing [26,26,26]

#### Reviewed books for:

- Routledge [22]
- Cambridge U Pr [23]

#### Served on editorial boards:

- Social Indicators Research: Associate Editor / Co-Editor<sup>2</sup> Jan 2024-??
- Political Behavior Jan-Dec 2019

#### Reviewed grant proposals:

- Polish National Science Foundation (NCN)
- Israel Science Foundation (ISF)

#### SERVICE TO STUDENTS:

Publications with students (all students unless indicated: “[not student]”):

<sup>1</sup>As a co-editor of Social Indicators Research, conducted through reviews (printed out, marked up, etc). Note: this is different from regular editing, where I just skim through the paper or read some parts only; and hence this extra work is listed here separately from editing as reviewing [starting oct2025 each such review marked with 2 last digits of year]

<sup>2</sup>This is not just sitting on the board of the journal and having an occasional board meeting, I am editing papers every week spending as of 2026  $1 + hrs/wk$  (earlier as i was less experienced it was  $2 + hrs/wk$ ) on skimming through the submitted manuscripts, inviting reviewers, reading reviews, and making editorial decisions.

10. Golden L [not student], Valente RR [not student], AOK, Mikhaeil E, Do Unpredictable and Unstable Work Hours Reduce Subjective Wellbeing? Social Indicators Research 2026
9. AOK, L Martinez [not student], E Mikhaeil; Happy Colombia, Unhappy Bogota (urban-rural happiness gradient in Colombia) Wellbeing, Space and Society 2026
8. E Mikhaeil, AOK; Well-Being and the authoritarian welfare state: sector of employment and human happiness in Arab countries; Public Organization Review 2025
7. MJ D'Italia, AOK; Constructing General Human Agency Indicators (GHAls) and a General Personal Agency Scale (GPAS); Social Indicators Research 2025
6. E Mikhaeil, AOK; Public-private job satisfaction differential: The case of Egypt; Public Organization Review 2025
5. AOK, BK Everett, E Mikhaeil; Happiness is In The Air if It Grows (Growing Places are Happier than Shrinking ones); Urban Affairs Review 2024
4. E Mikhaeil, AOK; Subjective well-being and urbanization in Egypt; Cities 2024
3. S Deb, AOK; Exploring the association of urbanization and subjective well-being in India; Cities 2023
2. FA Abarghuie, R Javadi, E Mirabi, AOK; Experience of urban thermal pleasure: A qualitative study to the understanding of thermal-scape quality; Cities 2022
1. R Stansfield [not student], E Aaronson, AOK; Police use of firearms: Exploring citizen, officer, and incident characteristics in a statewide sample; Journal of criminal justice 2021

Supervised dissertations for [year of graduation, first job]:

9. Isabelle Surielow [starting dissertation hours in Fall 2026]
8. Wei Chen [28??]
7. Erick W Udogu [27??]
6. Douglas N Zacher [27??]
5. Marlo Rossi [26??]
4. Michael D'Italia [24, RU-Camden] PDF
3. Shibin Yan [24, Walter Rand Institute for Public Affairs] PDF
2. Aditi Manke [21, Virginia Tech] PDF
1. Shourjya Deb [20, SIPRI] PDF

Supervised dissertations for less than the whole duration:

1. Helena Cabezas [Fall 25 only]

Served on dissertation committees:

10. Jason Schweitzer (Prevention Science) [27??]
9. Daniel Assamah [27??]
8. Monty Sanders [26??]
7. Brian Everett [27??]
6. Kate Cruz [22]
5. Charles Ian Hugh Forsyth (University of Otago NZ) [22]
4. Straso Jovanovski [19]
3. Spencer Clayton [18]
2. Rasheda Weaver [17]
1. Chris Wheeler [17]

Served on dissertation committees for less than the whole duration:

1. Yanan Li [stopped serving just before the proposal in 2023]

Supervised data science master thesis for [year of graduation]:

6. Nalluri Chandrahas [26]
5. Bhavyani Dodda [25]
4. Thanvitha Reddy Munagala [25]
3. Pavan Satya Venigalla [25]
2. Het Rakeshbhai Patel [25]
1. Raman Srinivas Naik [25]

Served on master thesis committees in social science:

2. Ethan Aaronson (crim) [20]
1. Rachel Wagner (psy) [20]

Independent/Directed Study [starting in 2023 counts shown (2014-present)]:

- fa25 x1
- sp25 x4
- fa24 x7
- sp24 x2
- fa23 x1

Researchers/Predocs/Postdocs/Lab Members/Visiting Scholars:

- hired 2 data scientists (Sep 2025-Jul 2026)
- planning to host Visiting Scholar, Professor Leonie C. Steckermeier ..... Fall 2026

#### DEPARTMENTAL AND CAMPUS-WIDE SERVICE:

At Rutgers-Camden served on the following departmental committees:

- Awards/Events, chair ..... 2013-16, 2021-22; 2023-25
- Awards/Events, member ..... 2012-13, 2019-20, 2023
- Strategic Planning, member ..... 2019-20
- PR/Website, chair ..... 2014-16
- PR/Website, member ..... 2012-13
- PhD, member ..... 2014-21
- Methods Qualifying Exam, member ..... 2012-21

Other Rutgers-Camden departmental service:

- held dissertation seminars for a group of doctoral candidates (in addition to one-on-one) ..... Fall 2025
- led Brown Bag Seminars ..... 2018-19; 2023-24
- As co-editor of a journal organized "Publish or Perish" workshop ..... Oct 2024

At Rutgers-Camden served on the following campus-wide committees:

- AI committee ..... Fall 2025
- Advancement and Promotions (A&P), member/ad hoc chair ..... 2020-25
- Advancement and Promotions (A&P), member ..... 2018-19
- Faculty Senate ..... 2018-25
- Faculty Life, chair ..... 2019-22
- Faculty Life, member ..... 2014-15

Other Rutgers-Camden campus-wide service:

- led creation and maintenance of data science lab and MS in data science and associated certificate programs 2018-22

- served as director for data science certificate: <https://graduateschool.camden.rutgers.edu/graduate-programs/#certificates> ..... 2022-??
- organized Geographic Information System (GIS) workshop attended by faculty from across the campus <https://theaok.github.io/shortGIS/> ..... 2017

## COMMUNITY SERVICE

provided data management for:

- Ohio Business Round Table (contracted by Miami University, Ohio; Contract Amount: \$7,000) ..... 2022
- 2018 Michigan Economic Competitiveness Study, Michigan Chamber of Commerce (contracted by Northwood University, Michigan; Contract Amount: \$5,000) ..... 2018

assisted following Rutgers-Camden students in their community projects as a faculty advisor:

- data science MS Ojobo Agbo Eje: "Camden Health Equity Data Visualization Initiative" project: [click for news item] [hired/supervised 4 students, budget \$30k] ..... 2024-26
- public affairs PhD Zach Chengcheng Yue at Walter Rand Institute: Covid-19 in Atlantic County ... Spring-Fall 2025
- public affairs PhD Marlo Rossi at Walter Rand Institute: Abortion in South Jersey; and other projects on abortion: [click for news item] ..... Fall 2025-Spring 2026
- MPA Olorunfunmi Adebajo's internship at the Camden County Community Development Office's Homelessness Assistance Unit: [click for news item], ..... 2024

other community service:

- made GIS maps dashboard (containing work by students in my GIS class): available on campus or via RU VPN only: <https://maps-dppa.camden.rutgers.edu/index.php?search=map&title=Special%3ASearch&profile=default&fulltext=1> ..... 2019-23
- assisted Hunterdon Central Regional High School student: happiness paper class project Fall 2025-Spring 2026
- organized Geographic Information System (GIS) workshop for The Walter Rand Institute for Public Affairs <https://theaok.github.io/shortGIS/vShortGeoda.pdf> ..... 2019

## MOST FREQUENT RECENT CONFERENCES

LASA (Latin American Studies Association) 2024

AAG (American Association of Geographers): 2015, 2016, 2017, covid19/none,

ISQOLS (International Society for Quality of Life Studies): 2015, 2017, 2018, covid19/none, 2023, 2024 (webinar), 2026 (regional)

ERSA (European Regional Science Association): 2015, 2017, 2018, covid19/none, 2023

## AFFILIATE SCHOLAR

- Australian Centre on Quality of Life (2016-present)
- CURE: Center for Urban Research and Education, Rutgers-Camden (2012-18)
- EHERO: Erasmus Happiness Economics Research Organization, Erasmus, the Netherlands (2013-16)
- Vistula University, Warsaw, Poland (2015-18)

### 3 papers: various social indicators

- 3.1 The Happiness-Energy Paradox: Energy Use is Unrelated to Subjective Well-Being (Okulicz-Kozaryn and Altman 2019)



# The Happiness-Energy Paradox: Energy Use is Unrelated to Subjective Well-Being

Adam Okulicz-Kozaryn<sup>1</sup> · Micah Altman<sup>2</sup>

Received: 3 April 2018 / Accepted: 27 February 2019 / Published online: 12 March 2019  
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## Abstract

Earth's per capita energy use continues to grow, despite technological advances and widespread calls for reduction in energy consumption. The negative environmental consequences are well known: resource depletion, pollution, and global warming. However many remain reluctant to cut energy consumption because of the widespread, although, implicit, belief that a nation's well being depends on its energy consumption. This article systematically examines the evidential support for the relationship between energy use and subjective well-being at the societal level, by integrating data from multiple sources, collected at multiple levels of government, and spanning four decades. This analysis reveals, surprisingly, that the most common measure of subjective well-being, life satisfaction, is unrelated to energy use – whether measured at the national, state or county level. The nil relationship between happiness and energy use is reminiscent of the well-known Easterlin Paradox, however the causal mechanisms responsible to each remain in question. We discuss the possible causes for the Happiness-Energy paradox and potential policy implications.

**Keywords** Energy use · Energy consumption · Energy intensity of economy · Sustainability · Happiness · Life satisfaction · Subjective well-being (SWB)

**Electronic supplementary material** The online version of this article (<https://doi.org/10.1007/s11482-019-09719-y>) contains supplementary material, which is available to authorized users.

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## Introduction

The environmental consequences of human energy consumption are one of the world's biggest problems. In particular, energy consumption is a key driver of climate change (MacKay 2008; Pachauri et al. 2014). For instance, 98% of US carbon dioxide emissions are due to energy consumption (Energy Information Administration 2008). Despite technological advances, Earth's per-capita energy use has increased about 40% over the past four decades, and continues to grow. An essential policy and ethical challenge is to find ways to be environmentally efficient at production of human well-being, to achieve good life for all within planetary boundaries (Dietz et al. 2009; O'Neill et al. 2018).

Energy is a strategic resource. Countries wage wars over energy sources, and much of politics is driven by energy. Many countries rely heavily on energy production and many use it as a political tool. Virtually all countries seek to obtain more energy sources. A recent example is fracking (Inman 2014). Yet, we need to consume less—for at least two obvious reasons: First, worldwide, most of the energy consumed is non-renewable, and this pattern of use will persist for some time (MacKay 2008). Second, energy consumption causes pollution, and pollution harms the environment, other species, and ourselves (MacKerron and Mourato 2009; Gandelman et al. 2012; Ferreira et al. 2013). Thus, natural resource depletion and pollution potentially cancel out the benefits of energy consumption.

Many argue that consumption must be reduced dramatically. A recent report by an Intergovernmental Panel on Climate Change is alarming (Pachauri et al. 2014 and <http://www.ipcc.ch>).

A major review estimated that global climate change could reduce global GDP by 20%. (Stern et al. 2006) Indeed, the threat is serious enough that claims for not growing the economy anymore or even “degrowing” are being strongly argued (Kallis 2011; Kallis et al. 2012). Some argue that reduction as high as a factor of ten is needed in affluent societies (Pretty 2013).

Moreover, it is generally believed that there is a fundamental tradeoff between societal energy consumption and self-interest at the individual and national level. It is believed that substantially reducing energy consumption requires individual sacrifices: It is widely (although not universally) believed that increased energy consumption is necessary both for overall national economic growth and to increase the directly consumed amenities (such as cooling, travel, lighting) that increase well-being (Jorgenson et al. 2014; Stern and Kander 2012; Dietz 2015; Carter 1977; Smil 2005). However recent research in the field of subjective well-being (SWB) offers theoretical reason for hope – the key point from SWB literature relevant here is that there is a consumption threshold beyond which SWB become satiated (Lamb and Steinberger 2017). In what follows we bridge the study of energy consumption and SWB and examine the empirical relationship between these at a global scale.

While energy consumption is useful, most energy consumption both pollutes and depletes natural resources (Arrow et al. 2004; Soytaş et al. 2007). How energy use affects human well-being on the whole remains an open question, which we examine in this research. Traditional approaches to examining the effects of development have measured societal benefits using the Gross Domestic Product (GDP) and its adjustments – per capita and purchasing power parity (Jorgenson 2014). Relying solely on

GDP as a measure of societal benefit has a broad range of shortcomings which are now becoming broadly recognized – and does not generally capture individual's evaluation of their quality of life (Giannetti et al. 2015). We evaluate the benefits and problems of energy use using both GDP and gold-standard measures of subjective well-being (SWB), life satisfaction.<sup>1</sup> Combining these measures reveals surprising relationships among energy consumption and societal benefit.

By combining data on energy consumption and SWB we find that the people in economically developed areas consuming more energy are not happier. This finding is consistent across time and multiple levels of spatial aggregation—it applies to patterns of energy consumption at the local, national, and global scale. Further, across both developed and developing countries with normal economies we observe a strong inverse relationship between SWB and the energy-intensiveness (the amount of energy per unit of GDP). This suggests that a national increase in energy consumption does not contribute to SWB (likely the opposite) except where such consumption leads to unusually great increases in GDP. By combining data on energy consumption and SWB we find that the people in economically developed areas that consume more energy are not happier. This finding is consistent across time and multiple levels of spatial aggregation—it applies to patterns of energy consumption at the local, national, and global scale.<sup>2</sup>

## Analysis and Results

### Selection of Measures and Data

There is a consensus that the survey items that we use are good measures of subjective well-being (Diener 2009; Oswald and Wu 2009; Stiglitz et al. 2009). All SWB measures come from surveys representative of given areas. Such measures are reasonably valid and reliable (Diener et al. 2013). One caveat is limited cross-cultural comparability (Diener and Suh 2003). Our cross-country results showed strong relationships and it is unlikely that the whole effect is due to measurement error, and we mostly used data within the US.

Unlike SWB, energy consumption measures used here are not very similar or directly comparable and differ across studies due to availability of measures. For the purposes of determining broad and durable patterns, we identified the existing data sources that have been systematically and continuously measured over periods of at least five years; have coverage of units within the target population (states, counties, nations) that is as complete as possible; and are of high quality.

<sup>1</sup> The concept of subjective well being (SWB) has been well-studied, for recent reviews see Diener (2009) and Diener et al. (2013). It is now widely recognized in this field of study that SWB is a broad multidimensional concept that unifies such popular concepts as (affective) happiness, and life satisfaction. Because it is a multidimensional concept, a complete measurement of all aspects of SWB requires collecting a large set of measures from each subject. Most broad-scale research in this field aims to measure the cognitive dimension of SWB, i.e., life satisfaction, because of its general domain (necessary for cross-comparisons), and established validity and reliability (see, Diener et al. 2013). We follow the standard practice in this area, and make use only empirical data and use life-satisfaction as our measure of SWB.

<sup>2</sup> As argued later, only developing nations could improve SWB through greater energy consumption, but across the developed world, the relationship between energy consumption and SWB is nil.

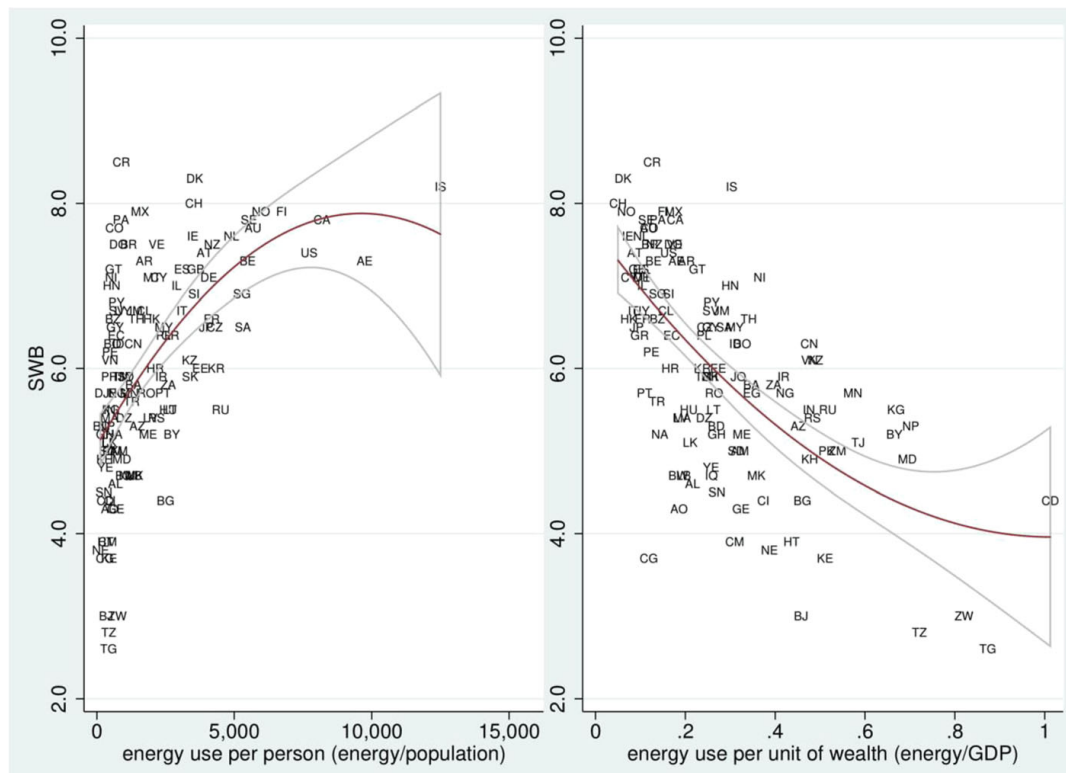
**Global Patterns** We started by examining the relationship between energy use and SWB across the world, by country. We used data from the World Database of Happiness ([http://www1.eur.nl/fsw/happiness/hap\\_nat/nat\\_fp.php?mode=8](http://www1.eur.nl/fsw/happiness/hap_nat/nat_fp.php?mode=8)). The SWB scale is based on multiple data sources and is for the most part based on responses to questions of the form: “All things considered, how satisfied are you with your life as a whole these days?” on a scale from 0 = “dissatisfied” to 10 = “satisfied.” The measure of total per capita energy consumption (kg of oil equivalent per capita) comes from the World Bank (<http://data.worldbank.org/indicator/EG.USE.PCAP.KG.OE>). All data were averaged over the years 2000–2009.

Data are plotted in Fig. 1. The basis for the received wisdom that increasing well-being requires increased energy consumption is illustrated by the figure on the left. Across countries there is a clear positive association between energy use and SWB. SWB generally increases alongside energy consumption, but the variance across countries is large and there are many outliers. For example, some countries that have high energy use, such as Russia (RU), are unhappy. In contrast, many countries such as Costa Rica (CR) or Mexico (MX) are able to reach the highest level of SWB while maintaining very low energy use. As energy use rises, its relationship with SWB flattens out,<sup>3</sup> especially after the threshold of 5000 kg of oil equivalent per capita. In general, developed countries have a greater SWB (Mazur 2011; Jorgenson 2014).

How does the relationship between energy use and SWB change if we take economic development into account? It is well known that the relationship between energy use and SWB is affected by level of development (e.g., Jorgenson and Givens 2015; Knight and Rosa 2011). As recently highlighted by Dietz et al. (2009), it is important to examine how efficient a nation is in producing human well-being. Surprisingly, we find that when we measure energy as a function of economic efficiency, the relationship reverses. This is shown in the second panel of Fig. 1, which displays on the x axis the energy intensity of gross domestic product (energy/GDP). Countries that consume less energy per unit of wealth are happier. Some of the happiest countries are the highly-developed Nordic countries (DK, FI, NO, SE). Their energy consumption is high (Fig. 1, panel 1), but relative to income, their energy consumption is among the lowest in the world (Fig. 1, panel 2). In a descriptive sense, high energy intensity means that a country requires a high cost to convert energy into GDP. Another way to put this is that some countries are more efficient (in the economic, not technical sense) at converting energy into wealth.

There are a number of plausible causal explanations for this observed pattern. One hypothesis consistent with the data is that energy-use is largely unrelated to SWB once basic needs are met, but is driven up by other factors (some of which are explored below). A second plausible hypothesis is that countries with lower energy-intensity of GDP employ more energy-efficient technologies: These technologies reduce energy input for the same output of goods or services, so that consumption can remain high at low levels of pollution. A third plausible hypothesis is that countries with lower energy-intensity of GDP have economies that are rely less on inherently energy-intensive industries. A fourth plausible hypothesis is that country-specific characteristics (e.g. as industrial development, financial development, air temperature, and population) simultaneously drive energy use up and well-being down.

<sup>3</sup> This is known as saturation, first mentioned with respect to human development by Martinez and Ebenhack (2008).



**Fig. 1** Energy is measured as kg of oil equivalent. Countries that consume more energy per person are happier (first panel). The positive relationship in first panel is spurious—taking into account the level of development, the relationship reverses—the more energy per unit of wealth, the less SWB (second panel). Quadratic fit shown with 95% confidence intervals. Energy use refers to use of primary energy before transformation to other end-use fuels (kg of oil equivalent). All data were averaged over the 2000–2009 period. Country codes are in Table S1 in supplementary material. Several outliers on energy, GDP, or energy/GDP were dropped: Ukraine, Mozambique, Turkmenistan, Uzbekistan, Ethiopia, Luxembourg, and Qatar. And two additional outliers Kuwait, and Trinidad and Tobago—they would change L-shaped relationship in the second panel to U-shaped because they have extreme leverage—compare with Fig. S2 using full sample in supplementary material. Also, the relationships between income, energy and SWB are further explored in Figs. S3–S8—examining these interrelationships is a topic for the future research—present study focuses on documenting the main relationship only

These hypotheses cannot be definitely tested with the available data, however examining individual countries favors the first hypothesis: The Netherlands (NL) is rich, energy efficient, and happy. But there are also outliers. For instance, Colombia (CO) is happy and energy efficient, but poor. In general, Latin America poses a puzzle for SWB researchers. Latin Americans are relatively poor but happy. They also use very little energy and have similar energy intensity to that of the US. Notably, all Latin American countries cluster at the top left in the first panel. Great SWB is possible using little energy. East European post-Soviet countries, on the other hand, cluster at the bottom and some at the left. Some countries are relatively unhappy despite low energy intensity of GDP such as Congo (CG). Some countries, on the other hand, are relatively happy despite high energy intensity such as Trinidad and Tobago (TT) and Kazakhstan (KZ).

**US States and Counties** To further probe the hypotheses discussed above, and to answer old question of whether more energy is needed to increase well-being if there is already a great deal of energy being consumed (Mazur and Rosa 1974), we turned to the US, since it is among the countries that use the most energy per capita. State and

county level SWB data come from the Behavioral Risk Factor Surveillance System (<http://www.cdc.gov/brfss>) using a similar measure of SWB: “In general, how satisfied are you with your life?” on a scale from 1 = “very dissatisfied” to 4 = “very satisfied.” State energy data came from the US Energy Information Administration (<http://www.eia.gov/state>) and is measured as total energy consumption per capita in the residential sector.<sup>4</sup> California’s residential electricity consumption per capita came from the Energy Consumption Data Management System (<http://www.ecdms.energy.ca.gov/elecbycounty.aspx>).<sup>5</sup> State level data were averaged over 2005–2010; and California data were averaged over 2006–2010. Results are shown in Fig. 2.

Energy-hungry states are not happier. There are two outliers, Hawaii and California, consuming much less energy in the residential sector than others. We zoom in on California counties in the second panel of the same figure. There is also a great deal of variation in energy use across California counties, and the relationship with SWB is also nil. We have experimented with energy intensity of GDP as we did earlier across countries, but in the case of the US subregions the results are not different.

**Over Time Movement** It is well-known that SWB is related to income in a cross-section, but not over time (Easterlin 1974; Easterlin et al. 2012). To account for this we supplemented our cross-sectional results with an exploration of SWB and energy consumption over time. We used the General Social Survey (<http://gss.norc.org>), which measures SWB as follows: “Taken all together, how would you say things are these days—would you say that you are very happy, pretty happy, or not too happy?” 1= “not to happy”, 2=“pretty happy”, 3=“very happy”. Energy consumption is measured as total energy use per capita in the residential sector, the same measure as used for states. Figure 3 shows SWB and energy use over time by census division. There is not much co-movement: the two series correlate at .2 only. Energy consumption is only weakly related to SWB.

High residential energy use in the US is often linked to energy inefficient suburbanization: sprawling car-dependent large housing (so called McMansions) (Duany et al. 2001). Energy use and pollution per capita are higher in sprawling suburbs than in compact cities (Meyer 2013). Such suburbs typically crop up in affordable and spacious South and indeed much of the country, and least in increasingly compact and expensive North East and West Coast<sup>6</sup>—Figure 3 agrees: New England and Pacific have their energy use per capita declining. Another problem for climate change is that the rest of the World may want to mimic this unsustainable lifestyle.<sup>7</sup>

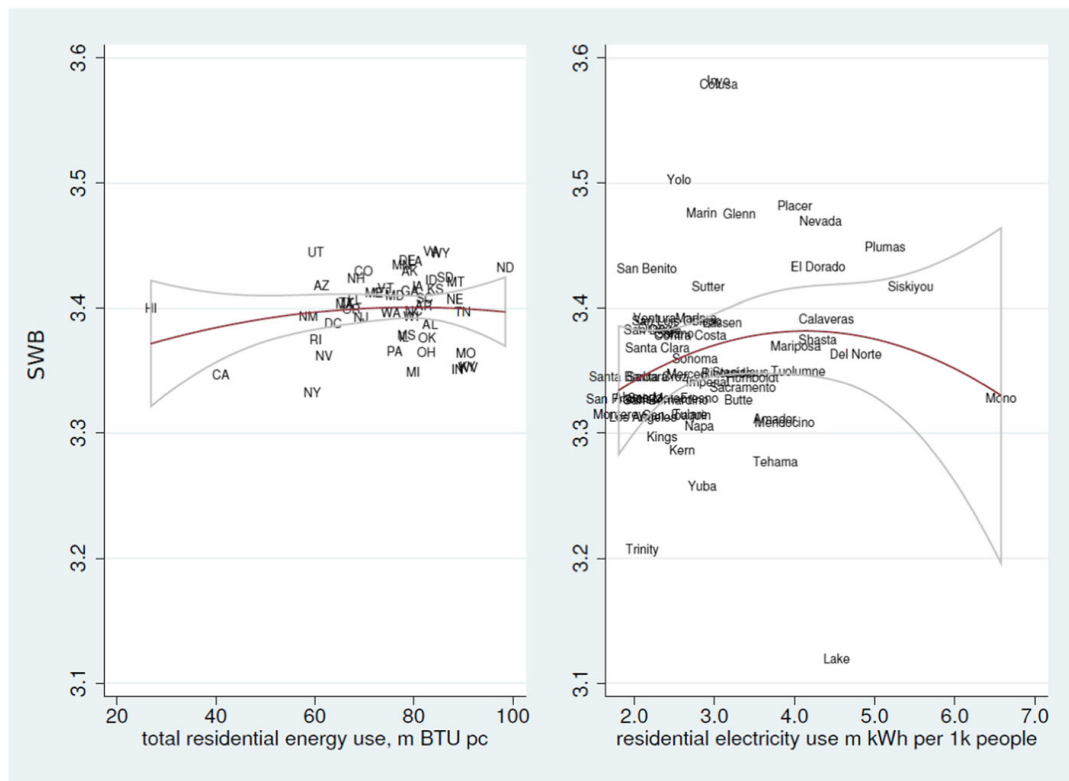
Americans continue to consume large amounts of energy as compared to other countries. With the notable exceptions of New England and the Pacific region, energy

<sup>4</sup> The series code is TERPB, and further information is available at [https://www.eia.gov/state/seds/sep\\_use/notes/use\\_technotes.pdf](https://www.eia.gov/state/seds/sep_use/notes/use_technotes.pdf)

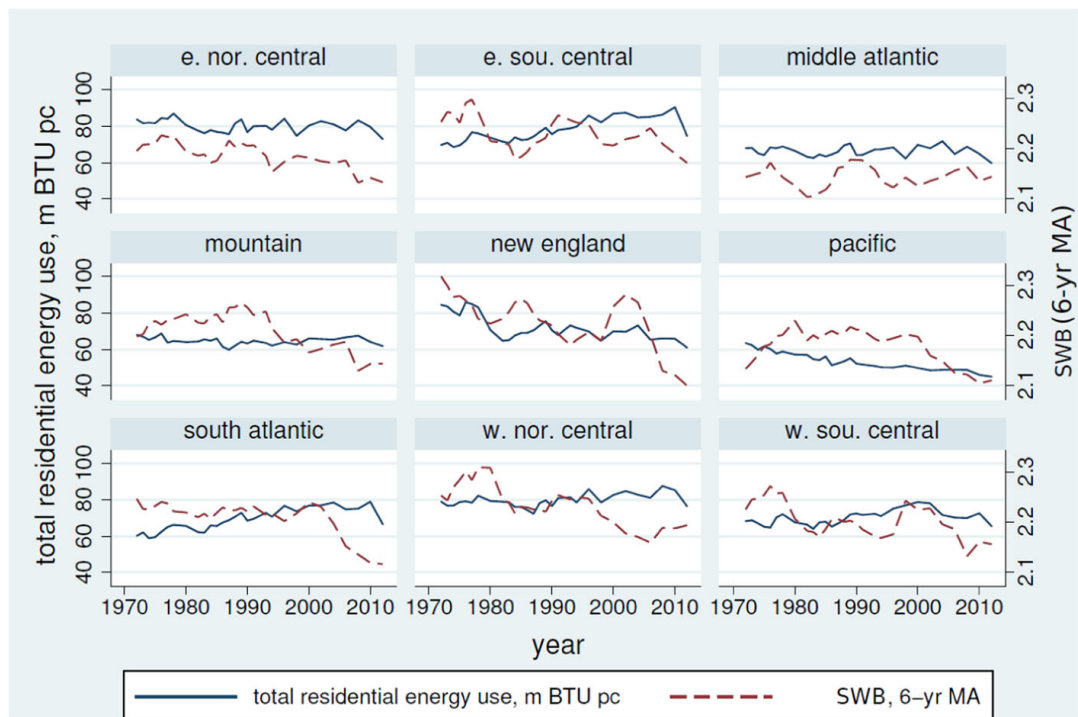
<sup>5</sup> Total residential electricity consumption has two components: (A) validated Quarterly Fuel and Energy Report (QFER) retail sales/delivery data and (B) self-generation. Electricity Consumption = retail sales + self-generation. Retail sales: electricity amounts delivered from the grid and consumed by the customer. Self-generation: electricity produced and consumed on-site by the customer (for example, parking lot roofs covered with PV panels). Self-generation only contributes to total consumption and not to total sales.

<sup>6</sup> Parts of California, notably Los Angeles, are often argued as examples of wasteful suburbanization, yet Los Angeles new homes are much smaller than average (<https://www.laweekly.com/news/how-the-size-of-the-typical-la-home-has-grown-over-the-years-7364336>).

<sup>7</sup> Notably, Central and Eastern Europe have been suburbanizing recently (Stanilov and Sýkora 2014).



**Fig. 2** SWB and total residential energy use across US states and residential electricity use across California counties. The relationship was also quite flat in terms of total energy consumption and its GDP intensity. State level data were averaged over 2005–2010, and California data were averaged over 2006–2010



**Fig. 3** SWB (6-yr moving average) and total energy consumption in residential sector per capita. Correlation is .2 only (for unsmoothed series)

use is not decreasing, and sometimes it is increasing. Americans do not spend any less on energy either: 5–10% of personal expenditure over past 50 years were on energy (BEA 2014).

## Discussion

The analysis above shows that energy use is apparently unrelated to SWB, especially in developed nations such as the US. Why? There are a number of potential causal explanations rooted in on existing theories of individual behavior, social behavior, and economics. Existing observed information does not allow us to definitively test each of these causal theories here. The discussion below aims to suggest productive areas for future by outlining four possible behavioral explanations, and several additional economic/industrial explanations.

One possibility is that bounded rationality plays a strong role: Decisions on how much to produce and consume based on expected SWB are tricky, humans are often predictably wrong and experience much less SWB than expected (Kahneman et al. 1997), and are wrong in their perceptions of energy consumption and savings (Attari et al. 2010; Dietz 2014). Thus it is perhaps not surprising that there is a discrepancy between expectation (more energy use, more SWB) and experience (more energy use, no more SWB) is not surprising.<sup>8</sup>

Second explanation is that energy consumption is a positional good: Consumption buys position, but because everyone is in competition with everyone else, the race cannot be won (Frank 2012; Kasser 2003). A related explanation is a “hedonic treadmill” (Brickman et al. 1978): It is possible that consumption does not increase well-being because of hedonic adaptation. More consumption would not make people happy if their basic needs are already satisfied.

A third potential explanation builds on Veblen’s concept of conspicuous or wasteful consumption (Veblen 2005a, b). Such consumption does not satisfy needs but simply aims to demonstrate that one is better than others. Much of such consumption wastes energy without satisfying human needs, and does not make us any happier (Csikszentmihalyi 1999; Frank 2004, 2005, 2012).

A fourth potential explanation based on the livability theory (Veenhoven 2014), is that consumption result in more well-being only up to the point that is used to satisfy basic human needs. Hence, energy used to alleviate extreme temperatures, provide food, shelter, basic transportation, etc, will increase human SWB (if it out-weighs the costs such as pollution). But energy consumed on non-essential human needs (arguably most energy consumed in developed nations) will result in little SWB, if any, and our findings are consistent with such an explanation.

In parallel with these potential behavioral explanations, industrial and economic factors could contribute to the relationship between energy use and SWB. One possibility that any direct positive effect of energy use on SWB is offset by an increase in pollution. (See, for example, Li et al. 2018 for a review of the effects of air pollution on

<sup>8</sup> While we believe the behavioral explanations are pivotal, there are other potential explanations. For instance, access to technology may affect the threshold energy level needed to meet core human needs.

SWB). If such an indirect effect exists, the relative dependence of a nation's economy on industries that are energy intensive and/or highly polluting, and the availability and deployment of energy efficient technologies in these industries, would be likely to mediate together the relationship between SWB and energy consumption (Stern 2011). These mediators in turn may be causally affected by country-specific characteristics such as economic history, population, and climate. Future research can focus on testing specific explanations—here we have focused on documenting the overall relationship.

## Conclusions

Again, while for simplicity we use term “energy”, it is important to keep in mind that we have used different measures of energy. At the country level, among developed countries, there is no relationship between energy use and development. Further, for all countries with normal economies, the less energy-intensive is a country's economy, the happier are its people. (Fig. 1, Panel 2). Across US states and California counties, energy consumption and SWB have no apparent relationship. Likewise, the changes over time in energy use are almost unrelated to changes in SWB across US Census regions. With this correlational study we aim to bring the relationship between energy use and well-being to wider audiences, and encourage more research in this area.

While correlation is not sufficient for causality, it is typically necessary – especially across relatively homogeneous units of observation (Mazur 2011). Although this is an observational study and does not demonstrate causality, the patterns observed suggest that in the US, and across the developed World, change in energy use will not cause significant change in SWB. Further it suggests that, even in developing countries, increases in energy consumption do not contribute to SWB except where such consumption leads to unusually great increases in GDP. These patterns contrasts with widely held assumption about the relationship between SWB and energy consumption. There are many possible explanations—and while the available data does not support a causal analysis, we offer some conjectures for future research.

We found that energy consumption is neither necessary for SWB, nor linked directly to it, and we speculate the relation between energy use and SWB is very similar to the relation between economic growth and SWB, i.e., the Easterlin Paradox (Easterlin et al. 2010): common wisdom has it that over time income should increase SWB, and so common wisdom has it that energy use should increase SWB, but neither does. Specifically, Easterlin Paradox states that at a point in time SWB varies directly with income, but over the long run (10 years or more) SWB does not increase when a country's income increases. Two countries that dramatically increased both GDP and energy use are Japan and more recently China. From 1960 till 1990 Japanese energy use per capita increased more than three fold; from 1990 to 2007 Chinese energy use per capita more than doubled, but SWB remained flat in both China and Japan.<sup>9</sup>

<sup>9</sup> Primary energy use (before transformation to other end-use fuels) in kilograms of oil equivalent, per capita from World Development Indicators accessed through <https://www.google.com/publicdata>. SWB data come from Frey and Stutzer (2002) and Easterlin et al. (2012)—note that Chinese SWB slightly increased in last several years. Also note that in the US both energy use per capita and SWB remain flat over past four decades (using GSS data and above measure of energy use). We provide additional analyses in supplementary material.

It is striking that there were only several attempts to relate energy consumption to SWB. In a small sample of 55 countries, one early study used a set of 27 indicators to measure quality of life (Mazur and Rosa 1974), which was later extended over time (Mazur 2011). Mazur (2013) confirmed his earlier findings: among industrial nations, already high in their energy and electricity consumption, further increases in per capita energy or electricity consumption are unrelated to changes in SWB. Another study is a recent article that analyzes cross national data (Winfrey 2013). But neither study explores energy intensity of GDP, nor variation at finer geographic representation than a country. This is an important contribution of this study: there is more heterogeneity across units at more aggregated levels (e.g., countries are less comparable than US states), and there is less precision (e.g., state level SWB and energy averages are less precise descriptions of individuals than county level averages). And it is well known that there may be different relationships at different levels of aggregation (e.g., Ashkanasy 2011), i.e. earlier cross-national findings may not hold at state or county levels. Furthermore, cross-sectional relationships often differ from time-series relationships (Easterlin et al. 2012), i.e. earlier cross-sectional findings needed confirmation in time-series analysis in present study.

We do not claim causality in this correlational study. Yet, as persuasively pointed out by labor economist Andrew Oswald (e.g., Blanchflower and Oswald 2011; Oswald 2014), correlational studies are not without merit despite what many economists think—many scientific breakthroughs were first discovered in observational studies, for instance, that smoking is related to cancer. It is often overlooked that experiments tend to suffer from many problems that are not inherent in observational studies such as lack of external validity, small sample size, artificial laboratory setting, and so forth, for discussion see Pawson and Tilley (1997). At the same time, we encourage research into the causal relationships between energy use and SWB.

There is a need for future research in this important area. There have been many calls to systematically collect SWB data, and we should collect energy use data for the same subjects. Such data would allow the exploration of the relationship at the individual level.

We conjecture that in the developed world, we can decrease energy consumption without much loss in SWB, if any. While people from India and China may need to consume more energy, it is possible that Americans could consume considerably less without reducing SWB. For example, Texans might reduce energy consumption by half, following the example of California, and remain content. We agree with Dietz et al. (2009) that we should strive for environmentally efficient well-being: achievement of human well-being at low level of environmental stress; and we agree with Steinberger and Roberts (2010) that high human development can be achieved at moderate energy and carbon levels.

Based on the our analysis, we suggest an interventions to decrease energy consumption. First, we may simply need to increase awareness of the empirical relationship between SWB and energy consumption: there is no evidence that increasing already substantial consumption does buy substantial SWB. Just as increasing income beyond a point does not result in much SWB (Kahneman and Deaton 2010), increasing energy consumption beyond a point does not result in much SWB either. In short, SWB can be achieved at low levels of energy consumption—human flourishing requires energy to satisfy basic needs only. We are hopeful that awareness and education can change behavior. Many other ways to curb consumption have been suggested (Dietz 2014, 2015; Asensio and Delmas 2015; Dumas 1987; Attari et al. 2010). Proposals to

consume less energy are urgently needed as market economy is often incentivizing people to consume more energy, for instance when new sources are discovered as with current natural gas bonanza in the US (Hopkins 2017).

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### 3.2 Religiosity and Trust: Evidence from the United States (Valente and Okulicz-Kozaryn 2021)



# Religiosity and Trust: Evidence from the United States

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## Abstract

**Background** Trust is one of the key driving forces behind human action and an important factor in shaping human interaction. Trust can improve economic growth, political and civic involvement, democratic stability, and subjective well-being. Yet, trust has been in decline for the last 60 years in the U.S.

**Purpose** This article tests the effect of several indicators of religiosity, including an index for both social and individual religiosity, on trust. Common religious doctrine instructs followers to place their trust solely in God, and can therefore be interpreted as a determinant of generalized trust. Thus, the purpose of this paper is to find out whether religious people are more likely to be distrustful of others and whether they are more likely to be misanthropic.

**Methods** We use the US General Social Survey (GSS, 1972–2018,  $n > 10k$ ), a large, recurring, and nationally-representative sample of U.S. adults. Using the GSS, we investigate the relationship between religiosity and trust (interpersonal and generalized) in a well-controlled model using OLS regressions. We examine both the effects of social religiosity (e.g. church attendance, membership at religious organization), and individual religiosity (e.g. belief in God, feeling of closeness to God, prayer), on trust and on misanthropy. Several additional robustness tests were conducted.

**Results** The findings demonstrate that while social religiosity or belonging (services attendance, church membership) predicts more trust, individual religiosity or believing (prayer, closeness and belief in God) predicts lower trust. Likewise, social religiosity lowers misanthropy, while individual religiosity promotes it. Furthermore, we show that it is important to consider individual and social religiosity simultaneously because they correlate and have opposite effects—this is an intriguing and not entirely obvious finding as most people would expect that religiosity in general, has a positive effect on trust.

**Conclusions and Implications** Our results indicate that religiosity is a substantial determinant of social trust and of misanthropy. The divergent results based on whether religiosity is social or individual in character is a new conceptual approach

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Extended author information available on the last page of the article

towards religiosity not previously undertaken in the literature. Ingroup favoritism and outgroup derogation theory explains our findings—connecting with God disrupts connection with humans.

**Keywords** Faith · Misanthropy · Religion · Trust · U.S. General Social Survey (GSS)

“It is better to trust in the Lord than to put confidence in man.”

Psalms 118:8

“Stop trusting in man, who has but a breath in his nostrils. Of what account is he?”

Isaiah 2:22

“Cursed is the one who trusts in man (...) but blessed is the one who trusts in the Lord.”

Jeremiah 17:5–7

## Introduction

The Bible and the Torah contain passages admonishing its followers to trust God only, and not to put their trust in men (e.g., Psalm 118:8(9), Jeremiah 17:5(7), Micah 7:5(6)). The Quran, likewise, contains many verses with the word *Tawakkul*, which implies the Islamic concept of perfect trust and reliance in God alone. This begs the question, are religious people more likely to be distrustful of others? Is the trustworthiness that others see in religious people reciprocated?

Forty years ago, Julian B. Rotter noted that “the entire fabric of our day-to-day living, of our social order, rests on trust<sup>1</sup>” (1971, p. 443). Fiske (2009) considers trust as one of the key driving forces behind human action—trust shapes human interaction. Trust can improve economic growth, interaction and coordination, commerce and trade, political and civic involvement, crime prevention, health, democratic stability, and subjective well-being (SWB) (e.g. see Berggren and Jordahl 2006; Hempel et al. 2012; Lount 2010; Kramer 1999; Sosis 2005; Uslaner 2002; Helliwell 2006; Berggren et al. 2008; Uslaner 2003). Yet, studies have shown that trust has been in decline for at least 60 years in the U.S. (Rotter 1971; Putnam 2001). In particular, a steady decline has been documented by the General Social Survey since it started to measure trust in 1972.

Religion exerts a profound influence on social conduct, and its effects on human society are exacerbated by the fact that the majority of the world population—85%—possesses some form of religious belief (Sedikides 2010). Specifically, common religious doctrine instructs followers to place their trust solely in God, and can therefore

<sup>1</sup> We will use term “trust” instead of generalized or interpersonal trust for simplicity.

be interpreted as a potential determinant of generalized trust (Iannaccone 1998; Tan and Vogel 2008).

In what follows, we investigate the relationship between *religiosity* and *trust*, and examine the effects of both individual and social religiosity simultaneously. Our results show that while social religiosity, or belonging (attending services, church membership), predicts more trust and less misanthropy,<sup>2</sup> individual religiosity, or believing (prayer, belief, closeness to God), predicts lower trust and higher levels of misanthropy. We begin with a brief excursus into the evolving literature on religiosity and trust, then present our model, documenting how we used the received literature to control for sources of individual variation, discuss results, and conclude by placing them within the in-group favoritism and out-group derogation theory, reinforced by the findings that a connection with God implies disrupted connection with humans.

## Previous Studies and Contribution

A large body of literature has established a number of factors that affect trust [e.g. see Camerer (2003) p. 86, and Hoffmann (2013) for a detailed overview]. However, until recently, religion was relatively unexplored in the trust literature, as well as in political science and economics (with some exceptions, e.g., Iannaccone 1998; Wald 2005; McCleary and Barro 2006; Wald and Wilcox 2006). This relative oversight impedes our understanding of the effects of religiosity on trust, given that people are often influenced by religious behavior, theologies, and beliefs, which in turn can significantly affect economic and political outcomes in many domains (Berggren and Bjornskov 2011). Furthermore, the linkage between *religiosity* and *trust* has been scantily considered. As stated by Dilmaghani (2017), few “studies have explored the influence of religious intensity, or degree of commitment to different dimensions of religiosity, on trust” (p. 49). Dilmaghani (2017) goes on to provide an important contribution to the literature by showing that high levels of religious commitment negatively correlates with trust in unknown others for Catholics, while the opposite is true for Protestants in Canada. Other imperative contributions include Berggren and Jordahl (2006) who found that religiosity and religious fractionalization lower trust, and the comprehensive study of Berggren and Bjornskov (2011) who using data for 109 countries found a robust, negative relationship between religiosity and trust, whose magnitude was positively related to the degree of fragmentation in a country.

Most studies focus on the connection between religious affiliation and trust (La Porta 1996; Fehr and Fischbacher 2002; Bellemare and Kröger 2007). These studies, which largely consist of cross-country studies, examine determinants of trust and its effect on people who belong to hierarchical religions (e.g., Islam, Orthodox Churches, the Catholic Church, and some other religions, as defined by Putnam

<sup>2</sup> This refers to people who have a general hatred, distrust, or contempt for the human species.

(1993)). The effect of belonging to a hierarchical religious organization<sup>3</sup> is negative (La Porta 1996; Zak and Knack 2001; Bjornskov 2007), although Brañas-Garza et al. (2009) using individual level data, found that Catholic affiliation in Latin America was positively associated with trust. Protestantism seems to have a positive effect on trust in studies using cross-sectional (Uslaner 2002) and individual-level data (Trautmüller 2009). A few studies, on the other hand, found no statistical significance effect of religious affiliation on trust (Bjornskov 2007; Alesina and Ferrara 2002), except for Hinduism and Buddhism, which have a positive effect in some specifications of trust (Bjornskov 2007). A more recent study differentiating between types of Christian denominations, found that affiliation in more conservative denominations lead to less trust, and affiliation with more liberal denominations entails more trust (Daniels and Von Der Ruhr 2010). Likewise, belonging to a main-line denomination predicts trust (Veenstra 2002; Welch et al. 2004, 2007; Berggren and Bjornskov 2009; Trautmüller 2009; Smidt 1999; Orbell et al. 1992).

Of the existing few quantitative studies analyzing the effect of *religiosity* on *trust*, findings are discordant. Schoenfeld (1978), who surveyed attitude of over 1500 U.S. respondents, found hints of a curvilinear relationship with high-attendance individuals and members of more fundamentalist churches displaying greater *mistrust* (Schoenfeld 1978; Hoffmann 2013). Welch et al. (2007), on the other hand, found no statistical association between religiosity<sup>4</sup> and trust. More recently, Tan and Vogel (2008) examined the relationship between individual religiosity and trust doing a randomized experiment using a trust game (see Berg 1995). They found that more religious trustees are trusted more, and such behavior is more pronounced in more religious trusters (Tan and Vogel 2008).

A growing number of studies have found that social religiosity increases trust (Smidt 1999; Veenstra 2002; Bègue 2002; Welch et al. 2004; Trautmüller 2009), while individual religiosity, or believing, decreases trust (Helliwell and Putnam 2004). People feel more connected to God when praying alone rather than in a group, and people with insecure and anxious attachments to others are likely to hold the strongest religious beliefs (Epley et al. 2008). Thus, religiosity seems to promote ingroup trust and outgroup distrust (Sosis 2005).

Our research contributes to the literature in several ways. First, this study measures religiosity using several indicators, including an index for both social (e.g., fellowship, church attendance, membership) and individual (e.g., prayer, belief in God) religiosity. Notice that we prefer to use the simpler “individual” and “social” religiosity terms as opposed to “extrinsic” and “intrinsic”—to be sure, both terminologies share similarities, but the latter suffers from some notable problems as outlined by Kirkpatrick and Hood (1990).<sup>5</sup>

<sup>3</sup> Defined as Catholic, Eastern Orthodox, or Muslim.

<sup>4</sup> In Welch et al. (2007), religiosity was measured by activity in religious congregations, belief in absolute morality, frequency of prayer, and belief in the sinfulness of human nature.

<sup>5</sup> The conceptualization of intrinsic and extrinsic religiosity can be traced back to Allport (1950)’ study of mature and immature religions, while the empirical measurement of the constructs were established by Wilson (1960) and Feagin (1964). Over the years, many criticism was leveled against the I–E framework (Hunt and Morton 1971; Hoge 1972; Kirkpatrick and Hood 1990), particularly the lack of conceptual clarity in the definitions, as well as the factorial structure, reliability, and construct validity of the I

To our knowledge, thus far, only Helliwell and Putnam (2004) differentiate between social and individual religiosity. Most studies measure religiosity specifically as denomination membership (Alesina and Ferrara 2000; Addai et al. 2013), or use a single or a few measures of religiosity.

Second, this study considers social and individual religiosity simultaneously since they are positively correlated and have opposite relationships with trust and misanthropy. Although Welch et al. (2007) considers both types of religiosity simultaneously, they do not address the differences between social and individual religiosity. Moreover, with the exception of Helliwell and Putnam (2004), other studies have not proposed that there can be opposite effects from individual and social religiosity. This study corroborates and provides empirical evidence for this claim.<sup>6</sup>

Another contribution of this research is to show that individual religiosity, or believing, is negatively related to interpersonal or generalized trust. Finally, our work also corroborates and expands on the findings of the GSS Topical Report No. 29 (Smith 1997) that using a smaller sample size provided bivariate correlation results on the relationship between religiosity and misanthropy.

We use ingroup favoritism outgroup derogation/prejudice theory (Tajfel 1982; Tajfel et al. 1971; Byrne 1971; McPherson et al. 2001) to conceptualize belief in God. It is perhaps unorthodox to do so—the theory has been so far used on humans only, but it can elucidate the relationship people develop with supernatural beings. An adherent clearly forms a very close relationship with God, who is, of course, very different from humans, and hence, there is a clear ingroup and outgroup. If a person forms a connection with God, it may be difficult for her to connect with humans, who not only are very different, but also very inferior to an omnipotent God. A related explanation is that a connection with God fulfills one's desire for control, a fundamental human motive (Yzerbyt and Demoulin 2010), and as a result, one does not need to put trust into humans. Adherent's ingroup is God (and perhaps other supernatural beings or religious figure such as the Prophet, Virgin Mary, etc.) and the outgroup is the human species. Ingroup bias has often been used to reflect ingroup favoritism, whereby favorable treatment, attitudes, or warmth are limited for the ingroup (Hamley et al. 2020), and this bond is formed at the expense of the outgroup. Indeed, religious writings, as quoted at the beginning, specifically prescribe trust in God, and distrust in humans.

Belief in “God,” or religiosity more generally, may be an outgrowth of the attachment system, whereby “God” and other believers in the same spiritual community are referred to as attachment figures [e.g., “God the father,” priests referred to as

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Footnote 5 (continued)

and E scales. Thus, we heed to Kirkpatrick and Hood (1990) suggestion that researchers should “pursue more promising methodological and theoretical directions” (p. 443), and simply classify social religiosity to refer to religious rituals and practices performed in fellowship with others, and individual religiosity to refer to one's personal belief and practices performed by one's self. We discuss these definitions in more detail in the next section.

<sup>6</sup> Smith (1997) has made a related suggestion: church attendance should reduce misanthropy and “fundamentalist beliefs, which emphasize the sinful nature of humans and a stern and authoritarian God” should increase misanthropy.

“fathers” or “patriarchs,” other members are referred to as “brothers” and “sisters” (see Kirkpatrick 2005; Rowatt and Kirkpatrick 2002)].<sup>7</sup>

Circles of trust are based on ingroup trust (rather binding, not bridging social capital). Individual religiosity will reduce (generalized) trust because there is only an adherent and her God in the circle. Other adherents may appear in the circle only as externalities—because they share the same God.

Social religiosity, on the other hand, explicitly adds other adherents in the circle. In other words, individual religiosity draws a circle that separates a person from mankind. Social religiosity draws a circle, which separates believers from nonbelievers of a particular god. Adherence to a religion separates a person from other religions. In fact, a recent meta-analysis suggests that religiosity promotes outgroup prejudice (Hall et al. 2010). Yet, the problem is not likely to be serious in a relatively religiously homogeneous society such as the U.S., where the vast majority of population is Christian. Hence, we propose that:

*Individual religiosity will decrease trust, and social religiosity will increase trust.*<sup>8</sup>

## Method

### Data

All variables come from the US General Social Survey (GSS) (<http://gss.norc.berkeley.edu/>). The GSS is a cross-sectional, nationally representative survey. It was administered almost every year until 1994 when it became biennial. The unit of analysis is at the individual level and data are collected in face-to-face, in-person interviews (Davis et al. 2007). The full dataset contains 64,813 observations pooled over 1972–2018, but the sample size used will vary depending on the variables used and missing data. All variables were recoded in such a way that a higher value means more.

### Trust

The outcome of interest, trust, involves confidence or faith that some other, upon whom we must depend, will not act in ways that occasion us painful consequences (Fiske 2009). We measure it with:

TRUST. “Generally speaking, would you say that most people can be trusted or that you can’t be too careful in dealing with people?” 1 = “cannot trust”, 2 = “depends”, 3 = “can trust”. We also measure trust as a binary variable, collapsing “depends” and “cannot trust” into one category in the supplementary material. Merging these two categories can result in lost of information, of course, but only slightly since fewer than 5% of respondents answered “depends.” Note that recent studies question

<sup>7</sup> We are grateful to an anonymous reviewer of an earlier version of this study for this suggestion.

<sup>8</sup> At least in relatively religiously homogeneous society such as the U.S.

the validity of this item (Sapienza et al. 2013), and hence, we also use an alternative measure, *misanthropy*.

Another potential problem with the trust measure is that it is mostly a binary concept and there are few responses in the intermediate category—such measure, of course, does not allow for much variance. This disadvantage is offset at least in two ways. We use a *misanthropy* scale that allows more variance, and we have a large sample size that adds statistical power.

## Misanthropy

The second outcome of interest is *misanthropy*. Following Smith (1997), *misanthropy* is measured as an index of three measures, *trust*, *fair* and *helpful*, as follow:

TRUST. “Generally speaking, would you say that most people can be trusted or that you can’t be too careful in dealing with people?” 1 = “cannot trust”, 2 = “depends”, 3 = “can trust”.

FAIR. “Do you think most people would try to take advantage of you if they got a chance, or would they try to be fair?” 1 = “take advantage”, 2 = “depends”, 3 = “fair”.

HELPFUL. “Would you say that most of the time people try to be helpful, or that they are mostly just looking out for themselves?” 1 = “lookout for self”, 2 = “depends”, 3 = “helpful”.

We used factor analysis with varimax rotation to produce an index, and we reversed it so that it measures *misanthropy*. Cronbach’s alpha is .67. Frequencies of these and all other variables are in the supplementary material.

## Religiosity

The definition and measurement of religiosity is important because it is a broad and fuzzy concept. Glock (1962) and later Glock and Stark (1965) contended that all religions have five universal dimensions: Ideological (belief), Intellectual (knowledge), Ritualistic (religious behavior), Experiential (religious experiences that affect the senses), and Consequential (the effects of the other four dimensions in the secular world). Although researchers have since questioned whether these dimensions are indeed “universal” and applicable to all religions most researchers accept the concept of religiosity as a multidimensional phenomenon (Glock and Stark 1965; Faulkner and Gordon 1966; King and Hunt 1972, 1975; Clayton and Gladden 1974), and many different dimensions have been proposed in the literature (e.g., Fukuyama 1961; Lenski 1961; King and Hunt 1972).

The literature is generally divided among two approaches in measuring religiosity. The first attempts to operationalize dimensions that have been conceptually created. This approach assumes the existence of certain dimension, and then selects constructs that are believed to measure these dimensions (Lenski 1961; Glock and Stark 1965; Fukuyama 1961; Allport and Ross 1967; Wilson 1960). The second approach involves analyzing empirically the relationship among sets of items from a large pool of religiosity indicators (King 1967; King and Hunt 1969, 1975, 1972).

In this paper, we combine both approaches. Following Neuberg et al. (2014) we initiate with the assumption that religion is more than just a set of beliefs, but also community practices, socialization functions, organizational structures, and other elements. Specifically, we define two dimensions of religiosity: social and individual. The literature on religiosity is replete with such a distinction. Dittes (1971) for example, made a distinction between religion's explicit (e.g. public, social, formalized) and subjective modes (e.g. personal attitudes, values, commitments). Greeley (1972) made a distinction between "meaning and belonging" while Davison (1975), separated religiosity modes as "private and public."

Social religiosity refers to social interaction such as church attendance and religious meetings. Individual religiosity, on the other hand, is about personal interaction with God, a feeling of closeness to God, which occurs through prayer, worship, or communion—for elaboration see Okulicz-Kozaryn (2010). Echoing Lambert et al. (2010), we note that it is difficult to find a behavior that is so widespread like prayer, and yet, so little studied (90% of Americans say they pray at least occasionally). The distinction between social and individual religiosity is also important for practical reasons—the two types of religiosity predict different outcomes: for instance only social religiosity predicts support for terrorism (Ginges et al. 2009), and only individual religiosity predicts low subjective well-being (SWB) (Okulicz-Kozaryn 2010).

Thus, social religiosity (belonging) is measured in this study with the following variables:

ATTEND. "How often do you attend religious services?" 0 = "never" to 8 = "more than once wk"

MEMBER. "Now we would like to know something about the groups or organizations to which individuals belong. Here is a list of various organizations. Could you tell me whether or not you are a member of each type?" "Church-affiliated groups" 1 = "yes", 0 = "no"

In addition, an index was created for social religiosity using these two variables, *attend* and *member*, to test the robustness of our findings (refer to the Supplemental Material).

Individual religiosity (believing) is measured with the following variables:

PRAY. "About how often do you pray?" 0 = "never" to 6 = "several times a day"

BELIEVE. "Please look at this card and tell me which statement comes closest to expressing what you believe about God." 1 = "don't believe" to 6 = "know god exists"

CLOSE. "How close do you feel to God most of the time? Would you say extremely close, somewhat close, not very close, or not close at all?" 1 = "does not believe" to 5 = "extremely close"

An index was created for individual religiosity using the variables, *pray* and *believe* (refer to the Supplemental Material). The variable *close* was not included in the index because of the large amount of missing data for that specific variable.

## Controls

We follow Welch et al. (2007) and control for personal characteristics such as: income, marital status, age, age<sup>2</sup>, education, gender, race, born in the U.S., and political leaning. Different denominations imply different levels of individualism/collectivism (Cohen and Hill 2007), potentially affecting trust, and hence we include dummies for main religious denominations, which helps ensure that effects are not due to specific denominations. In addition, we control for religious tradition, the *fund* variable on the GSS. Without these controls, the religiosity variables may be picking up differences in practice across traditions and denominations.

We also control for socio-economic class, unemployment, and occupational prestige, since higher social class is positively associated with religious service attendance, and negatively associated with prayer, thus ensure that it is not an unobserved aspect of social class that is driving the results for religiosity and trust.

Furthermore, we control for subjective wellbeing and health—the goal is to alleviate the possible problem of spuriousness. It may not be individual religiosity that causes lower trust, but the lack of success, unhappiness, or poor health that makes a person be both closer to God and farther from people.

Data were pooled over many years, and hence we include year dummies.<sup>9</sup> The U.S. is quite heterogeneous across its regions—the relationship between religiosity and trust may differ by region, thus, we include a dummy for the South. Refer to Tables 1, and 5 in the “Appendix” for sample details and descriptive statistics.

## Results

The measures of religiosity correlate moderately at about .3–.6 in Table 2. There is a higher correlation among measures within each category, social and individual, and smaller correlation between measures from each of the two categories. For instance, people may attend religious events for social reasons. All social religiosity measures indicate higher trust and lower misanthropy, and all individual religiosity measures indicate the opposite, lower trust and higher misanthropy. All correlations are significant at .01, except for MISANTHROPY and CLOSE, which were insignificant, but note that the sign is as expected. The variable CLOSE only has about 10k observations—if there were more observations, the correlation would be more significant. The correlations are statistically significant, but small substantively.

One of the key findings from our models is that the effect sizes become larger when both measures of religiosity, individual and social, are included in one model—this is expected based on the correlations, as there are moderate to high correlations among religiosity measures, and only small but opposite correlations between social religiosity and trust or misanthropy, and individual religiosity and

<sup>9</sup> We did some additional robustness tests and divided the dataset by decades to control for possible changes in society over time as it relates to religion. The results mostly concur to our main findings. See the “Appendix” for more details.

**Table 1** Descriptive statistics—GSS 1972–2018

| Variable        | Obs    | Mean     | S.D.     | Min     | Max      |
|-----------------|--------|----------|----------|---------|----------|
| Trust           | 41,258 | .3728731 | .4835746 | 0       | 1        |
| Misanthropy     | 39,774 | 2.006502 | .7572934 | 1       | 3.041185 |
| Attend          | 64,218 | 3.761469 | 2.729547 | 0       | 8        |
| Member          | 20,654 | .3553307 | .478625  | 0       | 1        |
| Pray            | 35,813 | 4.243766 | 1.645931 | 1       | 6        |
| Belief          | 24,654 | 5.077229 | 1.424088 | 1       | 6        |
| Close           | 10,009 | 4.081427 | .8554988 | 1       | 5        |
| South           | 64,813 | .3554071 | .4786402 | 0       | 1        |
| Real income     | 58,292 | .0317147 | .0294714 | .000227 | .162607  |
| Socio-class     | 61,330 | 2.449845 | .6618418 | 1       | 4        |
| Protestant      | 64,524 | .5752278 | .4943122 | 0       | 1        |
| Catholic        | 64,524 | .2429174 | .4288489 | 0       | 1        |
| Fundamentalist  | 62,271 | 1.964558 | .7586637 | 1       | 3        |
| Conservative    | 55,327 | .3389304 | .4733504 | 0       | 1        |
| Liberal         | 55,327 | .2748387 | .446437  | 0       | 1        |
| Married         | 64,786 | .5267959 | .4992853 | 0       | 1        |
| Unemployed      | 64,792 | .0336307 | .180278  | 0       | 1        |
| Prestige of job | 60,048 | 42.98771 | 12.97033 | 16      | 80       |
| Age             | 64,585 | 46.09883 | 17.53433 | 18      | 89       |
| Age2            | 64,585 | 2432.55  | 1772.483 | 324     | 7921     |
| Education       | 64,636 | 12.86961 | 3.176276 | 0       | 20       |
| Male            | 64,813 | .4414855 | .4965681 | 0       | 1        |
| Born U.S.       | 55,550 | .9080108 | .2890133 | 0       | 1        |
| Household white | 57,874 | .7968691 | .4023326 | 0       | 1        |
| swb             | 60,053 | 2.185736 | .6376671 | 1       | 3        |
| Health          | 47,589 | 2.991994 | .8476682 | 1       | 4        |
| Year            | 64,813 | 1994.939 | 13.46529 | 1972    | 2018     |

**Table 2** Pairwise correlation matrix. \* $p < 0.05$ , \*\* $p < 0.01$ . Source: GSS 1972–2018

| Variable | Member | Pray   | Believe | Close  | Trust    | Misanthropy |
|----------|--------|--------|---------|--------|----------|-------------|
| Attend   | 0.56** | 0.53** | 0.44**  | 0.43** | 0.04**   | − 0.09**    |
| Member   | .      | 0.37** | 0.26**  | 0.27** | 0.07**   | − 0.12**    |
| Pray     | .      | .      | 0.59**  | 0.59** | − 0.05** | 0.02**      |
| Believe  | .      | .      | .       | 0.61** | − 0.09** | 0.06**      |
| Close    | .      | .      | .       | .      | − 0.06** | 0.01        |
| Trust    | .      | .      | .       | .      | .        | − 0.72**    |

**Table 3** OLS regressions of trust. Beta (fully standardized) coefficients reported. *Source:* GSS 1972–2018

|                                   | a1        | a2        | a3        | a4        | a5        |
|-----------------------------------|-----------|-----------|-----------|-----------|-----------|
| Pray                              | − 0.06*** | − 0.09*** | − 0.02**  | − 0.03**  | − 0.04**  |
| Attend                            |           | 0.07***   | 0.04***   | 0.04***   | 0.05***   |
| South                             |           |           | − 0.07*** | − 0.07*** | − 0.06*** |
| Family income in \$1986, millions |           |           | 0.04***   | 0.04***   | 0.02      |
| Class                             |           |           | 0.06***   | 0.05***   | 0.04***   |
| Protestant                        |           |           | 0.05***   | 0.05***   | 0.04**    |
| Catholic                          |           |           | 0.02*     | 0.02*     | 0.03*     |
| Fundamentalist                    |           |           | − 0.07*** | − 0.07*** | − 0.06*** |
| Conservative                      |           |           | − 0.01    | − 0.01    | − 0.02*   |
| Liberal                           |           |           | 0.05***   | 0.05***   | 0.04***   |
| Married                           |           |           | 0.03***   | 0.01      | 0.02*     |
| Unemployed                        |           |           | − 0.01    | − 0.00    | − 0.00    |
| Occupational prestige             |           |           | 0.04***   | 0.04***   | 0.04***   |
| Age                               |           |           | 0.28***   | 0.31***   | 0.43***   |
| Age2                              |           |           | − 0.17*** | − 0.19*** | − 0.28*** |
| Educ                              |           |           | 0.19***   | 0.19***   | 0.18***   |
| Male                              |           |           | 0.04***   | 0.04***   | 0.04***   |
| Born in U.S.                      |           |           | 0.00      | 0.00      | 0.00      |
| White household                   |           |           | 0.11***   | 0.11***   | 0.12***   |
| SWB                               |           |           |           | 0.08***   | 0.09***   |
| Health                            |           |           |           |           | 0.05***   |
| _cons                             | ***       | ***       | ***       | ***       | ***       |
| Year dummies                      | Yes       | Yes       | Yes       | Yes       | Yes       |
| N                                 | 23213     | 23074     | 18066     | 17982     | 9616      |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

trust or misanthropy. Hence, if only one type of religiosity was included in the regression models, it would pick up the effect of the excluded dimension of religiosity, since the effects are opposite and cancel each other out.

We ran several analyses and robustness tests. On Table 3, we first considered only PRAY to measure individual religiosity, and ATTEND to measure social religiosity, because these two variables have fewer missing observations in comparison to the other variables (note that the results are substantively the same for other measures of religiosity presented in the Supplemental Material).<sup>10</sup> Also, these are OLS regressions, but robustness tests show that the results are substantively the same from discrete models, (all additional models and regression results are available in the Supplemental Material) and all standard errors are robust to correct for heteroskedascity.

<sup>10</sup> The supplemental material will be made available to readers in a Gitlab repository upon publication.

**Table 4** OLS regressions of misanthropy index. Beta (fully standardized) coefficients reported. *Source:* GSS 1972–2018

|                                   | b1      | b2        | b3        | b4        | b5        |
|-----------------------------------|---------|-----------|-----------|-----------|-----------|
| Pray                              | 0.02*** | 0.08***   | 0.04***   | 0.04***   | 0.04**    |
| Attend                            |         | − 0.11*** | − 0.07*** | − 0.06*** | − 0.06*** |
| South                             |         |           | 0.08***   | 0.08***   | 0.07***   |
| Family income in \$1986, millions |         |           | − 0.05*** | − 0.04*** | − 0.03*   |
| Class                             |         |           | − 0.09*** | − 0.08*** | − 0.07*** |
| Protestant                        |         |           | − 0.06*** | − 0.06*** | − 0.05*** |
| Catholic                          |         |           | − 0.04*** | − 0.04*** | − 0.05*** |
| Fundamentalist                    |         |           | 0.08***   | 0.08***   | 0.06***   |
| Conservative                      |         |           | 0.00      | 0.00      | 0.01      |
| Liberal                           |         |           | − 0.03*** | − 0.03*** | − 0.03**  |
| Married                           |         |           | − 0.04*** | − 0.02*   | − 0.01    |
| Unemployed                        |         |           | 0.00      | − 0.00    | − 0.00    |
| Occupational prestige             |         |           | − 0.04*** | − 0.04*** | − 0.04*** |
| Age                               |         |           | − 0.27*** | − 0.31*** | − 0.40*** |
| Age2                              |         |           | 0.10*     | 0.14***   | 0.21***   |
| Educ                              |         |           | − 0.19*** | − 0.19*** | − 0.17*** |
| Male                              |         |           | 0.02**    | 0.02**    | 0.01      |
| Born in the U.S.                  |         |           | − 0.01    | − 0.01    | − 0.00    |
| White household                   |         |           | − 0.12*** | − 0.11*** | − 0.12*** |
| SWB                               |         |           |           | − 0.12*** | − 0.12*** |
| Health                            |         |           |           |           | − 0.07*** |
| _cons                             | ***     | ***       | ***       | ***       | ***       |
| Year dummies                      | Yes     | Yes       | Yes       | Yes       | Yes       |
| N                                 | 22461   | 22327     | 17543     | 17463     | 9155      |

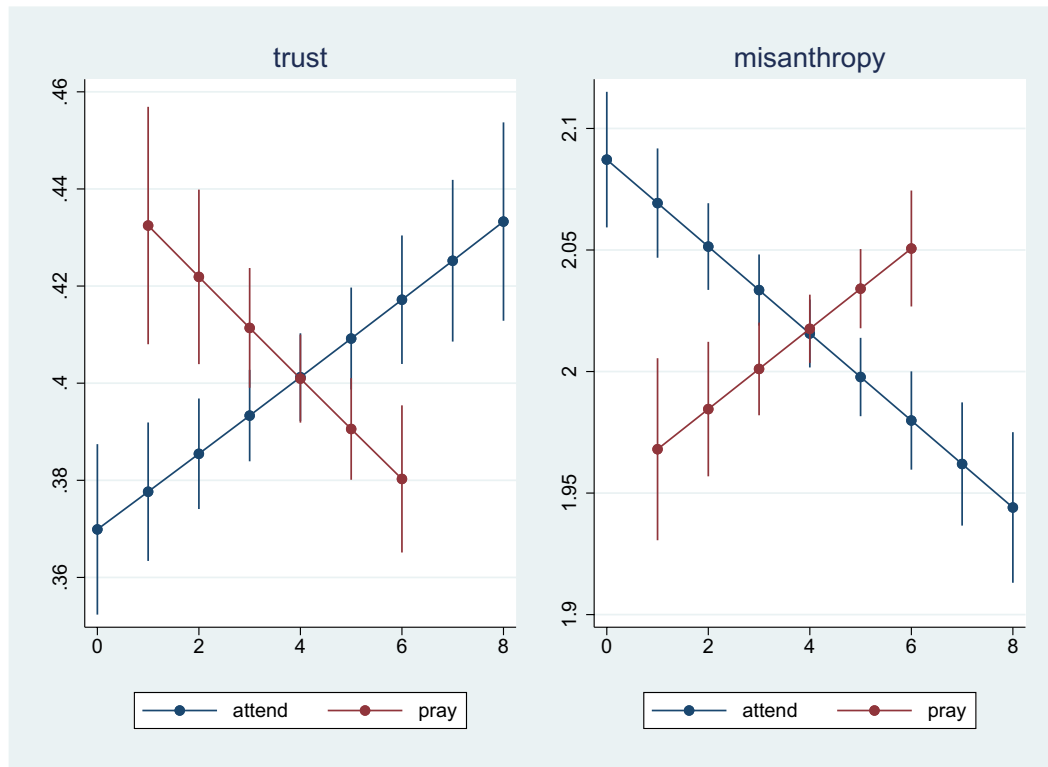
\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

We report beta coefficients in order to show that the effect size of religiosity is comparable to that of other predictors of trust in Table 3.

The coefficient on PRAY increases significantly from model a1 to a2 when controlling for ATTEND. An larger and similar effect is observed in the corresponding models b1 and b2 for the misanthropy dependent variable in Table 4—this is a classic suppression effect (MacKinnon et al. 2000).<sup>11</sup>

Next, we add a full set of controls following (Welch et al. 2007) in model a3. Then, we add robustness checks in the last two columns: in model a4 we control for health and in model a5 for subjective well-being (SWB), and the results persist. Therefore, it is not only the unhealthy or unhappy that are more religious and less trusting—we control for these factors.

<sup>11</sup> We are grateful to an anonymous reviewer of an earlier version of this study for this suggestion.



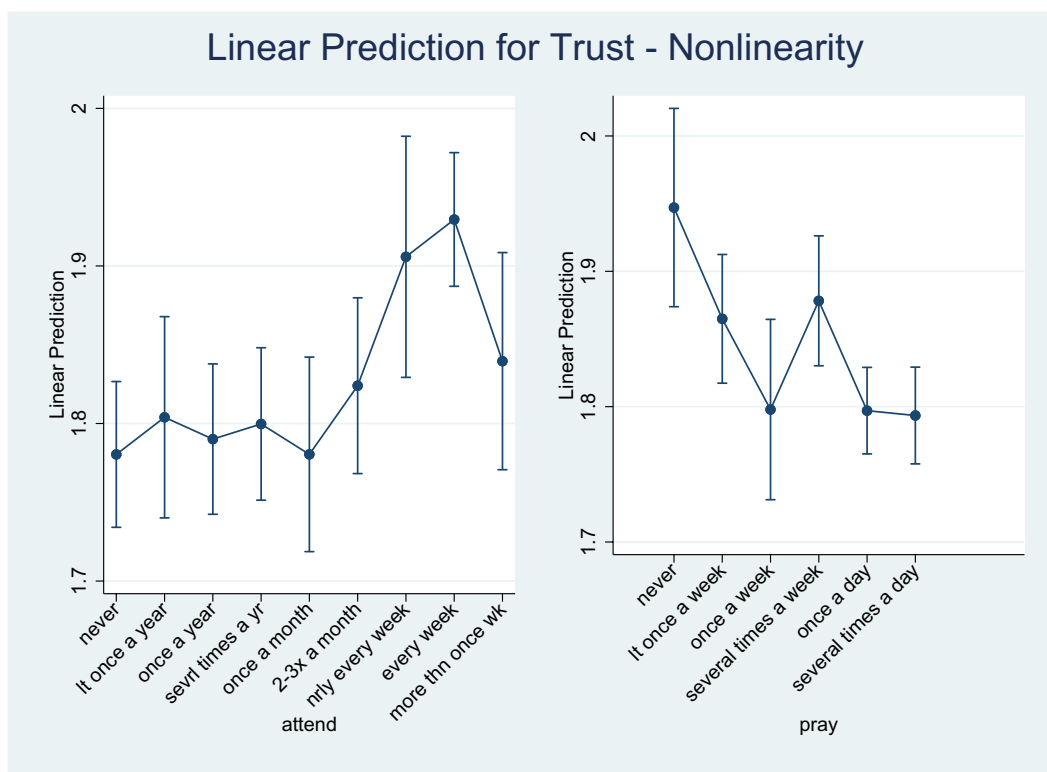
**Fig. 1** Predicted values from full models: a5 (but using binary trust in logistic regression: aLogit5 in supplementary material) and b5

These effects are not trivial. The effect sizes of ATTEND and PRAYER are at least about half that of income (a3) or even as large as income (a4), and about as large as being a liberal (a3 to a5)—relatively sizable magnitudes as compared to other predictors of trust. Irrespective of magnitude, it is interesting to find opposite effects depending on whether one believes or belongs.

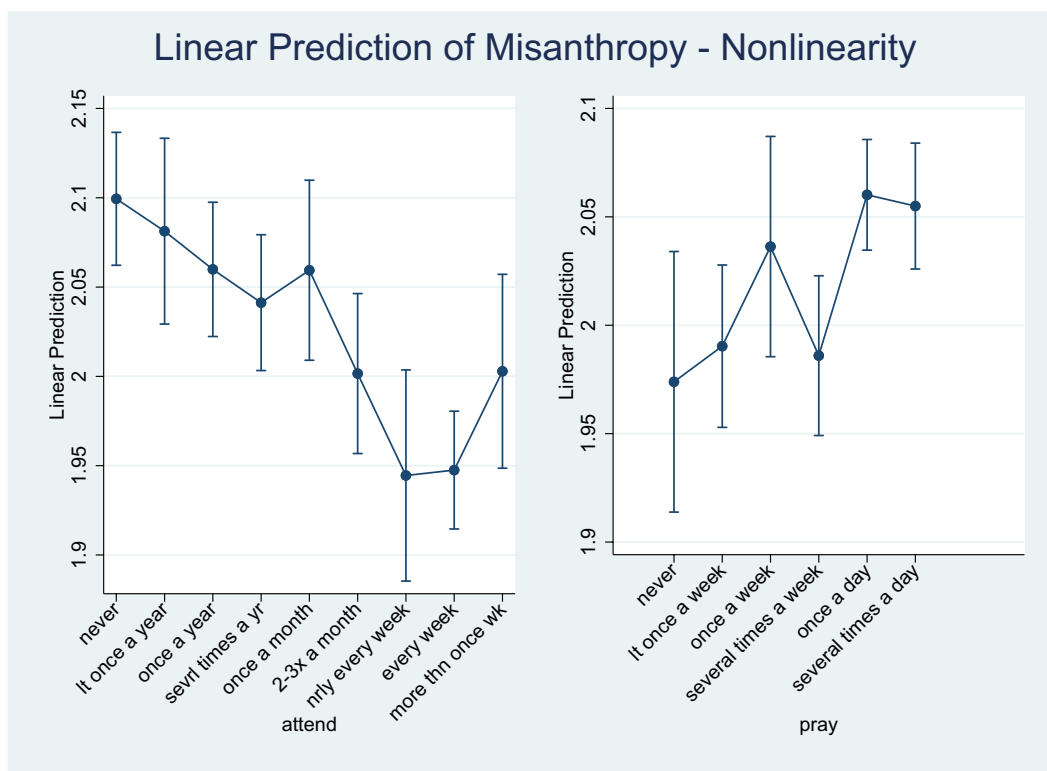
In Table 4, we regress MISANTHROPY on the same variables. Using misanthropy measures, the effect sizes are even larger—as large as having a liberal ideology. Strikingly, PRAYER increases MISANTHROPY by about as much as income decreases it in last full model.

Following Cumming (2014), it is more meaningful to interpret these coefficients by looking at 95% CI. Thus, we plot them in Fig. 1. The graphs show that increasing prayer from none (1) to more than once a day (6) is associated with a decrease in the probability of trusting others from approximately .44 to .38, and an increase from never attending religious services (0) to attending more than once a week (8) would result in about the same magnitude or boost, from .37 to .44. The differential effects of social and individual religiosity are as strong or even stronger in the second panel showing the effects on misanthropy.

Next we re-run the models using the social religiosity index and the individual religiosity index, and the results are robust as indicated in Tables 1 to 4 in the Supplemental Material.



**Fig. 2** Nonlinear relationship between attend, prayer and Trust (see “[Appendix](#)” for regressions)



**Fig. 3** Nonlinear relationship between attend, prayer and Misanthropy (see “[Appendix](#)” for regressions)

To follow Schoenfeld (1978), we also tested whether there was a curvilinear relationship between the religious measures and, trust and misanthropy, by re-running our model using dummies for the subcategories of attend and prayer.

As illustrated in Fig. 2, there's a nonlinear effect of prayer and trust—not so much for the rarely/seldom categories, but for the “daily” and “more often” categories, where we found the effect to be about 3–4 times stronger.<sup>12</sup> When examining attendance and trust we found that the strongest effect was in the “every week” which was several times stronger than the category “once a week.”

Per misanthropy, there were also clear nonlinearities as illustrated in Fig. 3. Those who prayed “daily” or “more often” show an effect several fold than the other categories, and were the only categories significant in more elaborate specifications (refer to the “Appendix”). Similar with the finding for trust, “once a week” was the strongest category, at least 2 times stronger than “more than once a week.” Lower categories also have an effect, but much smaller.

In addition to the main independent variables effects, the results for other variables in our model allow us to examine how the association between trust vary across socio-demographic characteristics including marital status, race, education and socioeconomic status. Our results are concurrent with previous findings. Trust varies across important sociodemographic characteristics, with married individuals, whites, and high-socioeconomic individuals (in terms of class, occupational prestige, education, income) having higher levels of trust (Brehm and Rahn 1997; Knack and Keefer 1997; Lindstrom 2012; Marschall and Stolle 2004; Welch et al. 2007; Bradshaw et al. 2019). We also found that individuals with greater life satisfaction and health have higher levels of trust consistent with the literature (Bradshaw and Ellison 2010; Ellison et al. 2014; Kent et al. 2018), whereas unemployed, conservative, fundamentalists, and people in the south being less trusting of others.

## Discussion

Our analysis shows that religiosity is a substantial determinant of social trust. Its effect, however, has two opposing parts: social religiosity furthers trust, whereas individual religiosity lowers it. Likewise, social religiosity lowers misanthropy, while individual religiosity promotes it. Religiosity is therefore a substantial determinant of trust and misanthropy as compared to other predictors. Although effect sizes are not large, the opposite effects of social and individual religiosity are intriguing and not entirely obvious—most people would expect that religiosity, in general, has a positive effect on the social fabric of society, including on trust.

Religious writings, as quoted at the beginning, specifically prescribe trust in God, and distrust in humans. Thus, we use ingroup favoritism outgroup derogation/prejudice theory (Tajfel 1982; Tajfel et al. 1971; Byrne 1971; McPherson et al. 2001) to conceptualize believing in God and explain why individual religiosity lowers trust. The adherent's ingroup is God, and the outgroup is other people. The ingroup bond

<sup>12</sup> See the “Appendix” for the regression results.

is formed at the expense of the outgroup. Arguably, belief in God brings self-sufficient orientation, making people free of human dependency—those who believe in God, transfer any need for dependency to God and, as a result, will distance themselves from depending on other people. In this way, God enhances individualism, but diminishes one's communal motivations.<sup>13</sup>

Studies have found that money and power also activate a self-sufficient orientation in which people value autonomy and prefer to be free of dependency (Vohs et al. 2006, 2008; Lammers et al. 2012). Power, in particular, provides a sense of independence to pursue and reach one's goals (Keltner et al. 2003; Inesi et al. 2011; Galinsky et al. 2015; Anderson and Berdahl 2002). Since for most believers God is all powerful, their belief might provide a sense of self-empowerment, resulting in the same self-sufficient behaviors observed in powerful people. This makes logical sense: If you have God who is all powerful on your side, why would you need to depend or rely on other people? To echo a Biblical verse, “If God is for us, who can be against us?” (Romans 8:31).

These findings contravene arguments that individual religiosity promotes pro-social behavior, such as charitable donations and volunteerism (Monsma 2007; Brooks 2007). We reconcile our results by arguing that pro-social behavior does not relate to increased attitudinal or cognitive trust. Norenzayan and Shariff (2008) propose possible mechanisms and explanations. Trust is not a manifest variable as volunteering or charity; it can be concealed without judgment or penalty. As a result, while individual religiosity can increase volunteering, charity, and trustworthiness, it does not increase trust. Religiously induced pro-social behavior and reputation sensitivity, do not need to translate into generalized or interpersonal trust. In fact, the motivation behind volunteering and charity could be to impress or boost one's reputation, or a result of self-deception—believers make trusting impression, but actually trust less. The believer projects an overly positive image of themselves in evaluative contexts, possibly to avoid shame or guilt.

Other studies have suggested that religion may actually create a divide in society, particularly if religious people see others as morally inferior (Berggren and Bjornskov 2009). This may result in lower levels of social trust (Uslaner 2012), despite any pro social norm that may be part of a religion ideology.

It is important to underline that individuals may not belong to any religion tradition, but still feel a connection to God or to the divine (believing without belonging) (Davie 2006)—another reason to support the use of both dimensions of religiosity. Furthermore, our results show the importance of analyzing individual and social religiosity simultaneously since we find that they correlate and have opposite effects on trust.

Our results confirm that religious involvement, or social religiosity, is positively related to social trust (Meuleman and Billiet 2011). Previous studies have shown that regular attendance of a religious service has a positive effect on the level of trust

<sup>13</sup> This may be specially advantageous under exceptionally adverse conditions, such as during war times or terminal illnesses, where dependency or trust in others is limited—when one faces eminent death, trust in God becomes more beneficial than reliance on people.

(Daniels and Von Der Ruhr 2010; Mencken et al. 2009; Uslaner 2012; Welch et al. 2004). By attending religious services and being an affiliated member of a religious group, people are more exposed to religious ideologies that promote social trust and respect for human beings (Halman and Pettersson 2001). For example, the Bible promotes the story of the ‘Good Samaritan’ (Luke 10:25–37), one of the five pillars in Islam (*Zakat*) is to give alms (charity) to the poor, and *Tzedakah* (charity) to the needy is also an obligation in Judaism. Thus, by being exposed to these ideologies of social solidarity and respect for others, one will further social trust. In addition to religious ideology, why does social religiosity predict trust? Religiosity is often a “social glue” (Gervais et al. 2011)—in many places, within the US, arguably especially in remote and rural areas, religiosity may be the major or even the only form of social engagement.

These results have practical and policy implications. Trust is a desirable outcome, an end in itself. Accordingly, efforts should be made to increase it, which is especially important today, since trust has been on decline in the U.S. for several decades (Putnam 2001). It will become increasingly difficult to live in a society without trusting people, because the economy and everyday life requires collaboration, sharing and trusting in others in a constant and daily basis (e.g., Botsman 2012)—trust helps with communication, lowers transaction cost and risk (Sosis 2005).

We are not saying that in an effort to increase trusting behavior, policies should be implemented to make people less religious. On the contrary, in a democratic society, religious freedom is an essential freedom and no attempts should be made to diminish it. At the same time, however, the negative relationship between religiosity and trust needs to be highlighted and discussed. Why are religious people less trusting? Does it make sense to trust in a divine being, but not in your “neighbor”? Isn’t this idea counter to most religious ideologies that promote social solidarity? Perhaps, by making people aware of this phenomenon, which could be a result of unconscious bias, their behavior could change.

## Limitations

This is a cross-sectional observational study, and hence we do not claim causality. There may be a problem of reverse causality: trust could affect religiosity, and not the other way round—e.g. lack of trust in humans may cause increased trust in God [(Epley et al. 2008) for example, found that loneliness increases belief in God]. Since the data is cross-sectional, we cannot properly address this issue, but we tried to by using additional controls for satisfactions with friendship and other domains, and the results were similar (refer to the Supplemental Material Tables 9 and 10). In addition, there are theoretical reasons to expect that religiosity causes lower trust: strong ingroup religiosity is formed at the expense of outgroup trust (Sosis 2005). In other words, ingroup religiosity promotes outgroup prejudice (Hall et al. 2010).

Additionally, a higher misanthropy index cronbach’s alpha (higher than .7) would have been better. However, alpha values that are borderline with .7 are also acceptable (Hair et al. 2006; Taber 2017) when you are measuring a latent variable with

three items as we do in this case. However, it's likely that the correlations are attenuated over what they would have been if the scale had higher reliability.

## Conclusion and Implications

This study finds evidence that social religiosity, or belonging, predicts more trust, whereas individual religiosity, or believing, results in lower trust. These divergent results based on whether religiosity is social or individual in character is a new conceptual approach towards religiosity not previously undertaken in the literature.

What mechanisms could be driving these results? Research by Hempel et al. (2012) found that once theological conservatism was directly modeled, there was very little difference in the average level of generalized trust and followers of different denominations. This provides some initial clues, by suggesting that a certain theological dispensation (theological conservatism) may affect the content of prayer and overall belief systems towards God versus humanity. Thus, future research, should examine whether our findings are being driven by a characteristic of all denominational forms or by a certain theological dispensation that affects the content of prayer, and overall belief system towards God versus humanity.

The ingroup favoritism and outgroup derogation theory explains our findings—particularly, given our results that misanthropy is negatively related to trust. Personal connection to God (individual religiosity) results in more misanthropy or hatred towards humankind, whereas having more social interaction in a religious setting (social religiosity) yields lower levels of misanthropy—once the ingroup expands to include more than just God, but also people, hatred for the outgroup (humans) seems to diminish. Future research should aim to design a more reliable scale to measure misanthropy using additional items when creating this index.

We focused on individual (believing) and social (belonging) religiosity, but there are other typologies worth investigating. For instance Smith (1997, p. 183) proposes that “fundamentalist beliefs, which emphasize the sinful nature of humans and a stern and authoritarian God, should be more misanthropic than those with a liberal religious orientation, which emphasizes human goodness and a compassionate and caring God.”

Future research could further examine individual denominations and expand our research by deploying interaction terms to examine denomination versus social, and denomination versus individual relationships (refer to the “Appendix”). In addition, future research may also investigate whether trust will be lower for religious adherents in countries with more fractionalized/polarized religion. It would be informative to analyze the impact of religious diversity on social trust in other countries and how it differs from our findings.

## Appendix

See Table 5.

**Table 5** Variables description. *Source:* GSS 1972–2018

| Variable              | Description—GSS variable in parenthesis                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trust                 | “Generally speaking, would you say that most people can be trusted or that you can’t be too careful in dealing with people?” (trust)                                                                                                                                                                                                                                          |
| Fair                  | “Do you think most people would try to take advantage of you if they got a chance, or would they try to be fair?” (fair)                                                                                                                                                                                                                                                      |
| Helpful               | “Would you say that most of the time people try to be helpful, or that they are mostly just looking out for themselves?” (helpfuL)                                                                                                                                                                                                                                            |
| misanthropy           | Misanthropy index: trust, fair, and helpful                                                                                                                                                                                                                                                                                                                                   |
| Attend                | “How often do you attend religious services?” (attend)                                                                                                                                                                                                                                                                                                                        |
| Member                | “Now we would like to know something about the groups or organizations to which individuals belong. Here is a list of various organizations. Could you tell me whether or not you are a member of each type?” - Church-affiliation groups<br>1=yes, 0=no”                                                                                                                     |
| Pray                  | “About how often do you pray?” 0=never to 6=several times a day (pray)                                                                                                                                                                                                                                                                                                        |
| Believe               | “Please look at this card and tell me which statement comes closest to expressing what you believe about God.” 1=does not believe to 5=extremely close (god)                                                                                                                                                                                                                  |
| Close                 | “How close do you feel to God most of the time? Would you say extremely close, somewhat close, not very close, or not close at all?” (neargod)                                                                                                                                                                                                                                |
| Family income         | Family income in \$1986, millions (income in 1972–2006 surveys in constant dollars (base = 1986)). Since this variable is based on categorical data, income is not continuous, but Based on categorical mid-points and imputations. For details see GSS Methodological Report No. 64. (realinc)                                                                               |
| Socio-economic class  | “If you were asked to use one of four names for your social class, which would you say you belong in: the lower class, the working class, the middle class, or the upper class?” (class)                                                                                                                                                                                      |
| Unemployed            | “Last week were you working full time, part time, going to school, keeping house, or what?” “Unemployed, laid off, looking for work” (wrkstat)                                                                                                                                                                                                                                |
| Occupational prestige | Prestige of respondents occupation (prestg10)                                                                                                                                                                                                                                                                                                                                 |
| Protestant            | “What is your religious preference? Is it Protestant, Catholic, Jewish, some other religion, or no religion?” (relig)                                                                                                                                                                                                                                                         |
| Catholic              | “What is your religious preference? Is it Protestant, Catholic, Jewish, some other religion, or no religion?” (relig)                                                                                                                                                                                                                                                         |
| Fundamentalist        | Fundamentalism/Liberalism of Respondents religion (fund)                                                                                                                                                                                                                                                                                                                      |
| Conservative          | “We hear a lot of talk these days about liberals and conservatives. I’m going to show you a seven-point scale on which the political views that people might hold are arranged from Extremely liberal–point 1–to extremely conservative–point 7. Where would you place yourself on this scale?” -if slightly conservative, conservative, or extremely conservative (polviews) |
| Liberal               | “We hear a lot of talk these days about liberals and conservatives. I’m going to show you a seven-point scale on which the political views that people might hold are arranged from Extremely liberal–point 1–to extremely conservative–point 7. Where would you place yourself on this scale?” -if slightly liberal, liberal, or extremely liberal (polviews)                |
| Married               | marital status: “Are you currently–married, widowed, divorced, separated, or have you never been married?” Note: variable recoded to 1 if married, 0 otherwise (marital)                                                                                                                                                                                                      |
| Age                   | age of respondent (age)                                                                                                                                                                                                                                                                                                                                                       |
| Education             | highest year of school completed (educ)                                                                                                                                                                                                                                                                                                                                       |
| Male                  | male (sex)                                                                                                                                                                                                                                                                                                                                                                    |

**Table 5** (continued)

| Variable         | Description—GSS variable in parenthesis                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Born in the U.S. | “Were you born in this country?” (born)                                                                                                                   |
| White household  | “Race of household” (hhrace)                                                                                                                              |
| Health           | “Would you say your own health, in general, is excellent, good, fair, or poor?”                                                                           |
| swb              | Subjective wellbeing “Taken all together, how would you say things are these days—would you say that you are very happy, pretty happy, or not too happy?” |

All variables were recoded so that the higher value means more, or in the case of dummy variables, one means “yes” and 0 means “no”

## Nonlinearity

See Tables 6 and 7.

**Table 6** OLS regressions of trust (nonlinearity). Beta (fully standardized) coefficients reported. *Source:* GSS 2000–2018

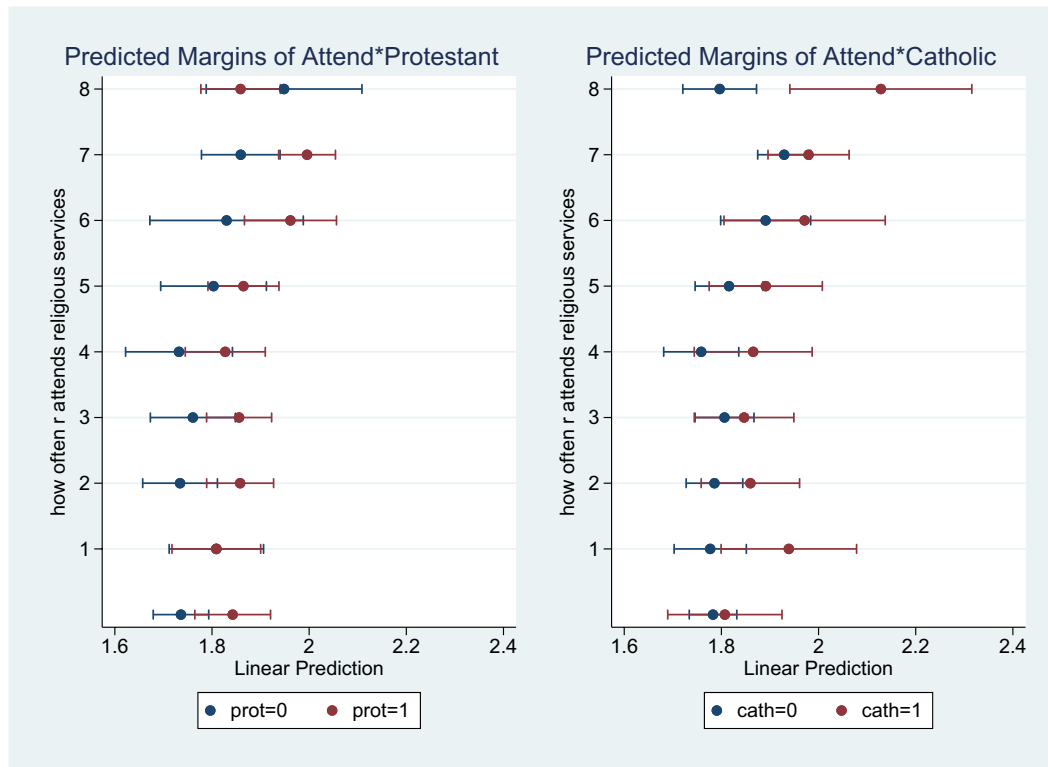
|                                   | A1        | A2        | A3        | A4        | A5        |
|-----------------------------------|-----------|-----------|-----------|-----------|-----------|
| Pray (ref: never)                 |           |           |           |           |           |
| It once a week                    | − 0.05*** | − 0.06*** | − 0.03**  | − 0.04*** | − 0.03*   |
| Once a week                       | − 0.05*** | − 0.06*** | − 0.03*** | − 0.03*** | − 0.04**  |
| Several times a week              | − 0.04*** | − 0.06*** | − 0.02    | − 0.02    | − 0.03    |
| Once a day                        | − 0.11*** | − 0.14*** | − 0.06*** | − 0.07*** | − 0.07*** |
| Several times a day               | − 0.10*** | − 0.15*** | − 0.05*** | − 0.05*** | − 0.07*** |
| Attend (ref: never)               |           |           |           |           |           |
| It once a year                    |           | 0.02*     | 0.01      | 0.01      | 0.01      |
| Once a year                       |           | 0.01      | 0.00      | 0.00      | 0.01      |
| Sevrl times a yr                  |           | 0.03***   | 0.01      | 0.01      | 0.01      |
| Once a month                      |           | 0.01      | 0.00      | 0.00      | 0.00      |
| 2–3x a month                      |           | 0.01      | 0.01      | 0.01      | 0.01      |
| nrly every week                   |           | 0.04***   | 0.02*     | 0.02*     | 0.03*     |
| Every week                        |           | 0.10***   | 0.05***   | 0.05***   | 0.06***   |
| More thn once wk                  |           | 0.02*     | 0.02      | 0.01      | 0.02      |
| South                             |           |           | − 0.07*** | − 0.07*** | − 0.06*** |
| Family income in \$1986, millions |           |           | 0.05***   | 0.04***   | 0.02      |
| Class                             |           |           | 0.06***   | 0.05***   | 0.04***   |
| Protestant                        |           |           | 0.06***   | 0.06***   | 0.05**    |
| Catholic                          |           |           | 0.03**    | 0.03**    | 0.04**    |
| Fundamentalist                    |           |           | − 0.07*** | − 0.07*** | − 0.06*** |
| Conservative                      |           |           | − 0.01    | − 0.01    | − 0.02*   |
| Liberal                           |           |           | 0.05***   | 0.05***   | 0.04***   |
| Married                           |           |           | 0.03***   | 0.01      | 0.02      |
| Unemployed                        |           |           | − 0.01    | − 0.00    | − 0.00    |
| Occupational prestige             |           |           | 0.04***   | 0.04***   | 0.04***   |
| Age                               |           |           | 0.28***   | 0.31***   | 0.43***   |
| Age squared                       |           |           | − 0.17**  | − 0.20*** | − 0.28*** |
| Highest year of school completed  |           |           | 0.23***   | 0.23***   | 0.22***   |
| Male                              |           |           | 0.04***   | 0.04***   | 0.04***   |
| Born in the U.S.                  |           |           | 0.00      | 0.00      | 0.00      |
| White household                   |           |           | 0.11***   | 0.11***   | 0.12***   |
| SWB                               |           |           |           | 0.08***   | 0.09***   |
| Health                            |           |           |           |           | 0.05***   |
| N                                 | 23,213    | 23,074    | 18,066    | 17,982    | 9616      |
| Year dummies                      | Yes       | Yes       | Yes       | Yes       | Yes       |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

**Table 7** OLS regressions of misanthropy (nonlinearity). Beta (fully standardized) coefficients reported. Source: GSS 2000–2018

|                                   | B1      | B2        | B3        | B4        | B5        |
|-----------------------------------|---------|-----------|-----------|-----------|-----------|
| Pray (ref: never)                 |         |           |           |           |           |
| It once a week                    | 0.04**  | 0.05***   | 0.02      | 0.02      | 0.01      |
| Once a week                       | 0.02*   | 0.04***   | 0.02*     | 0.02*     | 0.02      |
| Several times a week              | 0.02    | 0.05***   | 0.01      | 0.01      | 0.00      |
| Once a day                        | 0.06*** | 0.12***   | 0.06***   | 0.06***   | 0.05*     |
| Several times a day               | 0.05*** | 0.12***   | 0.05**    | 0.05***   | 0.05*     |
| Attend (ref: never)               |         | 0.00      | 0.00      | 0.00      | 0.00      |
| It once a year                    |         | − 0.02*   | − 0.01    | − 0.01    | − 0.01    |
| Once a year                       |         | − 0.03*** | − 0.02*   | − 0.02    | − 0.02    |
| Sevrl times a yr                  |         | − 0.04*** | − 0.02*   | − 0.02*   | − 0.03*   |
| Once a month                      |         | − 0.02**  | − 0.02    | − 0.01    | − 0.01    |
| 2-3x a month                      |         | − 0.03*** | − 0.03*** | − 0.03**  | − 0.04**  |
| Nrly every week                   |         | − 0.06*** | − 0.04*** | − 0.04*** | − 0.04*** |
| Every week                        |         | − 0.14*** | − 0.08*** | − 0.07*** | − 0.08*** |
| More thn once wk                  |         | − 0.05*** | − 0.04*** | − 0.03**  | − 0.03*   |
| South                             |         |           | 0.08***   | 0.08***   | 0.07***   |
| Family income in \$1986, millions |         |           | − 0.05*** | − 0.04*** | − 0.03*   |
| Class                             |         |           | − 0.09*** | − 0.08*** | − 0.07*** |
| Protestant                        |         |           | − 0.07*** | − 0.06*** | − 0.05**  |
| Catholic                          |         |           | − 0.04*** | − 0.04*** | − 0.05**  |
| Fundamentalist                    |         |           | 0.08***   | 0.07***   | 0.06***   |
| Conservative                      |         |           | 0.00      | 0.00      | 0.01      |
| Liberal                           |         |           | − 0.03*** | − 0.03*** | − 0.03**  |
| Married                           |         |           | − 0.04*** | − 0.02*   | − 0.01    |
| Unemployed                        |         |           | 0.00      | − 0.00    | − 0.00    |
| Occupational prestige             |         |           | − 0.04*** | − 0.04*** | − 0.04*** |
| Age                               |         |           | − 0.28*** | − 0.32*** | − 0.41*** |
| Age squared                       |         |           | 0.10**    | 0.14***   | 0.21***   |
| Highest year of school completed  |         |           | − 0.19*** | − 0.19*** | − 0.17*** |
| Male                              |         |           | 0.02**    | 0.02**    | 0.01      |
| Born in the U.S.                  |         |           | − 0.01    | − 0.01    | − 0.00    |
| White household                   |         |           | − 0.12*** | − 0.11*** | − 0.12*** |
| SWB                               |         |           |           | − 0.12*** | − 0.12*** |
| Health                            |         |           |           |           | − 0.07*** |
| N                                 | 22,461  | 22,327    | 17,543    | 17,463    | 9155      |
| Year dummies                      | Yes     | Yes       | Yes       | Yes       | Yes       |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err



**Fig. 4** Predicted Margins of Interaction terms from full models: Trust, 95% CI

### Interaction Terms—Differences Across Denominations

In order to see whether the differences across faiths (i.e. denominations or belonging) in how religiosity affects trust and/or misanthropy, we ran additional models and tested for the interaction effects between belonging and denomination (protestant and catholic). As shown in Fig. 4 and Table 8, the effect of belonging was only significant for Catholics who attend church more than once a week, we did not find any other significant interactions (Fig. 4, Table 8).

### Results by Decades

We ameliorate to some degree the problem of pooling over time by employing year dummies in our main analyses, but to run additional robustness test we re-ran everything separated by the 1980s, then 1990s, and 2000s/2010s. The analyses do not go back to the 1970s, since the prayer question started only in 1983. The results mostly concur—for the 1990s, once full controls are included, the main independent variables lose significance. This could be a reflection of different conditions across time.

**Table 8** OLS regressions of trust and misanthropy index. Beta (fully standardized) coefficients reported. Source: GSS 1972–2018

|                           | Trust (prot) | Misanthropy (prot) | Trust (cath) | Misanthropy (cath) |
|---------------------------|--------------|--------------------|--------------|--------------------|
| Attend (ref: never)       |              |                    |              |                    |
| It once a year            | 0.020        | – 0.015            | – 0.002      | 0.002              |
| Once a year               | – 0.001      | – 0.008            | 0.001        | – 0.019            |
| Sevrl times a yr          | 0.008        | – 0.025            | 0.008        | – 0.026            |
| Once a month              | – 0.001      | – 0.027            | – 0.007      | – 0.001            |
| 2-3x a month              | 0.020        | – 0.052**          | 0.010        | – 0.022            |
| Nrly every week           | 0.021        | – 0.042*           | 0.025*       | – 0.039*           |
| Every week                | 0.050*       | – 0.068***         | 0.059***     | – 0.071***         |
| More thn once wk          | 0.057*       | – 0.049*           | 0.004        | – 0.024            |
| Protestant = 1            | 0.055*       | – 0.053*           |              |                    |
| Interactions              |              |                    |              |                    |
| It once a year*prot       | – 0.021      | 0.011              |              |                    |
| Once a year*prot          | 0.005        | – 0.019            |              |                    |
| Sevrl times a yr*prot     | – 0.003      | – 0.002            |              |                    |
| Once a month*prot         | – 0.002      | 0.017              |              |                    |
| 2-3x a month*prot         | – 0.011      | 0.021              |              |                    |
| Nrly every week*prot      | 0.005        | – 0.004            |              |                    |
| Every week*prot           | 0.010        | – 0.010            |              |                    |
| More thn once wk*prot     | – 0.048      | 0.018              |              |                    |
| Catholic = 1              |              |                    | 0.011        | – 0.012            |
| Interactions              |              |                    |              |                    |
| It once a year*cath       |              |                    | 0.018        | – 0.023            |
| Once a year*cath          |              |                    | 0.009        | – 0.009            |
| Sevrl times a yr*cath     |              |                    | 0.003        | – 0.008            |
| Once a month*cath         |              |                    | 0.013        | – 0.028            |
| 2-3x a month              |              |                    | 0.008        | – 0.030*           |
| Nrly every week*cath      |              |                    | 0.006        | – 0.014            |
| Every week*cath           |              |                    | 0.006        | – 0.017            |
| More thn once wk*cath     |              |                    | 0.031**      | – 0.019            |
| _cons                     | (***)        | (***)              | (***)        | (***)              |
| n                         | 9616         | 9155               | 9616         | 9155               |
| Year dummies and controls | Yes          | Yes                | Yes          | Yes                |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

**1980s**

See Fig. 5, Tables 9 and 10.



**Fig. 5** Predicted values from full models A2dec5 and B2dec5 by decade: 1980s, 95% CI. See tables below

**Table 9** OLS regressions of trust for the **1980s**. Beta (fully standardized) coefficients reported. *Source*: GSS 1983–1988

|                                   | A2dec1    | A2dec2    | A2dec3    | A2dec4    | A2dec5    |
|-----------------------------------|-----------|-----------|-----------|-----------|-----------|
| Pray                              | − 0.05*** | − 0.10*** | − 0.04*   | − 0.04*   | − 0.06**  |
| Attend                            |           | 0.10***   | 0.08***   | 0.07***   | 0.06**    |
| South                             |           |           | − 0.07*** | − 0.07*** | − 0.06**  |
| Family income in \$1986, millions |           |           | 0.03      | 0.03      | 0.03      |
| Class                             |           |           | 0.01      | − 0.00    | − 0.01    |
| Protestant                        |           |           | 0.05      | 0.05      | 0.08*     |
| Catholic                          |           |           | 0.04      | 0.04      | 0.07*     |
| Fundamentalist                    |           |           | − 0.07*** | − 0.07*** | − 0.06**  |
| Conservative                      |           |           | − 0.01    | − 0.01    | − 0.03    |
| Liberal                           |           |           | 0.03      | 0.03      | 0.02      |
| Married                           |           |           | 0.04*     | 0.02      | − 0.00    |
| Unemployed                        |           |           | 0.01      | 0.02      | 0.01      |
| Occupational prestige             |           |           | 0.03      | 0.03      | 0.02      |
| Age                               |           |           | 0.39***   | 0.45***   | 0.53***   |
| Age squared                       |           |           | − 0.28**  | − 0.34*** | − 0.41*** |
| Highest year of school completed  |           |           | 0.20***   | 0.20***   | 0.18***   |
| Male                              |           |           | 0.04*     | 0.04*     | 0.04*     |
| Born in the U.S.                  |           |           | 0.01      | 0.00      | 0.01      |
| White household                   |           |           | 0.18***   | 0.17***   | 0.20***   |
| SWB                               |           |           |           | 0.11***   | 0.12***   |
| Health                            |           |           |           |           | 0.05*     |
| N                                 | 5481      | 5470      | 4307      | 4249      | 2836      |
| Year dummies                      | Yes       | Yes       | Yes       | Yes       | Yes       |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

**Table 10** OLS regressions of misanthropy index for the **1980s**. Beta (fully standardized) coefficients reported. *Source*: GSS 1983–1988

|                                   | B2dec1 | B2dec2   | B2dec3    | B2dec4    | B2dec5    |
|-----------------------------------|--------|----------|-----------|-----------|-----------|
| Pray                              | – 0.00 | 0.07***  | 0.04*     | 0.05**    | 0.06**    |
| Attend                            |        | – 0.13** | – 0.11*** | – 0.09*** | – 0.09*** |
| South                             |        |          | 0.08***   | 0.08***   | 0.08***   |
| Family income in \$1986, millions |        |          | – 0.06*** | – 0.05**  | – 0.05*   |
| Class                             |        |          | – 0.03*   | – 0.02    | – 0.03    |
| Protestant                        |        |          | – 0.05    | – 0.05    | – 0.06    |
| Catholic                          |        |          | – 0.04    | – 0.05    | – 0.06    |
| Fundamentalist                    |        |          | 0.05**    | 0.04*     | 0.04*     |
| Conservative                      |        |          | 0.01      | 0.02      | 0.01      |
| Liberal                           |        |          | – 0.01    | – 0.01    | – 0.02    |
| Married                           |        |          | – 0.03    | – 0.00    | 0.02      |
| Unemployed                        |        |          | – 0.01    | – 0.01    | – 0.02    |
| Occupational prestige             |        |          | – 0.04*   | – 0.04*   | – 0.03    |
| Age                               |        |          | – 0.33*** | – 0.41*** | – 0.50*** |
| Age squared                       |        |          | 0.17      | 0.24**    | 0.32**    |
| Highest year of school completed  |        |          | – 0.21*** | – 0.20*** | – 0.17*** |
| Male                              |        |          | 0.02      | 0.02      | 0.01      |
| Born in the U.S.                  |        |          | 0.00      | 0.00      | – 0.02    |
| White household                   |        |          | – 0.21*** | – 0.20*** | – 0.20*** |
| SWB                               |        |          |           | – 0.13*** | – 0.14*** |
| Health                            |        |          |           |           | – 0.06**  |
| N                                 | 5408   | 5398     | 4268      | 4212      | 2810      |
| Year dummies and controls         | Yes    | Yes      | Yes       | Yes       | Yes       |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

**1990s**

See Fig. 6, Tables 11 and 12.



**Fig. 6** Predicted values from full models A3dec5 and B3dec5 by decade: 1990s, 95% CI. See tables below

**Table 11** OLS regressions of trust for the **1990s**. Beta (fully standardized) coefficients reported. *Source:* GSS 1990–1998

|                                   | A3dec1   | A3dec2    | A3dec3   | A3dec4   | A3dec5  |
|-----------------------------------|----------|-----------|----------|----------|---------|
| Pray                              | − 0.06** | − 0.08*** | − 0.02   | − 0.02   | 0.00    |
| Attend                            |          | 0.04*     | 0.01     | 0.01     | − 0.03  |
| South                             |          |           | − 0.05** | − 0.06** | − 0.03  |
| Family income in \$1986, millions |          |           | 0.04     | 0.05     | − 0.01  |
| Class                             |          |           | 0.03     | 0.03     | 0.05    |
| Protestant                        |          |           | 0.08*    | 0.08*    | 0.07    |
| Catholic                          |          |           | 0.04     | 0.04     | 0.02    |
| Fundamentalist                    |          |           | − 0.05*  | − 0.05*  | − 0.03  |
| Conservative                      |          |           | 0.03     | 0.03     | 0.07*   |
| Liberal                           |          |           | 0.07**   | 0.07**   | 0.10**  |
| Married                           |          |           | − 0.01   | − 0.01   | 0.03    |
| Unemployed                        |          |           | − 0.01   | − 0.01   | 0.00    |
| Occupational prestige             |          |           | 0.02     | 0.01     | 0.04    |
| Age                               |          |           | 0.28*    | 0.28*    | 0.52**  |
| Age squared                       |          |           | − 0.17   | − 0.17   | − 0.36* |
| Highest year of school completed  |          |           | 0.22***  | 0.21***  | 0.20*** |
| Male                              |          |           | 0.05*    | 0.05*    | 0.04    |
| Born in the U.S.                  |          |           | 0.02     | 0.02     | 0.00    |
| White household                   |          |           | 0.14***  | 0.13***  | 0.15*** |
| SWB                               |          |           |          | 0.03     | 0.05    |
| Health                            |          |           |          |          | 0.01    |
| N                                 | 3388     | 3326      | 2515     | 2500     | 1018    |
| Year dummies                      | Yes      | Yes       | Yes      | Yes      | Yes     |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

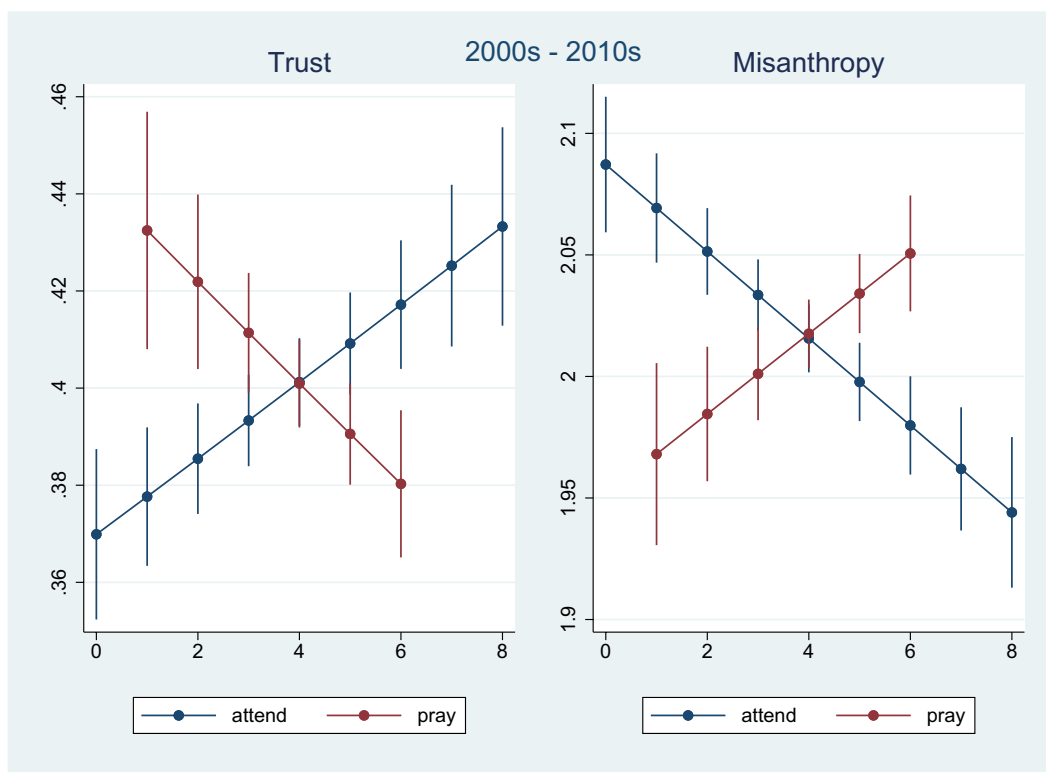
**Table 12** OLS regressions of misanthropy index for the **1990s**. Beta (fully standardized) coefficients reported. *Source*: GSS 1990–1998

|                                   | B3dec1 | B3dec2    | B3dec3    | B3dec4    | B3dec5    |
|-----------------------------------|--------|-----------|-----------|-----------|-----------|
| Pray                              | 0.02   | 0.07**    | 0.05*     | 0.04      | 0.03      |
| Attend                            |        | − 0.08*** | − 0.05*   | − 0.04    | 0.00      |
| South                             |        |           | 0.08***   | 0.09***   | 0.06      |
| Family income in \$1986, millions |        |           | − 0.03    | − 0.03    | − 0.01    |
| Class                             |        |           | − 0.06**  | − 0.05*   | − 0.05    |
| Protestant                        |        |           | − 0.14*** | − 0.14*** | − 0.16**  |
| Catholic                          |        |           | − 0.09**  | − 0.10**  | − 0.11*   |
| Fundamentalist                    |        |           | 0.08***   | 0.08***   | 0.06      |
| Conservative                      |        |           | − 0.01    | − 0.01    | − 0.07*   |
| Liberal                           |        |           | − 0.03    | − 0.03    | − 0.09**  |
| Married                           |        |           | − 0.03    | − 0.01    | − 0.01    |
| Unemployed                        |        |           | 0.01      | 0.00      | − 0.01    |
| Occupational prestige             |        |           | − 0.01    | − 0.01    | − 0.04    |
| Age                               |        |           | − 0.46*** | − 0.48*** | − 0.55*** |
| Age squared                       |        |           | 0.31**    | 0.33**    | 0.36*     |
| Highest year of school completed  |        |           | − 0.22*** | − 0.22*** | − 0.20*** |
| Male                              |        |           | 0.04*     | 0.04*     | 0.03      |
| Born in the U.S.                  |        |           | − 0.03    | − 0.03    | − 0.02    |
| White household                   |        |           | − 0.17*** | − 0.16*** | − 0.15*** |
| SWB                               |        |           |           | − 0.07*** | − 0.12*** |
| Health                            |        |           |           |           | − 0.01    |
| N                                 | 3348   | 3286      | 2488      | 2474      | 1008      |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

**2000–2010s**

See Fig. 7, Tables 13 and 14.



**Fig. 7** Predicted values from full models A4dec5 and B4dec5 by decade: 2000s-2010s, 95% CI. See tables below

**Table 13** OLS regressions of trust for the **2000s and 2010s**. Beta (fully standardized) coefficients reported. *Source*: GSS 2000–2018

|                                   | A4dec1    | A4dec2    | A4dec3    | A4dec4    | A4dec5    |
|-----------------------------------|-----------|-----------|-----------|-----------|-----------|
| Pray                              | − 0.06*** | − 0.09*** | − 0.02**  | − 0.03**  | − 0.04**  |
| Attend                            |           | 0.07***   | 0.04***   | 0.04***   | 0.05***   |
| South                             |           |           | − 0.07*** | − 0.07*** | − 0.06*** |
| Family income in \$1986, millions |           |           | 0.04***   | 0.04***   | 0.02      |
| Class                             |           |           | 0.06***   | 0.05***   | 0.04***   |
| Protestant                        |           |           | 0.05***   | 0.05***   | 0.04**    |
| Catholic                          |           |           | 0.02*     | 0.02*     | 0.03*1    |
| Fundamentalist                    |           |           | − 0.07*** | − 0.07*** | − 0.06*** |
| Conservative                      |           |           | − 0.01    | − 0.01    | − 0.02*   |
| Liberal                           |           |           | 0.05***   | 0.05***   | 0.04***   |
| Married                           |           |           | 0.03***   | 0.01      | 0.02*     |
| Unemployed                        |           |           | − 0.01    | − 0.00    | − − 0.00  |
| Occupational prestige             |           |           | 0.04***   | 0.04***   | 0.04***   |
| Age                               |           |           | 0.28***   | 0.31***   | 0.43***   |
| Age squared                       |           |           | − 0.17*** | − 0.19*** | − 0.28*** |
| Highest year of school completed  |           |           | 0.19***   | 0.19***   | 0.18***   |
| Male                              |           |           | 0.04***   | 0.04***   | 0.04***   |
| Born in the U.S.                  |           |           | 0.00      | 0.00      | 0.00      |
| White household                   |           |           | 0.11***   | 0.11***   | 0.12***   |
| SWB                               |           |           |           | 0.08***   | 0.09***   |
| Health                            |           |           |           |           | 0.05***   |
| N                                 | 23213     | 23074     | 18066     | 17982     | 9616      |
| Year dummies                      | Yes       | Yes       | Yes       | Yes       | Yes       |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

**Table 14** OLS regressions of misanthropy index for the **2000s and 2010s**. Beta (fully standardized) coefficients reported. *Source*: GSS 2000–2018

|                                   | B4dec1  | B4dec2    | B4dec3    | B4dec4    | B4dec5    |
|-----------------------------------|---------|-----------|-----------|-----------|-----------|
| Pray                              | 0.02*** | 0.08***   | 0.04***   | 0.04***   | 0.04**    |
| Attend                            |         | – 0.11*** | – 0.07*** | – 0.06*** | – 0.06*** |
| South                             |         |           | 0.08***   | 0.08***   | 0.07***   |
| Family income in \$1986, millions |         |           | – 0.05*** | – 0.04*** | – 0.03*   |
| Class                             |         |           | – 0.09*** | – 0.08*** | – 0.07*** |
| Protestant                        |         |           | – 0.06*** | – 0.06*** | – 0.05*** |
| Catholic                          |         |           | – 0.04*** | – 0.04*** | – 0.05*** |
| Fundamentalist                    |         |           | 0.08***   | 0.08***   | 0.06***   |
| Conservative                      |         |           | 0.00      | 0.00      | 0.01      |
| Liberal                           |         |           | – 0.03*** | – 0.03*** | – 0.03**  |
| Married                           |         |           | – 0.04*** | – 0.02*   | – 0.01    |
| Unemployed                        |         |           | 0.00      | – 0.00    | – 0.00    |
| Occupational prestige             |         |           | – 0.04*** | – 0.04*** | – 0.04*** |
| Age                               |         |           | – 0.27*** | – 0.31*** | – 0.40*** |
| Age squared                       |         |           | 0.10*     | 0.14***   | 0.21***   |
| Highest year of school completed  |         |           | – 0.19*** | – 0.19*** | – 0.17*** |
| Male                              |         |           | 0.02**    | 0.02**    | 0.01      |
| Born in the U.S.                  |         |           | – 0.01    | – 0.01    | – 0.00    |
| White household                   |         |           | – 0.12*** | – 0.11*** | – 0.12*** |
| SWB                               |         |           |           | – 0.12*** | – 0.12*** |
| Health                            |         |           |           |           | – 0.07*** |
| N                                 | 22,461  | 22,327    | 17,543    | 17,463    | 9155      |
| Year dummies                      | Yes     | Yes       | Yes       | Yes       | Yes       |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , robust std err

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### 3.3 Constructing General Human Agency Indicators (GHAls) and a General Personal Agency Scale (GPAS) (D'Italia and Okulicz-Kozaryn 2025)



# Constructing General Human Agency Indicators (GHAI)s and a General Personal Agency Scale (GPAS)

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## Abstract

Despite its importance for the social sciences, *human agency* remains an ambiguous and underoperationalized construct. After engaging prior research to articulate clear criteria for defining agency and synthesize a multidimensional conceptual framework for human agency, this study develops and validates preliminary General Human Agency Indicators (GHAI)s to measure subconstructs within that framework. Utilizing the Midlife in the United States (MIDUS) dataset, we aggregated a list of 30 survey items previously used in agency research and conducted an iterative process of exploratory factor analysis (EFA) and item elimination to reduce that list to a set of 9–13 items with a strong, consistent factorial structure. Using confirmatory factor analysis (CFA), we identified two bifactor models that demonstrated good fit: a nine-item General Personal Agency scale (GPAS) and a nine-item GHAI)s tool combining six items from the GPAS with three measuring agency achievement. Initial evidence for the construct validity of the tools was produced through tests of internal consistency and correlational analysis, indicating that the proposed GPAS and GHAI)s effectively measure personal agency, intrinsic agency, instrumental agency, and agency achievement.

**Keywords** Human agency · Personal agency · Capabilities approach · Self-determination Theory · MIDUS

## 1 Introduction

### 1.1 Human Agency, Underoperationalization, and Mismeasurement

*Human agency* (or *agency*) refers to a person's capacity to enact control over their lives and engage their physical and social environments to pursue self-determined goals (Ahearn, 2001; Kabeer, 1999; Kotan, 2010; Sen, 1985). A fundamental mechanism through which the social world is constructed and navigated; agency is a critical topic for the social

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sciences. A literature review by Cavazzoni et al. (2022) provides an overview of how agency has been explored in fields as diverse as psychology, sociology, social geography, global development, women's empowerment, childhood studies, and research on morality and sexuality.

Agency has particular importance for fields that engage human well-being. For example, Bhattacharyya (1995) defines the discipline of community development as the cultivation of social relations characterized by solidarity and agency and, in the Capabilities Approach to development, it is discussed as the enactment of human freedom, the expansion of which is the ultimate purpose of development activities (Sen, 1985, 1999). Human agency is a topic of interest to the study of subjective well-being, as well-being is produced when individuals achieve self-determined goals and functionings (Comim, 2005; Kotan, 2010). Research demonstrates a strong positive relationship between agency and life satisfaction (Graham & Nikolova, 2013; Hojman & Miranda, 2018; Wang, 2015), experienced across cultures as a consistent sequence: (1) as socioeconomic opportunities expand, people place greater value on freedom, (2) as valuation of freedom increases, so does the influence of agency on life satisfaction, (3) life satisfaction increases commensurate to the increased effect of agency (Welzel & Inglehart, 2010). Agency has also been discussed in the context of its relationships with empowerment (Chavis & Wanderman, 1990; Kieffer, 1984), prosocial behaviors (Christoph et al., 2014), community engagement (Peterson et al., 2008), volunteerism (Cicognani et al., 2015), and democratic participation (Sen, 1999).

Despite and, in some sense, because of its significance across disciplines, definitions of human agency abound, with no clear consensus regarding how the construct should be operationalized (Cavazzoni et al., 2022; Ibrahim & Alkire, 2007). Agency has been conceptualized as an internal attitude (Alsop et al., 2006; Pleeging et al., 2021), one's capacity for action (Cavazzoni et al., 2022; Giddens, 1984; Graham & Nikolova, 2013; Onyx & Bullen, 2000; Smith et al., 2000), or some combination thereof (Bryan et al., 2014; Hitlin & Elder, 2006; Kabeer, 1999; Kotan, 2010; Sen, 1985; Williams & Merten, 2014). Some extend the term to explicitly include interactions with one's environment (Bentley-Edwards, 2016; Christensen & Hooker, 2000; Horvath, 1998; Krauss et al., 2014) and relationships with others, either alongside (Alsop et al., 2006; Bandura, 2018; Bhattacharyya, 1995; Narayan et al., 2007; Yount et al., 2020) or independently from them (Lautamo et al., 2021; Salem et al., 2020; Steckermeier, 2019). Agency has also been used to describe both individual and collective capacity to participate in and transform existing sociocultural structures and norms (Bhattacharyya, 1995; Harvey, 2002; Lautamo et al., 2021; Veronese et al., 2019b; Zimmerman et al., 2019). A table expanding on Cavazzoni et al.'s (2022) original survey of previous agency definitions is included as Appendix A.

Disagreement over how agency is defined inevitably leads to challenges related to its measurement (Cavazzoni et al., 2022). A critical example of this is how agency is operationalized alongside communion as one of the "Big Two" traits in personality research (Gebauer et al., 2014). In personality research, agency refers to one's extraversion and openness to experience (DeYoung, 2006; Digman, 1997; Erdle et al., 2009; Paulhus & John, 1998) and their desire for independence, individuation, and "agentic contrast" (p. 454)—that is, how one articulates their identity by differentiating themselves from others. The agency personality construct has been useful for exploring personality (Gebauer et al., 2014) and for organizing psychological characteristics related to social behaviors (Wiggins, 1991), social values (Trapnell & Paulhus, 2012), self-enhancement strategies (Campbell et al., 2002), and developmental goals (Charles & Carstensen, 2010). However, while the agency personality construct shares some similarities with human agency as it has been engaged elsewhere in the social sciences, in that it reflects a mode of enacting control and over oneself and one's environment, none of

the characteristics attributed to the agency personality trait are necessary for the self-determination, pursuit, or achievement of goals per se, which is an essential component of human agency. Rather, the agency personality trait is better understood as something akin to individualism, complementing the collectivistic orientation of the communion trait.

Agency has also been assessed using proxy measures like income, education, employment opportunities, and access to resources (Alkire, 2008; Alsop et al., 2006; Bhattacharyya, 1995; Kabeer, 1999). However, this approach has been criticized on the grounds that proxy measures are poor indicators of what people value most (Anand et al., 2009; Helliwell & Barrington-Leigh, 2010). Socioeconomic proxies are merely means to the greater end of advancing human freedom, and one's ability to convert them into desired outcomes is influenced by personal, social, political, cultural, and environmental factors (Alkire, 2008; Sen, 1999). Proxy measures also risk creating unobserved or confounding variable bias, as multiple proxies might produce similar effects on agency; this risk is compounded in fields like poverty analysis where proxies are already in regular use and limits the capacity for research to probe interactions between agency and other topics of interest (Alkire, 2008). Similarly, tools like Vallacher and Wegner's (1989) Behavior Identification Form (BIF) and Yount et al.'s (2020) Women's Agency Scale 61 (WAS-61) use behaviors and measures for perceived influence over domains and activities to assess agency. While these measures were useful for the authors' specific purposes of assessing action identity and women's agency, respectively, they do not allow for subjective valuation of the relevant activities, which is important in agency research (Alkire, 2005). This effectively causes them to function as proxy indicators, subject to the limitations thereof.

Several attempts have been made to develop direct, subjective indicators for human agency, examples of which are shown in Table 1 alongside sample items from the BIF. Some researchers have deployed single-item indicators (e.g., Graham & Nikolova, 2013; Hojman & Miranda, 2018), eliminating opportunities to the dimensionality of the construct. Even when they consist of multiple items, direct measures of agency are often unidimensional and conflated with concepts like autonomy, freedom, internal locus of control, purposeful choice, and self-efficacy (c.f. Alsop et al., 2006; Bandura, 2018; Cavazzoni et al., 2022; Graham & Nikolova, 2013; Hojman & Miranda, 2018; Inglehart et al., 2008; Veenhoven, 2000; Verme, 2009; Welzel & Inglehart, 2010). While these concepts are likely important for informing our understanding of human agency and how it functions, they are previously defined psychosocial constructs; used in isolation as unidimensional indicators, they too function as proxy measures. Unidimensional assessment of agency also fails to address a preponderance of literature that conceptualizes it as multidimensional (c.f. Alkire, 2008; Bandura, 2001; Kotan, 2010; Sen, 1999; Smith et al., 2000; Yount et al., 2020). Scholars like Lautamo et al. (2021), Smith et al. (2000), and Yount et al. (2020) have successfully developed direct, subjective, multidimensional indicators to measure aspects of human agency, but none of these measures on their own are intended to measure general human agency but, rather, subconstructs defined by the authors that do not necessarily engage conceptualizations or latent structures proposed by others.

## 2 Conceptualizing Human Agency

There is presently an "urgent need" to develop a shared understanding of human agency and common tools for its measurement (Cavazzoni et al., 2022, p. 1148). Exploration of previous agency research produces several insights regarding its common themes and characteristics. In their analysis, Cavazzoni et al. (2022) define agency as "people's ability to

**Table 1** Sample measures from previous agency research

|                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single-item measures                                                  | I feel free to decide for myself how to lead my life.<br>(Hojman & Miranda, 2018)                                                                                                                                                                                                                                                                                                                                |
|                                                                       | How satisfied/dissatisfied are you with “Your freedom to choose what you do with your life”.<br>(Graham & Nikolova, 2013)                                                                                                                                                                                                                                                                                        |
|                                                                       | Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means “none at all” and 10 means “a great deal” to indicate how much freedom of choice and control you feel you have over the way your life turns out.<br>(Inglehart et al., 2008; Welzel & Inglehart, 2010) |
|                                                                       | I feel free to decide for myself how to lead my life.<br>(Hojman & Miranda, 2018)                                                                                                                                                                                                                                                                                                                                |
| Behavior Identification Form (BIF) (Vallacher & Wegner, 1989)         | How satisfied/dissatisfied are you with “Your freedom to choose what you do with your life”.<br>(Graham & Nikolova, 2013)                                                                                                                                                                                                                                                                                        |
|                                                                       | <i>Your task is to choose the identification, a or b, that best describes the behavior for you. Simply place a check mark in the space beside the identification statement that you pick</i>                                                                                                                                                                                                                     |
|                                                                       | Making a list<br><i>a. Getting organized*</i><br><i>b. Writing things down</i>                                                                                                                                                                                                                                                                                                                                   |
|                                                                       | Reading<br><i>a. Following lines of print</i><br><i>b. Gaining knowledge*</i>                                                                                                                                                                                                                                                                                                                                    |
| Personal Agency Scale Items (Smith et al., 2000)                      | Joining the Army<br><i>a. Helping the Nation’s defense*</i><br><i>b. Signing up</i><br>*Higher-level alternative                                                                                                                                                                                                                                                                                                 |
|                                                                       | I get what I want or need by relying on my own efforts and ability                                                                                                                                                                                                                                                                                                                                               |
|                                                                       | I control what happens to me by making choices in my best interest                                                                                                                                                                                                                                                                                                                                               |
|                                                                       | Using the right resources or tools helps me to achieve my goals                                                                                                                                                                                                                                                                                                                                                  |
| Assessment tool for perceived agency (ATPA-22) (Lautamo et al., 2021) | I feel that different areas of my daily life are balanced                                                                                                                                                                                                                                                                                                                                                        |
|                                                                       | I am satisfied with the amount of daily activities I manage to do                                                                                                                                                                                                                                                                                                                                                |
|                                                                       | I feel that I have a suitable amount to do on a daily basis                                                                                                                                                                                                                                                                                                                                                      |
| Measurement model of agency (Hitlin & Elder, 2006)                    |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Planfulness</i>                                                    | When you have a problem to solve, one of the first things you do is get as many facts about the problem as possible                                                                                                                                                                                                                                                                                              |
| <i>Optimism</i>                                                       | How likely is it that you will go to college?                                                                                                                                                                                                                                                                                                                                                                    |
| <i>Self-efficacy</i>                                                  | When get what you want, it’s usually because you worked hard for it                                                                                                                                                                                                                                                                                                                                              |

**Table 1** (continued)

|                                                       |                                                                                               |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Interpersonal agency scale items (Smith et al., 2000) | I achieve my goals by knowing when to ask others for help                                     |
|                                                       | I accomplish my goals by letting others know my needs and wants                               |
|                                                       | I get what I want or need by seeking the advice of others                                     |
| Collective agency (Yount et al., 2020)                | I feel like I have a pretty good understanding of the important issues that face my community |
|                                                       | I am often a leader in groups                                                                 |
|                                                       | I can usually organize people to get things done                                              |

exert control over one's life and pursue goals" (p. 1126). Similarly, Kabeer (1999) defines it as "the ability to define one's goals and act upon them." (p. 438). Broadly, these definitions align with a preponderance of those used in other studies, as shown in Appendix A, and reflect two aspects of agency, control and goal pursuit, that are commonly encountered in discussions on the topic.

Other scholars have sought to articulate the fundamental characteristics of human agency. Alkire (2008) identifies five key features of Sen's (1999) conceptualization of human agency:

*"(i) agency is exercised with respect to goals the person values; (ii) agency includes effective power as well as direct control; (iii) agency may advance wellbeing or may address other-regarding goals; (iv) to identify agency also entails an assessment of the value of the agent's goals; (v) the agent's responsibility for a state of affairs should be incorporated into his or her evaluation of it" (p. 6).*

Kotan (2010) describes agency as involving "(a) action, power and causality, (b) purposiveness and (c) the determination of objectives" (p. 369); these three characteristics are further reduced to two: "The ability to act to influence or affect the state of the world" and "The ability to judge and reflect upon goals and situations and to determine one's own goals and objectives as reasons for action" (Kotan, 2010, p. 370). Burger and Walk (2016) also identify three elements of agency—perceived control, commitment to self-determined goals, and self-efficacy—that parallel Kotan's. This pattern that emerges when these sets of characteristics are compared suggests that any operationalization of human agency should, at a minimum, engage one's enactment of *control* over one's life through the *self-determination* and pursuit of values, goals, and desired outcomes, their capacity to *influence* circumstances to achieve those goals and outcomes, and their *ability* to perform behaviors and leverage resources to exert the influence necessary to achieve desired ends.

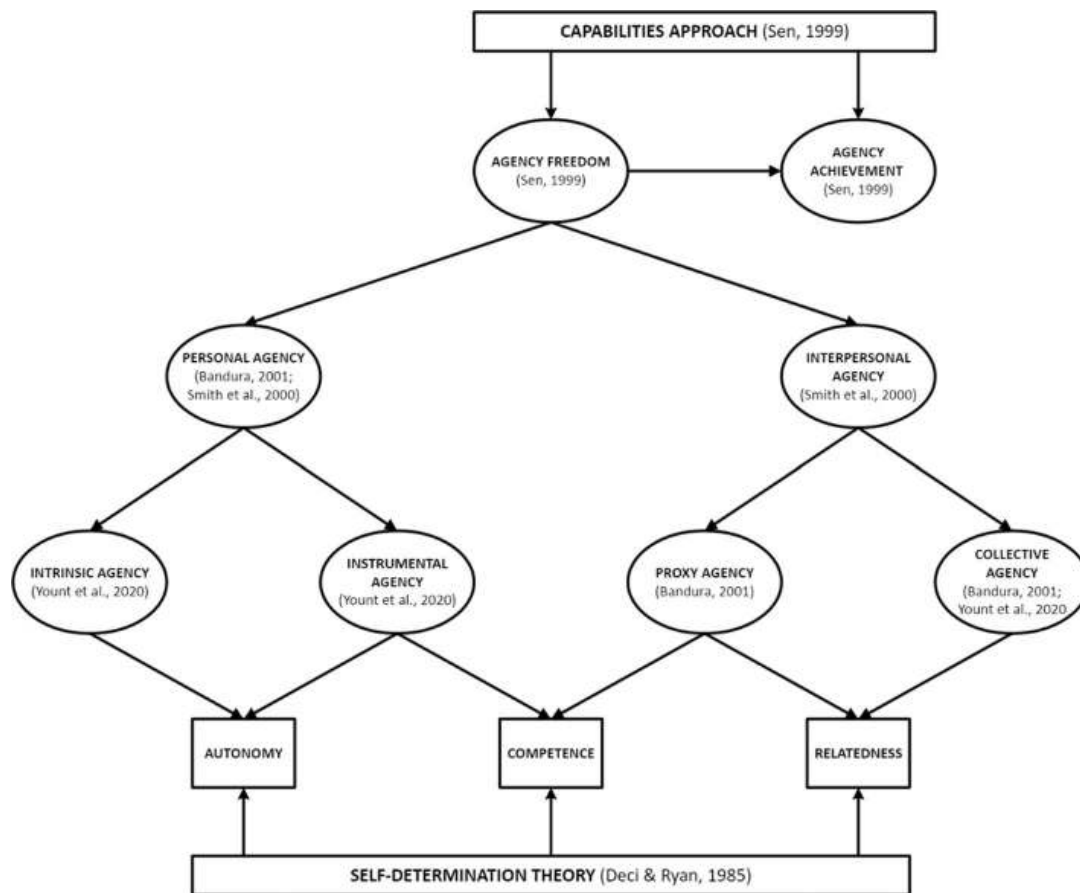
Many scholars have also explored various dimensions of human agency. Sen (1999) asserts that agency necessarily involves two elements: *agency freedom*, the capacity or potential to pursue goals, and *agency achievement*, one's success in achieving desired outcomes. Agency freedom is a prerequisite for agency achievement, as it reflects the conditions under which one strives to attain their goals (Sen, 1999). In practice, research typically focuses only on agency freedom, although some scholars have utilized well-being indicators to approximate agency achievement (Hojman & Miranda, 2018; Okulicz-Kozaryn, 2015; Veenhoven, 2000). Agency freedom is itself comprised of both personal

and social competencies (Cavazzoni et al., 2022) and involves one's ability to advance personal goals either on their own or in cooperation with others (Narayan & Petesch, 2007). Smith et al. (2000) articulates these domains as *personal agency*, one's capacity to achieve goals through individual efforts, and *interpersonal agency*, the ability to engage others to cooperatively pursue desired outcomes. As with agency freedom, scholarship tends to focus on personal agency, rather than interpersonal agency.

Bandura (2001) describes personal agency as the ability to make choices, plan actions, and perform those actions effectively, which he distinguishes from what could be considered two subdimensions of interpersonal agency: *proxy agency*, one's ability to leverage social relationships to pursue personal goals, and *collective agency*, the capability of groups to produce, pursue, and attain shared goals. Similarly, Yount et al. (2020) engage the concept of collective agency and articulate alongside it two dimensions subsumable under personal agency: *intrinsic agency*, one's internal motivations, perceptions, and attitudes related to the determination and pursuit of goals, and *instrumental agency*, the array of strategies one has available to enact their freedom, achieve goals, and establish control over their life. Reciprocally, proxy and collective agency (Bandura, 2001) could be considered subdimensions of Smith et al.'s interpersonal agency. These various categorizations of agency are mutually complementary and can be organized into a cohesive conceptual framework, shown in Fig. 1. Agency achievement is produced by agency freedom, which is comprised of both personal and interpersonal dimensions. Personal agency is made up of intrinsic and instrumental subdimensions, while interpersonal agency represents one's capacity to engage others to achieve personal (proxy agency) or common goals (collective agency).

The proposed framework for agency aligns with the basic human needs defined by Self-Determination Theory (SDT) (Deci & Ryan, 1985). SDT posits that all people possess inherent psychological needs for autonomy, competence, and relatedness; pursuing fulfillment of these needs is fundamental to human experience, necessary for well-being (Ryan & Deci, 2001) and contributes to personal growth, intrinsic motivation, vitality and "aliveness" (Ryan & Frederick, 1997, p. 530), and concordance of goals with personal interests and values (Sheldon & Elliot, 1999). Autonomy is reflected in both one's intrinsic and instrumental agency, competence is represented by instrumental and proxy agency, and both proxy and collective agency engage one's relatedness to others. Alkire (2005, 2008) has previously discussed theoretical connections between human agency and SDT in her exploration of subjective quantitative agency measures, noting that indicators for both autonomy and competence are important for the study of agency and that agency necessarily interacts with one's relatedness, as one may possess motivations and goals that involve other individuals and their well-being. Alkire (2005) also makes connections between agency and well-being, specifically, Ryff's (1989) multidimensional model of Psychological Well-Being (PWB); this supports Sen's position that agency freedom and achievement precede and contribute to well-being freedom and achievement.

The present framework also conforms to the elements of control, self-determination, influence, and ability that are common across discussions of the characteristics of agency. One's capacity for self-determination is manifested through both intrinsic and collective agency, and their ability to pursue self-determined ends is exercised through both instrumental and proxy agency. Subsumed under agency freedom, the dimensions of personal and interpersonal agency represent the totality of one's capability to enact control by influencing their circumstances to achieve those ends. As it was produced by interweaving strands of previous agency



**Fig. 1** Conceptual framework of human agency

research, aligns with existing theorizations of how agency functions, and engages the essential agency characteristics of control, self-determination, influence, and ability, we assert that the proposed conceptual framework for human agency is a useful starting point for developing a common operationalization of the construct.

However, without effective indicators for measurement, even the best conceptual framework remains confined to the realm of abstract theory. Now that essential characteristics of human agency have been identified and the concept has been organized into a multidimensional framework that synthesizes existing scholarship on the topic, we now turn our attention to the development of preliminary General Human Agency Indicators (GHAI) sufficient for assessing agency and its subdimensions. Such measures may contribute to a deeper understanding of agency, its effects on human functioning, and how individuals interact with their environments, circumstances and resources to pursue well-being. This, in turn, could have significant implications for global and community development, the formulation of public policy, implementation of empowerment activities, and social science research (Cavazzoni et al., 2022; Sen, 1999).

### 3 Methods

#### 3.1 Study Design, Data, and Item Selection

We utilized secondary data analysis with waves II (MIDUS II) and III (MIDUS III) of The Midlife in the United States (MIDUS) dataset, which were collected from 2004 to 2006 and 2013 to 2014, respectively (Ryff et al., 2017, 2019). MIDUS is a nationally representative dataset that includes information about a variety of psychological and socioeconomic subjects. Data was collected through an initial phone interview, with respondents selected by random digit dialing, followed by respondents' completion of a self-administered questionnaire. MIDUS II contains 4,032 observations of "non-institutionalized, English-speaking adults in the coterminous United States, aged 35 to 86" (MIDUS II documentation) who completed both the phone interview and questionnaire; MIDUS III includes 2,732 adults, now aged 40 to 94, who completed the interview and questionnaire and had participated in the MIDUS II study.

Using Cavazzoni et al.'s (2022) analysis as a starting point, we aggregated a list of survey tools and items used in previous scholarship on human agency or one of its sub-dimensions. We then searched the MIDUS codebook for items that were a) identical to an item used in prior research, b) similar to a previously used item, or c) the reverse of such an item. Next, we evaluated items to ensure they reflected at least one of the agency characteristics of control, self-determination, influence, or ability and conformed to the characteristics of subjectivity discussed by Alkire's (2005): subjective measures should a) represent the perceptions and valuations of the subject, b) allow for both positive and negative assessments, and c) emphasize overarching or enduring perceptions and valuations instead of "fleeting emotional states" (p. 222). We also assessed items for alignment with Alkire's (2008) categorization of agency measures, which include complementary global and multidimensional measures, measures of effective power or direct control, measures of the advancement of well-being and other valued outcomes, and measures of autonomy and ability. Items that were not reflective of agency characteristics, did not meet subjectivity criteria, or fit into at least one category of existing agency measure were removed from the study. This selection method allowed us to establish preliminary face and content validity for the selected measures as, instead of developing novel items, which inherently involves some level of subjective bias, we instead relied on those that had already been validated in peer-reviewed research and conformed to existing standards of subjectivity and established categories of agency measures. The selection process also allowed us to assess items from multiple prior agency measures simultaneously, enabling us to probe for potential relationships between items previously used to measure similar but heretofore disjoint concepts like perceived, personal, interpersonal, and intrinsic agency.

#### 3.2 Data Analysis

Multiple steps were taken to ensure the appropriateness of selected items for factor analysis. We calculated Variance Inflation Factor scores and item tolerance to test for multicollinearity, and employed tests for univariate and bivariate normality with Mardia's (1970) tests for multivariate skewness and kurtosis, Henze and Zirkler's (1990) consistent test, and the Doornik and Hansen (2008) omnibus test to assess distribution of data. We also examined Pearson's  $r$  and Spearman rank correlations between items, calculated

the Kaiser–Meyer–Olkin Measure of Sampling Adequacy (KMO) (Kaiser, 1970), and conducted Bartlett’s (1950) Test of Sphericity. As a final preparation for factor analysis, data was then subdivided into two samples for use exploratory (EFA) and confirmatory factor analysis (CFA).

As data was non-normal, we were unable to utilize maximum likelihood (ML) modeling, the typical precursor to CFA (Costello & Osborne, 2005), as it relies on assumptions of normality (Fabrigar et al., 1999). Instead, iterated principal axis factoring (IPF) was used. IPF is a robust estimation method that produces more accurate estimates than principal axis factoring (StataCorp, 2021a, 2021b) and is not affected by non-normal data (Costello & Osborne, 2005; Fabrigar et al., 1999). IPF requires that the number of factors to extract is specified in advance (StataCorp, 2021a, 2021b); therefore, Horn’s (1965) Parallel Analysis (PA) and Velicer’s (1976) minimum average partial method (MAP) were used to determine the appropriate number of factors to extract and retain. PA has consistently been shown to be one of the most accurate methods for determining latent structure (Hayton et al., 2004); however, a combination of decision-making rules produces more reliable interpretations than relying on a single method (Slocum-Gori & Zumbo, 2011) and, because PA sometimes recommends retaining too many factors, MAP is particularly complementary due to its tendency to underreport the number of factors to retain (Hayton et al., 2004).

In the social sciences, it is generally assumed that factors are correlated (Costello & Osborne, 2005). Therefore, after evaluating the dimensionality of data, we rotated factors using direct oblimin, which allows for both orthogonal and oblique solutions to be produced. Rotated solutions were examined for latent factorial structure. Following criteria asserted by Comrey and Lee (2013) and Tabachnick and Fidell (2013), we considered items with “fair” or better loadings ( $\lambda \geq 0.45$ ) to be salient on a given factor and those with “poor” or better loadings ( $\lambda \geq 0.32$ ) to be cross-loaded. We removed items from the dataset that were not consistently salient or cross-loaded, then re-estimated the dimensionality of remaining items with PA and MAP. This process of extraction, rotation, and data reduction was repeated until all remaining items demonstrated a consistent, interpretable latent factorial structure. To identify potential alternative models, we then reviewed the solutions generated using relaxed thresholds of  $\lambda \geq 0.4$  for salience (Gorsuch, 1983; Hinkin, 1995, 1998; Stevens, 1992) and a difference between loadings of  $< 0.2$  (Hinkin, 1998) to indicate cross-loading.

We assessed the fit of selected EFA models through CFA using asymptotic distribution free (ADF) estimation, which does not require normal data (StataCorp, 2021a, 2021b). We examined model fit by calculating two absolute fit indices, the root mean square error of approximation (RMSEA) and standardized root mean square residual (SRMR), alongside the Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI), which evaluate relative fit. Results of model chi-square tests were ignored because both non-normality of data (McIntosh, 2007) and large sample size made it likely that the test would reject the model (Bentler & Bonnet, 1980). We interpreted absolute fit indices using cutoffs of  $\leq 0.05$  for good fit (Byrne, 2013; Fabrigar et al., 1999) and  $\leq 0.08$  to indicate acceptable fit (Fabrigar et al., 1999; Hu & Bentler, 1999), and relative fit indices using cutoffs of  $\geq 0.95$  for good fit and Byrne’s (1994) less conservative threshold of  $\geq 0.9$  for acceptable fit. Because agency freedom and, therefore, personal agency is a precondition of agency achievement, a final model using the best-fitting GHAI was estimated to test the predictive effects of personal agency on agency achievement.

As the final step in our analysis, we performed several tests of preliminary construct validity on the best-fitting models. We assessed internal consistency and content,

substantive, and structural validity by calculating Cronbach's  $\alpha$ , average interitem correlations using both unstandardized and standardized items for the GPAS and all proposed GHAI scales and subscales, and analysis of pairwise Pearson's  $r$  correlations between individual scale items. Next, we tested convergent and concurrent validity by calculating pairwise Pearson's  $r$  correlations between GPAS and GHAI scales, correlates of agency, and other associated concepts: life satisfaction (Comim, 2005; Hojman & Miranda, 2018; Wang, 2015), self-acceptance and self-esteem (Azizli et al., 2015; Serdiuk et al., 2018; Skinner et al., 1996), purpose in life and autonomy (Alkire, 2005; Serdiuk et al., 2018), and positive relations with others and social integration (Christoph et al., 2014; Veenhoven, 2004). As agency is central to the human experience and can arguably be influenced by a diverse variety of factors, it was challenging to identify variables to test for discriminant validity; therefore, we compared coefficients between the GHAI and agency correlates with those between GHAI scales and the "Big Two" agency trait (Gebauer et al., 2014), given the demonstrable theoretical distinctions between the two constructs described above. Finally, to evaluate the predictive validity of the proposed tools, we calculated pairwise Pearson's  $r$  correlations between GPAS and GHAI scores in MIDUS II with those of agency correlates and associated concepts in MIDUS III.

## 4 Results

### 4.1 Item Selection

30 items were selected that matched or closely resembled those previously deployed in tools measuring human agency. 25 items were worded differently than, but conceptually aligned with, previous items; seven of these similar items were reverse measures of items used by Black (2016), Lautamo et al. (2021), and in the Basic Psychological Need Satisfaction Scale (Deci & Ryan, 2000; Gagné, 2003). The other five items selected were exact matches for items deployed by Nestadt, et al. (2022). No items were identified that were representative of those used previously to assess interpersonal, proxy, or collective agency (Smith et al., 2000; Yount et al., 2020); therefore, the remainder of the study focused predominantly on identifying measures for personal agency, its subdimensions, and agency achievement.

All selected items met Alkire's (2005) criteria for subjectivity and conformed to at least one established category of agency measures (2008). Most items were global, rather than pertaining to a specific life domain; however, indicators for direct control, effective power, autonomy, ability, and the advancement of well-being or other goals were identified, indicating that the initial list of selected measures engaged the agency construct in ways reflective of the breadth of prior research. All items also reflected at least one of the agency characteristics of control, self-determination, influence, or ability. Relationships between previously used agency items and those selected for the current study are summarized in Table 2.

Most items selected were included in the MIDUS dataset as scale items measuring other constructs. Twelve items were selected from three scales representing Ryff's (1989) PWB constructs of autonomy (5 of 7), environmental mastery (6 of 7, with the final item relating to fitting in with one's community), and purpose in life (1), although the single item selected from this latter construct was included in data collection but excluded from the final scale. All 12 items from Lachman and Weaver's (1998) perceived control scale were

**Table 2** Selected agency items and associated MIDUS constructs (bold items included in final GPAS and GHAI)

| Previously used agency item                                                                                                               | Used in                    | MIDUS item                                                                                            | MIDUS Construct |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|-----------------|
| I generally feel free to express my ideas and opinions                                                                                    | Gagné, (2003)              | I AM NOT AFRAID TO VOICE MY OPINIONS, EVEN WHEN THEY ARE IN OPPOSITION TO THE OPINIONS OF MOST PEOPLE | Autonomy        |
| I can easily express my thoughts and opinions to other people                                                                             | Lautamo et al., (2021)     |                                                                                                       |                 |
| When I disagree with others, I tell them                                                                                                  | Beyers et al., (2003)      |                                                                                                       |                 |
| When I act against the will of others, I usually get nervous                                                                              | Beyers et al., (2003)      | MY DECISIONS ARE NOT USUALLY INFLUENCED BY WHAT EVERYONE ELSE IS DOING                                |                 |
| I have a strong tendency to comply with the wishes of others                                                                              |                            |                                                                                                       |                 |
| I make up my own mind about doing good or bad things                                                                                      | Black, (2016)              |                                                                                                       |                 |
| No one can make me do something I know to be wrong                                                                                        |                            |                                                                                                       |                 |
| My actions in most situations are based on what other people tell me is the right thing to do                                             |                            |                                                                                                       |                 |
| In most cases, I can make my own decisions about what is right or wrong in a situation                                                    |                            |                                                                                                       |                 |
| How true would it be to say that your actions with respect to [the domain] are motivated by and reflect your own values and/or interests? | Ibrahim and Alkire, (2007) | I JUDGE MYSELF BY WHAT I THINK IS IMPORTANT, NOT BY THE VALUES OF WHAT OTHERS THINK IS IMPORTANT      |                 |
| I often change my mind after listening to others                                                                                          | Beyers et al., (2003)      | I TEND TO BE INFLUENCED BY PEOPLE WITH STRONG OPINIONS                                                |                 |
| I often agree with others, even if I'm not sure                                                                                           |                            | IT'S DIFFICULT FOR ME TO VOICE MY OWN OPINIONS ON CONTROVERSIAL MATTERS                               |                 |
| I generally feel free to express my ideas and opinions                                                                                    | Gagné, (2003)              |                                                                                                       |                 |
| I can easily express my thoughts and opinions to other people                                                                             | Lautamo et al., (2021)     |                                                                                                       |                 |

**Table 2** (continued)

| PSYCHOLOGICAL WELL-BEING (RYFF, 1989)                                                                 |                            |                                                                                                            |                                            |
|-------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Previously used agency item                                                                           | Used in                    | MIDUS item                                                                                                 | MIDUS Construct                            |
| I feel like I am free to decide for myself how to live my life                                        | Gagné, (2003)              | IN GENERAL, I FEEL I AM IN CHARGE OF THE SITUATION IN WHICH I LIVE                                         | Environmental Mastery                      |
| How true is the following statement for you?:<br>I feel free to decide for myself how to lead my life | Hojman and Miranda, (2018) |                                                                                                            |                                            |
| I make active choices about what to do daily                                                          | Lautamo et al., (2021)     |                                                                                                            |                                            |
| I can influence my living situation in satisfying ways                                                | Lautamo et al., (2021)     | <b>I HAVE DIFFICULTY ARRANGING MY LIFE IN A WAY THAT IS SATISFYING TO ME</b>                               |                                            |
| I do tasks that give me a feeling of competence or satisfaction                                       |                            |                                                                                                            |                                            |
| I can take care of my everyday tasks independently                                                    | Lautamo et al., (2021)     | <b>THE DEMANDS OF EVERYDAY LIFE OFTEN GET ME DOWN</b>                                                      |                                            |
| I can influence my living situation in satisfying ways                                                | Lautamo et al., (2021)     | <b>I HAVE BEEN ABLE TO BUILD A LIVING ENVIRONMENT AND A LIFESTYLE FOR MYSELF THAT IS MUCH TO MY LIKING</b> |                                            |
| I manage to perform in my environment                                                                 | Lautamo et al., (2021)     | <b>I AM QUITE GOOD AT MANAGING THE MANY RESPONSIBILITIES OF MY DAILY LIFE</b>                              |                                            |
| I am satisfied with the amount of daily activities I manage to do                                     |                            |                                                                                                            |                                            |
| I feel that I have a suitable amount to do on a daily basis                                           |                            |                                                                                                            |                                            |
| I control things by managing my affairs properly                                                      | Smith et al., (2000)       |                                                                                                            |                                            |
| I do tasks that I feel challenge me appropriately                                                     | Lautamo et al., (2021)     | <b>I OFTEN FEEL OVERWHELMED BY MY RESPONSIBILITIES</b>                                                     |                                            |
| I energetically pursue my goals                                                                       | Snyder (1991)              | <b>I AM AN ACTIVE PERSON IN CARRYING OUT THE PLANS I SET FOR MYSELF</b>                                    | Purpose in Life (removed from final scale) |
| I meet the goals that I set for myself                                                                |                            |                                                                                                            |                                            |

**Table 2** (continued)

| Previously used agency item                                                                 | Used in                | MIDUS item                                                         | MIDUS Construct       |
|---------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------|-----------------------|
| Perceived control (Lachman & Weaver, 1998)                                                  |                        |                                                                    |                       |
| There is little I can do to change many of the important things in my life                  | Nestadt et al. (2022)  | <b>THERE IS LITTLE I CAN DO TO CHANGE THE IMPORTANT</b>            | Perceived Constraints |
| I often feel helpless in dealing with the problems of life                                  | Nestadt et al. (2022)  | <b>I OFTEN FEEL HELPLESS IN DEALING WITH THE PROBLEMS OF LIFE</b>  |                       |
| In my daily life, I frequently have to do what I am told                                    | Gagné (2003)           | <b>OTHER PEOPLE DETERMINE MOST OF WHAT I CAN AND CANNOT DO</b>     |                       |
| Luck, more than what you do, is responsible for whether things turn out for the best        | Black (2016)           | <b>WHAT HAPPENS IN MY LIFE IS OFTEN BEYOND MY CONTROL</b>          |                       |
| I am the one responsible for my own behavior, good or bad                                   |                        |                                                                    |                       |
| I feel responsible for the consequences of my actions                                       |                        |                                                                    |                       |
| There is not much opportunity for me to decide for myself how to do things in my daily life | Gagné (2003)           | <b>THERE ARE MANY THINGS THAT INTERFERE WITH WHAT I WANT TO DO</b> |                       |
| I do tasks that I feel are important to me                                                  | Lautamo et al., (2021) |                                                                    |                       |
| I have little control over the things that happen to me                                     | Nestadt et al. (2022)  | <b>I HAVE LITTLE CONTROL OVER THE THINGS THAT HAPPEN TO ME</b>     |                       |
| When things don't turn out as I expected, it seems like someone else took control of things | Black (2016)           |                                                                    |                       |

**Table 2** (continued)

| Previously used agency item                                                     | Used in                                                          | MIDUS item                          | MIDUS Construct                                             |
|---------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------|
| Perceived control (Lachman & Weaver, 1998)                                      |                                                                  |                                     |                                                             |
| There is really no way I can solve some of the problems I have                  | Nestadt et al. (2022)                                            | <b>THERE IS REALLY NO WAY I CAN</b> |                                                             |
| I can handle problems and pressure (by discussing them or taking actions)       | Lautamo et al., (2021)                                           | <b>SOLVE THE PROBLEMS I HAVE</b>    |                                                             |
| I can solve daily challenges in a reasonable way                                |                                                                  |                                     |                                                             |
| I can think of many ways to get out of a jam                                    | Snyder (1991)                                                    |                                     |                                                             |
| There are lots of ways around any problem                                       |                                                                  |                                     |                                                             |
| Even when others get discouraged, I know I can find a way to solve the problem  |                                                                  |                                     |                                                             |
| Even when others want to quit, I know that I can find ways to solve the problem | Poteat et al. (2018); Veronese et al. (2019a), (b), (2020a), (b) |                                     |                                                             |
| When I have a problem, I can come up with lots of ways to solve it              |                                                                  |                                     |                                                             |
| Sometimes I feel that I am being pushed around in life                          | Nestadt et al. (2022)                                            |                                     | <b>I SOMETIMES FEEL I AM BEING PUSHED AROUND IN MY LIFE</b> |
| I feel pressured in my life                                                     | Gagné, (2003)                                                    |                                     |                                                             |

Table 2 (continued)

| Previously used agency item                                                                                                                                                                   | Used in                                                                         | MIDUS item                                                                                             | MIDUS Construct                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Perceived control (Lachman & Weaver, 1998)                                                                                                                                                    |                                                                                 |                                                                                                        |                                                                                            |
| Once I decide on a goal, I do whatever I can to achieve it                                                                                                                                    | Smith et al., (2000)                                                            | <b>I CAN DO JUST ABOUT ANYTHING I REALLY SET MY MIND TO</b>                                            | Personal Mastery                                                                           |
| I can think of many ways to get the things in life that are important to me                                                                                                                   | Poteat et al. (2018); Snyder (1991); Veronese et al. (2019a), (b), (2020a), (b) | <b>WHEN I REALLY WANT TO DO SOMETHING, I USUALLY FIND A WAY TO SUCCEED AT IT</b>                       |                                                                                            |
| When you get what you want, it's usually because you worked hard for it                                                                                                                       | Hitlin & Elder (2006); Williams and Merton (2014)                               | <b>WHETHER OR NOT I AM ABLE TO GET WHAT I WANT IS IN MY OWN HANDS</b>                                  |                                                                                            |
| I get what I want or need by relying on my own efforts and ability                                                                                                                            | Smith et al., (2000)                                                            |                                                                                                        |                                                                                            |
| I control what happens to me by making choices in my best interest                                                                                                                            |                                                                                 |                                                                                                        |                                                                                            |
| Some people believe they can decide their own destiny, while others think they do not have control over their destiny. Please, to what extent do you believe you can decide your own destiny? | Victor et al. (2013)                                                            | <b>WHAT HAPPENS TO ME IN THE FUTURE MOSTLY DEPENDS ON ME</b>                                           |                                                                                            |
| Other items                                                                                                                                                                                   |                                                                                 |                                                                                                        |                                                                                            |
| Previously used agency item                                                                                                                                                                   | Used in                                                                         | MIDUS item                                                                                             | MIDUS Construct                                                                            |
| Once I decide on a goal, I do whatever I can to achieve it                                                                                                                                    | Smith et al., (2000)                                                            | <b>WHEN THINGS DON'T GO ACCORDING TO MY PLANS, MY MOTTO IS, "WHERE THERE'S A WILL, THERE'S A WAY."</b> | Selective Primary Control (Heckhausen & Schulz, 1993; Wrosch, Heckhausen, & Lachman, 2000) |
| When I have a problem, I can come up with lots of ways to solve it                                                                                                                            | Poteat et al. (2018); Veronese et al. (2019a), (b), (2020a), (b)                | <b>WHEN FACED WITH A BAD SITUATION, I DO WHAT I CAN TO CHANGE IT FOR THE BETTER</b>                    |                                                                                            |
| There are lots of ways around any problem                                                                                                                                                     | Snyder (1991)                                                                   |                                                                                                        |                                                                                            |

**Table 2** (continued)

| Other items                                                                                                                                                                                | Used in                                                    | MIDUS item                                                                                                                                                                     | MIDUS Construct                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Previously used agency item                                                                                                                                                                |                                                            |                                                                                                                                                                                |                                                         |
| I can solve daily challenges in a reasonable way                                                                                                                                           | Lautamo et al., (2021)                                     | EVEN WHEN I FEEL I HAVE TOO MUCH TO DO, I FIND A WAY TO GET IT ALL DONE                                                                                                        |                                                         |
| I find it difficult to decide what I want                                                                                                                                                  | Beyers et al. (2003)                                       | I KNOW WHAT I WANT OUT OF LIFE                                                                                                                                                 | Self-Directedness and Planning (Prenda & Lachman, 2001) |
| When people ask me what I want, I immediately know the answer                                                                                                                              |                                                            |                                                                                                                                                                                |                                                         |
| How much control do you feel you have in making personal decisions that affect your everyday activities?                                                                                   | Ibrahim and Alkire, (2007)                                 | At present, how much control do you have over your LIFE IN GENERAL? A lot, some, a little, or none at all?                                                                     | None                                                    |
| “Please use this scale where ‘1’ means “none at all” and ‘10’ means “a great deal” to indicate how much freedom of choice and control you feel you have over the way your life turns out.” | Inglehart et al., (2008); Veenhoven, (2000); Verme, (2009) | Using a 0 to 10 scale where 0 means “no control at all” and 10 means “very much control,” how would you rate the amount of control you have over your life overall these days? |                                                         |

selected, reflecting the subscales of perceived constraints and personal mastery. Three of five items were selected from a scale representing selective primary control (Heckhausen & Schulz, 1993; Wrosch, Heckhausen, & Lachman, 2000) and one of three items was selected from Prenda and Lachman's (2001) self-directedness and planning scale. Only two selected items, both related to one's level of control over their life, were not part of an existing scale.

Altogether, findings from the item selection process produced initial evidence of the content validity of selected items. All items met multiple established criteria for agency measures and were selected from scales measuring constructs that align conceptually with agency (Alkire, 2005, 2008; Burger & Walk, 2016; Kotan, 2010). However, some items demonstrated questionable content validity; specifically, items from the PWB autonomy scale appeared to emphasize assertiveness, individuation, and agentic contrast. As previously discussed, these characteristics are typically unique to discussions of agency as a personality trait (Gebauer et al., 2014) and are not requisite characteristics for the self-determination and pursuit of goals.

## 4.2 Data Management and Descriptive Statistics

Descriptive statistics for the full sample and subsamples for MIDUS II and MIDUS III datasets are summarized in Appendix B alongside findings from tests for multivariate normality described below. Values for seven items were reversed so that larger values consistently reflected stronger agentic perceptions. VIF scores were less than 2.5 and tolerance values were greater than 0.4 for all items in both the full sample and subsamples, indicating a low risk of multicollinearity among items.

Data in all samples violated assumptions of normality. Mean scores and standard deviations indicated that, on average, respondents reported moderate to high levels scores for each item, and cursory examination of item skewness and kurtosis suggested that all items demonstrated a left-tailed distribution. Tests of univariate and bivariate normality for all items were significant at the level of  $p < 0.0001$ , confirming that items were non-normally distributed. Similarly, Mardia's tests for multivariate skewness and kurtosis, Henze-Zirkler's consistent test, and the Doornik-Hansen omnibus test were all significant at the level of  $p < 0.0001$ , leading to a rejection of the null hypothesis that items possess multivariate normality.

## 4.3 Suitability of Data for Factor Analysis and Determination of Factors to Retain

The average interitem correlation was  $r = 0.26$  in both MIDUS II and III. Item-rest Pearson's correlations ranged from  $r = 0.31$ - $0.69$  and  $r = 0.3$ - $0.67$  for unstandardized and standardized items, respectively, in MIDUS II, and from  $r = 0.26$ - $0.7$  and  $r = 0.27$ - $0.68$  in MIDUS III, indicating positive relationships between individual items and remaining items in the series. All pairwise Spearman rank correlations between items were significant ( $p < 0.05$ ) and positive, although relationships varied in strength from negligible to strong in both MIDUS II ( $\rho = 0.08$ - $0.64$ ) and MIDUS III ( $\rho = 0.09$ - $0.63$ ). Overall, correlational analyses demonstrated relationships between selected items, providing preliminary evidence of their suitability for factor analysis.

Five items were frequently associated with coefficients less than 0.2 in both datasets, although they also demonstrated moderate to strong relationships with some items. Two were related to selective primary control ("When things don't go according to plan, my

motto is ‘Where there’s a will there’s a way’”, “Even when feel I have too much to do, I find a way to get it all done”), two were from the PWB autonomy scale (“I am not afraid to voice my opinions, even when they are in opposition to the opinions of most people”, “I tend to be influenced by people with strong opinions”), and the final item, “At present, how much control do you feel you have over your life in general?”, was independent from any scale.

The KMO index was 0.944 for both waves of data, exceeding the suggested cutoff of 0.6 (Kaiser, 1970). Bartlett’s Test of Sphericity (Bartlett, 1954) was significant in both MIDUS II ( $\chi^2=38,616.06$ ,  $p<0.001$ ) and III ( $\chi^2=27,390.5$ ,  $p<0.001$ ). These findings corroborated those from correlational analyses, indicating study data was suitable for factor analysis.

#### 4.4 Exploratory Factor Analysis

For both MIDUS II and III, PA indicated that five components be retained while MAP recommended that two factors be extracted from the data; to fully explore the factorial structure of the selected items, initial extractions included solutions within this range. Five-factor rotations produced four factors with at least two salient loadings ( $\lambda \geq 0.45$ ) in both waves, which were interpretable as the constructs from which several items were derived: perceived constraints and personal mastery (Lachman & Weaver, 1998), selective primary control (Heckhausen & Schulz, 1993; Wrosch, Heckhausen, & Lachman, 2000), and the PWB autonomy construct (Ryff, 1989). However, seven items cross-loaded on factors in at least one wave of data, and seven were never salient on any factor in either sample, indicating a possible overextraction of factors. Four-factor solutions resulted in three salient factors consistent across waves that could be understood to represent perceived constraints, the PWB autonomy construct, and a combination of selective primary control and personal mastery, with five items cross-loading and six never being salient on a factor.

Three-factor models were the first to demonstrate a clear structure of two consistent factors, the first of which reflected a combination of perceived constraints and environmental mastery (Ryff, 1989) and the second a combination of selective primary control and personal mastery. While only one item from the PWB autonomy construct cross-loaded in MIDUS II, it was not salient on any factor in MIDUS III; in total, ten items were never salient on any factor in either sample. This latent structure was maintained in two-factor models, with single items selected from the constructs of self-directedness and planning (Prenda & Lachman, 2001), purpose in life (Ryff, 1989), and environmental mastery loading onto the second factor, one item cross-loading in MIDUS III only, and nine items never demonstrating salience. However, in the two-factor solution, factors also appeared to be interpretable as personal agency, one’s attitudes and perceptions regarding their ability to enact control over their life and pursue self-determined goals, and agency achievement, one’s ability to succeed in these capacities.

Multifactor solutions consistently produced a ratio of first-to-second eigenvalues greater than five, the only exception being the MIDUS III five-factor extraction (1:2  $\lambda=4.92$ ). This suggested the presence of a general dominant factor which explained between 69 and 85 percent of variance in MIDUS II solutions and 67 to 84 percent of variance in MIDUS III (in five- and two-factor rotations, respectively). Therefore, a one-factor solution was also extracted despite not being recommended by PA or MAP. 20 items were consistently salient on this general factor, with items from Lachman and Weaver’s (1998) perceived constraints scale and Ryff’s (1989) environmental mastery scale typically loadings that were good ( $\lambda \geq 0.55$ ) or better (Tabachnick & Fidell, 2013). No items from the selective primary

control or PWB autonomy scales were consistently salient, neither was one item from the personal mastery scale (“What happens in the future mostly depends on me”) and the non-scale item “At present, how much control do you have over your life in general?”. Factor loadings and other information for initial extractions with one through three factors are provided in Appendix C.

Because of the quantity of non-salient and cross-loaded items, extracted latent factors were difficult to explain beyond the scales that items represented; however, two- and three-factor models indicated the presence of an alternative factorial structure. To improve interpretability of these factors, items were eliminated if they were not consistently salient ( $\lambda \geq 0.45$ ) or cross-loaded on the two factors that emerged from two- and three-factor solutions, as were those that were not salient on the single-factor model in both waves. A total of 10 items were removed, which included all items from the selective primary control and PWB autonomy scales. We then conducted IPF extractions on the remaining items; this process of reduction and re-estimation was repeated until all remaining items were consistently salient or cross-loaded in both multifactor and single-factor solutions. Six iterations of extraction, rotation, and reduction were conducted in total; findings from the process are summarized in Table 3.

In the third and fourth iterations of extractions, clear latent structures began to emerge. Two-factor solutions produced factors interpretable as personal agency and agency achievement. Agency achievement persisted as a latent variable in three-factor extractions, while items reflecting personal agency subdivided into factors interpreted as intrinsic and instrumental agency. However, strict application of reduction criteria eliminated the agency achievement factor at the conclusion of the fourth round of extractions. The nine items that remained at the end of the data reduction process comprised a General Personal Agency Scale (GPAS) explainable as a single latent variable, personal agency, or as a combination of intrinsic and instrumental agency subfactors. Factor loadings for this final round of extractions are found in Table 4.

While the purpose of data reduction was to simplify the latent structure of the selected items to produce a measure for personal agency, the loss of agency achievement indicators resulted in a substantive reduction of the potential explanatory power of the emerging GHAI. Further, the agency achievement factor demonstrated moderate to strong correlations with personal, intrinsic, and instrumental agency in the third and fourth iterations of extraction, suggesting the presence of positive relationships between the latent variables. Therefore, solutions from the fourth round of extractions, the last that included at least three items consistently salient on agency achievement, were reexamined using relaxed factor loading cutoffs of  $\lambda \geq 0.4$  for salience and  $\lambda \geq 0.2$  for cross-loading items, which are commonly encountered in EFA (Gorsuch, 1983; Hinkin, 1995, 1998; Stevens, 2002). Based on these less conservative cutoffs, all items were consistently salient on a single factor in one- and two-factor models. In the three-factor solutions, the item “I often feel helpless in dealing with the problems of life” cross-loaded onto intrinsic and instrumental agency, and “I have been able to build a living environment and a lifestyle for myself that is much to my liking” cross-loaded onto instrumental agency and agency achievement in MIDUS II and III. Therefore, both the nine-item GPAS and the 13-item GHAI measuring personal agency and agency achievement scale from the fourth iteration of extractions were retained for CFA. Two alternative models based on the 13-item GHAI were also developed: an 11-item version that included cross-loaded items and only the three highest loading items for intrinsic agency, and a nine-item tool that eliminated cross-loaded items, leaving only the three highest loading items for intrinsic agency, instrumental agency, and agency achievement.

**Table 3** Summary of data reduction process

| Iteration     | 1                                          | 2                                       | 3                                         | 4*                                      | 5                                       | 6* |
|---------------|--------------------------------------------|-----------------------------------------|-------------------------------------------|-----------------------------------------|-----------------------------------------|----|
| # of items    | 30                                         | 20                                      | 17                                        | 13                                      | 11                                      | 9  |
| PA factors    | 5                                          | 3                                       | 3                                         | 3                                       | 2                                       | 2  |
| MAP factors   | 2                                          | 2                                       | 2                                         | 1                                       | 1                                       | 1  |
| Items removed | Control over life in general at present    | Know what I want out of life            | In charge of situation in which I live    | Actively carry out plans I set for self | Able to build lifestyle to my liking    |    |
|               | Where there's a will there's a way         | Whether I get what want is in own hands | Good managing daily responsibilities      | When really want something, find way    | Do just about anything I set my mind to |    |
|               | Do what can to change for better           | Rate control over life                  | Many things interfere with what I want do |                                         |                                         |    |
|               | Even when feel too much, get it all done   |                                         | Feel pushed around in life                |                                         |                                         |    |
|               | Not afraid to voice opinions in opposition |                                         |                                           |                                         |                                         |    |
|               | Decisions not influenced by others doing   |                                         |                                           |                                         |                                         |    |
|               | Judge self by what I think is important    |                                         |                                           |                                         |                                         |    |
|               | Influenced by people with strong opinions  |                                         |                                           |                                         |                                         |    |
|               | Difficult voice opinion on controversial   |                                         |                                           |                                         |                                         |    |
|               | Happens to me in future depends on me      |                                         |                                           |                                         |                                         |    |

\*Solutions were retained for CFA

**Table 4** Factor loadings for 9-item personal agency scale (GPAS)

| Single factor                              |             |             | 2-Factor   |      |  | Loadings on individual factors |             |             |                     |      |      |             |             |       |       |
|--------------------------------------------|-------------|-------------|------------|------|--|--------------------------------|-------------|-------------|---------------------|------|------|-------------|-------------|-------|-------|
| Personal agency                            |             |             | Uniqueness |      |  | Intrinsic agency               |             |             | Instrumental agency |      |      | Uniqueness  |             |       |       |
| M2                                         | M3          |             | M2         | M3   |  | M2                             | M3          |             | M2                  | M3   |      | M2          | M3          |       |       |
| Item                                       |             |             |            |      |  |                                |             |             |                     |      |      |             |             |       |       |
| Little can do to change important things   | <b>.634</b> | <b>.671</b> | .599       | .549 |  | <b>.604</b>                    | <b>.694</b> |             | .577                | .509 |      | <b>.659</b> | <b>.710</b> | .566  | .496  |
| Others determine what I can and cannot do  | <b>.631</b> | <b>.637</b> | .602       | .594 |  | <b>.555</b>                    | <b>.623</b> |             | .594                | .570 |      | <b>.634</b> | <b>.660</b> | .598  | .564  |
| What happens in life is beyond my control  | <b>.636</b> | <b>.680</b> | .596       | .538 |  | <b>.715</b>                    | <b>.768</b> |             | .528                | .467 |      | <b>.683</b> | <b>.719</b> | .533  | .483  |
| Little control over things happen to me    | <b>.676</b> | <b>.708</b> | .543       | .499 |  | <b>.747</b>                    | <b>.785</b> |             | .471                | .428 |      | <b>.718</b> | <b>.744</b> | .485  | .446  |
| Really no way I can solve problems I have  | <b>.698</b> | <b>.748</b> | .513       | .441 |  | <b>.760</b>                    | <b>.689</b> |             | .442                | .420 |      | <b>.743</b> | <b>.766</b> | .447  | .413  |
| Helpless dealing with problems of life     | <b>.756</b> | <b>.801</b> | .429       | .359 |  | .430                           | .421        | .392        | .444                | .361 |      | <b>.694</b> | <b>.742</b> | .519  | .446  |
| Difficult arranging life in satisfying way | <b>.643</b> | <b>.668</b> | .586       | .553 |  |                                |             | <b>.595</b> | <b>.639</b>         | .471 |      |             |             | .500  | .461  |
| Demands of everyday life often get me down | <b>.599</b> | <b>.629</b> | .641       | .604 |  |                                |             | <b>.761</b> | <b>.780</b>         | .434 |      |             |             | .479  | .447  |
| Overwhelmed by my responsibilities         | <b>.594</b> | <b>.585</b> | .647       | .658 |  |                                |             | <b>.738</b> | <b>.721</b>         | .480 | .517 |             |             | .497  | .528  |
| Eigenvalue                                 | 3.845       | 4.206       |            |      |  | 3.913                          | 4.268       | .585        | .555                |      |      | 2.852       | 3.148       | 1.995 | 2.118 |
| Proportion of variance                     | 1.000       | 1.000       |            |      |  | .870                           | .885        | .130        | .115                |      |      | 1.000       | 1.000       | 1.000 | 1.000 |

Additional extractions and rotations were conducted for each alternative model and for each subscale in all models. The factorial structure and loadings for the 11- and 9-item GHAs corresponded with the 13-item model; the only difference being that the item “I have been able to build a living environment and a lifestyle for myself that is much to my liking” cross-loaded on personal agency and agency achievement in two-factor rotations of the 11-item agency models. Rotations of individual factors conformed to findings produced by full-model rotations, with cross-loaded variables from full models consistently salient on each of the latent factors they had previously cross-loaded onto, indicating that these items might influence the relationships between factors. Factor loadings for the GPAS and 9-item GHAs, which demonstrated the best model fit during CFA, are found in Table 5 alongside loadings for their individual scales; loadings for 13- and 11-item agency models and individual scales are included in Appendix D.

Of the 13 items included in final models, five were similar to items used by Lautamo et al. (2021), four were exact matches to those employed by Nestadt et al. (2022), and three were like items included in the agentic pathways section of Snyder's (1991) Hope Scale. Other items were like those deployed by Black (2016), Gagné (2003), Poteat et al. (2018), and Veronese, et al. (2019a, b), 2020a, b). Final items included four from Ryff's (1989) environmental mastery scale and the single item excluded from her purpose in life scale; the remaining items were from Lachman and Weaver's (1998) perceived constraints and personal mastery scales. The intrinsic agency factor was comprised entirely of perceived constraints items and instrumental agency of environmental mastery items; agency achievement included a combination of items reflecting environmental mastery, personal mastery, and purpose in life. This distribution of items supports the face, content, and structural validity of the selected models, as agency represents the interaction of one's capacities to enact control over life through the pursuit and achievement of self-determined goals—that is, to overcome perceived constraints through mastery of one's environment to shape one's circumstances according to one's purpose and desired ends.

#### 4.5 Confirmatory Factor Analysis

Based on findings from EFA, a total of sixteen path models were specified and estimated. Five models failed to converge without modification due to issues related to sample size and the number of fitted parameters in the model. When this occurred, it was resolved by removing paths for items that cross-loaded during EFA; covariance between intrinsic and instrumental agency was also eliminated from models when it was found to be insignificant. Relationships between all remaining variables and constructs were statistically significant with practically significant effect sizes. Fit index values for all final, converged CFA models are shown in Table 6.

CD values indicated that models accounted for between 86 and 98 percent of variance in MIDUS II models and between 88 and 99 percent of variance in MIDUS III models. Among single factor models, only the GPAS consistently demonstrated acceptable absolute fit ( $RMSEA$  and  $SRMR \leq 0.08$ ), suggesting that personal agency and agency achievement are distinct yet interrelated constructs. All two-factor models were shown to have acceptable absolute fit or better, and absolute indices for all three-factor and bifactor models indicated good fit. Findings from relative fit indices were less consistent. Only the bifactor 9-item personal agency and three-factor and bifactor 9-item agency models consistently demonstrated acceptable fit ( $\geq 0.9$ ) or better, although 13- and 11-item bifactor agency models produced some evidence of acceptable relative fit in MIDUS II. CFI

**Table 5** Factor loadings for 9-item personal agency and agency achievement scales (GHAI)

| Item                                       | Single factor   |       |            | 2-Factor   |                  |       | 3-Factor            |      |                    |       |            |      |
|--------------------------------------------|-----------------|-------|------------|------------|------------------|-------|---------------------|------|--------------------|-------|------------|------|
|                                            | Agency          |       |            | Uniqueness |                  |       | Personal agency     |      | Agency achievement |       | Uniqueness |      |
|                                            | M2              | M3    | M3         | M2         | M3               | M3    | M2                  | M3   | M2                 | M3    | M2         | M3   |
| What happens in life is beyond my control  | .588            | .641  | .654       | .589       | .569             | .621  | .645                | .584 | .658               | .763  | .549       | .447 |
| Little control over things happen to me    | .657            | .675  | .569       | .545       | .641             | .669  | .554                | .533 | .847               | .802  | .340       | .381 |
| Really no way I can solve problems I have  | .660            | .710  | .565       | .495       | .584             | .668  | .570                | .497 | .602               | .586  | .510       | .466 |
| Difficult arranging life in satisfying way | .651            | .692  | .576       | .521       | .604             | .668  | .571                | .515 |                    | .569  | .660       | .452 |
| Demands of everyday life often get me down | .594            | .629  | .647       | .605       | .685             | .703  | .580                | .548 |                    | .751  | .756       | .442 |
| Overwhelmed by my responsibilities         | .589            | .587  | .653       | .655       | .695             | .677  | .577                | .593 |                    | .753  | .699       | .515 |
| Actively carry out plans I set for self    | .479            | .442  | .771       | .804       |                  |       | .403                | .399 |                    |       | .421       | .444 |
| Do just about anything I set my mind to    | .488            | .476  | .762       | .774       |                  |       | .665                | .544 |                    | .667  | .554       | .609 |
| When really want something, find way       | .536            | .453  | .712       | .795       |                  |       | .794                | .848 |                    | .800  | .829       | .336 |
| Eigenvalue                                 | 3.092           | 3.217 |            |            | 3.166            | 3.286 |                     | .690 | .790               | 3.248 | 3.359      | .727 |
| Proportion of variance                     | 1.000           | 1.000 |            |            | .821             | .806  |                     | .179 | .194               | .719  | .722       | .161 |
|                                            |                 |       |            |            |                  |       |                     |      |                    |       | .172       | .105 |
| Loadings on individual factors             |                 |       |            |            |                  |       |                     |      |                    |       |            |      |
| Item                                       | Personal agency |       | Uniqueness |            | Intrinsic agency |       | Instrumental agency |      | Agency achievement |       | Uniqueness |      |
|                                            | M2              | M3    | M2         | M3         | M2               | M3    | M2                  | M3   | M2                 | M3    | M2         | M3   |
| What happens in life is beyond my control  | .609            | .657  | .630       | .568       | .673             | .743  |                     |      |                    |       | .547       | .449 |
| Little control over things happen to me    | .682            | .695  | .535       | .517       | .814             | .789  |                     |      |                    |       | .338       | .378 |
| Really no way I can solve problems I have  | .657            | .714  | .569       | .490       | .684             | .716  |                     |      |                    |       | .532       | .488 |
| Difficult arranging life in satisfying way | .637            | .682  | .594       | .535       |                  |       | .655                | .702 |                    |       | .571       | .507 |

**Table 5** (continued)

| Loadings on individual factors             |                |             |            |      |                  |         |                     |             |                    |             |            |      |
|--------------------------------------------|----------------|-------------|------------|------|------------------|---------|---------------------|-------------|--------------------|-------------|------------|------|
| Item                                       | Personalagency |             | Uniqueness |      | Intrinsic agency |         | Instrumental agency |             | Agency achievement |             | Uniqueness |      |
|                                            | M2             | M3          | M2         | M3   | M2               | M3      | M2                  | M3          | M2                 | M3          | M2         | M3   |
| Demands of everyday life often get me down | <b>.630</b>    | <b>.655</b> | .603       | .572 |                  |         | <b>.741</b>         | <b>.754</b> |                    |             | .451       | .432 |
| Overwhelmed by my responsibilities         | <b>.629</b>    | <b>.624</b> | .604       | .611 |                  |         | <b>.742</b>         | <b>.710</b> |                    |             | .450       | .496 |
| Actively carry out plans I set for self    |                |             |            |      |                  |         |                     |             | .490               | .503        | .760       | .747 |
| Do just about anything I set my mind to    |                |             |            |      |                  |         |                     |             | <b>.670</b>        | <b>.642</b> | .551       | .589 |
| When really want something, find way       |                |             |            |      |                  |         |                     |             | <b>.797</b>        | <b>.785</b> | .366       | .384 |
| Eigenvalue                                 | 2.46556        | 2.70721     |            |      | 1.58276          | 1.68515 | 1.52809             | 1.56467     | 1.3236             | 1.28093     |            |      |

Bold numbers indicate a loading of good or better ( $\lambda \geq .55$ ); table does not include loadings  $< .32$

**Table 6** Fit indices for CFA models

## 9-Item personal agency models (GPAS)

|        | Single factor |             | 2-Factor    |             | Bifactor    |             |
|--------|---------------|-------------|-------------|-------------|-------------|-------------|
|        | M2            | M3          | M2          | M3          | M2*         | M3*         |
| RMSEA  | <b>.073</b>   | <b>.072</b> | <b>.043</b> | <b>.047</b> | <b>.041</b> | <b>.036</b> |
| PCLOSE | .000          | .000        | .934        | .675        | <b>.949</b> | <b>.977</b> |
| CFI    | .779          | .784        | <b>.930</b> | <b>.915</b> | <b>.951</b> | <b>.962</b> |
| TLI    | .705          | .713        | <b>.899</b> | .878        | <b>.907</b> | <b>.928</b> |
| SRMR   | <b>.072</b>   | <b>.063</b> | <b>.030</b> | <b>.033</b> | <b>.025</b> | <b>.024</b> |
| CD     | .893          | .904        | .950        | .954        | .974        | .974        |

## 9-Item personal agency and agency achievement models (GHAI)

|        | Single Factor |      | 2-Factor    |             | 3-Factor    |             | Bifactor (covariance) |             | Bifactor (directional) |             |
|--------|---------------|------|-------------|-------------|-------------|-------------|-----------------------|-------------|------------------------|-------------|
|        | M2            | M3   | M2          | M3          | M2          | M3          | M2                    | M3*         | M2                     | M3          |
| RMSEA  | .088          | .084 | <b>.069</b> | <b>.070</b> | <b>.040</b> | <b>.041</b> | <b>.030</b>           | <b>.035</b> | <b>.030</b>            | <b>.035</b> |
| PCLOSE | .000          | .000 | .000        | .000        | <b>.971</b> | .936        | <b>1.000</b>          | <b>.988</b> | <b>1.000</b>           | <b>.988</b> |
| CFI    | .679          | .732 | .821        | .828        | <b>.943</b> | <b>.946</b> | <b>.973</b>           | <b>.966</b> | <b>.973</b>            | <b>.966</b> |
| TLI    | .572          | .643 | .743        | .752        | <b>.911</b> | <b>.916</b> | <b>.949</b>           | <b>.938</b> | <b>.949</b>            | <b>.938</b> |
| SRMR   | .110          | .108 | <b>.060</b> | <b>.060</b> | <b>.030</b> | <b>.035</b> | <b>.025</b>           | <b>.038</b> | <b>.025</b>            | <b>.038</b> |
| CD     | .858          | .875 | .948        | .956        | .975        | .979        | .975                  | .982        | .974                   | .964        |

## 13-Item personal agency and agency achievement models (GHAI)

|        | Single Factor |             | 2-Factor    |             | 3-Factor    |             | Bifactor     |             |
|--------|---------------|-------------|-------------|-------------|-------------|-------------|--------------|-------------|
|        | M2            | M3          | M2          | M3          | M2          | M3          | M2           | M3*         |
| RMSEA  | <b>.067</b>   | <b>.064</b> | <b>.058</b> | <b>.058</b> | <b>.042</b> | <b>.043</b> | <b>.036</b>  | <b>.042</b> |
| PCLOSE | .000          | .000        | .002        | .012        | <b>.996</b> | <b>.960</b> | <b>1.000</b> | <b>.974</b> |
| CFI    | .624          | .669        | .718        | .733        | .866        | .859        | <b>.911</b>  | .874        |
| TLI    | .549          | .602        | .657        | .674        | .826        | .817        | .867         | .825        |
| SRMR   | .106          | .100        | <b>.076</b> | <b>.077</b> | <b>.047</b> | <b>.055</b> | <b>.035</b>  | <b>.055</b> |
| CD     | .909          | .915        | .963        | .961        | .983        | .984        | .985         | .989        |

## 11-Item personal agency and agency achievement models (GHAI)

|        | Single Factor |             | 2-Factor    |             | 3-Factor    |             | Bifactor    |             |
|--------|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|        | M2            | M3          | M2          | M3          | M2          | M3          | M2*         | M3          |
| RMSEA  | <b>.075</b>   | <b>.073</b> | <b>.063</b> | <b>.065</b> | <b>.045</b> | <b>.048</b> | <b>.036</b> | <b>.047</b> |
| PCLOSE | .000          | .000        | .000        | .000        | .885        | .630        | <b>.999</b> | .717        |
| CFI    | .654          | .690        | .760        | .766        | .888        | .881        | <b>.942</b> | .893        |
| TLI    | .568          | .612        | .693        | .694        | .842        | .832        | <b>.900</b> | .840        |
| SRMR   | .105          | .106        | <b>.050</b> | <b>.075</b> | <b>.046</b> | <b>.055</b> | <b>.032</b> | <b>.058</b> |
| CD     | .893          | .900        | .955        | .958        | .978        | .981        | .983        | .988        |

\*Reduced from full EFA model

Bold statistics indicate good fit; italics indicate acceptable fit

and TLI values are influenced by correlations between items and the number of estimated parameters, which may explain why relative fit of models are not significantly different from independent ones. Further, non-normal data can inflate absolute fit index values and underestimate relative fit index values (Finney & DiStefano, 2006), so models may fit better than values indicate.

Overall, the bifactor models for the GPAS and 9-item GHAI fit best, consistently demonstrating RMSEA and SRMR values below  $\leq 0.05$ , CFI and TLI indices above 0.9, and explaining between 96 and 98 percent of variance in the data. As agency achievement is theorized to be produced from agency freedom (Sen, 1999), of which personal agency is a component, a final 9-item GHAI model was fitted that included a direct regression path from personal agency to agency achievement; this model consistently indicated that the predictive effect of personal agency on agency achievement was statistically and practically significant. Path diagrams for 9-item personal agency and 9-item agency models are displayed in Fig. 2, and diagrams for 13- and 11-item agency models are found in Appendix D.

#### 4.6 Internal Consistency and Construct Validity

Unstandardized and standardized Cronbach's alpha coefficients for most modeled scales and subscales were greater than 0.7; only the three-item agency achievement scale included in the 9-item GHAI showed questionable consistency in MIDUS II ( $\alpha=0.672$ ) and III ( $\alpha=0.666$ ). All final scales were comprised of items with more than five ordered categories and were therefore treated as ordinal approximations of continuous variables; pairwise Pearson's  $r$  correlations were calculated to test strength and directionality of relationships, rather than Spearman rank coefficients. Average interitem correlations for all scales were of moderate or greater effect size ( $r \geq 0.3$ ), and all pairwise Pearson's  $r$  correlations among the 13 items included across measures were positive and significant at the level of  $p \leq 0.05$ , ranging from  $r=0.2$  to  $0.55$  in MIDUS II and  $r=0.2$ - $0.58$  in MIDUS III. Correlations between personal agency and agency achievement items were typically weaker than those between items within individual factors. Correlations between factors in each identified model consistently demonstrated significant, positive relationships of moderate to strong effect ( $r=0.44$ - $0.67$  in MIDUS II and  $r=0.38$ - $0.71$  in MIDUS III); in the GHAI, the relationship between intrinsic and instrumental agency factors was consistently stronger than that of either factor with agency achievement, mirroring the latent structure identified during factor analysis. Together, these findings provide evidence for the content, substantive, and structural validity of the proposed scales. Internal consistency statistics and correlations between agency items and factors are summarized in Appendix E.

Pairwise correlations between the GPAS and GHAI and correlates of agency showed weak to strong positive relationships between all measures in MIDUS II and III. Again, relationships between personal, intrinsic, and instrumental agency were consistently stronger than those between agency achievement and personal agency, further supporting the structural validity of the proposed indicators. Correlations among the GPAS and GHAI scales were typically larger than those between agency measures and agency correlates. Of agency correlates, the GPAS and GHAI consistently demonstrated the most substantive relationships with self-acceptance, self-esteem, and purpose in life. Overall, agency measures showed mostly moderate to strong significant positive relationships with agency correlates, supporting both the convergent and concurrent validity of the GHAI.

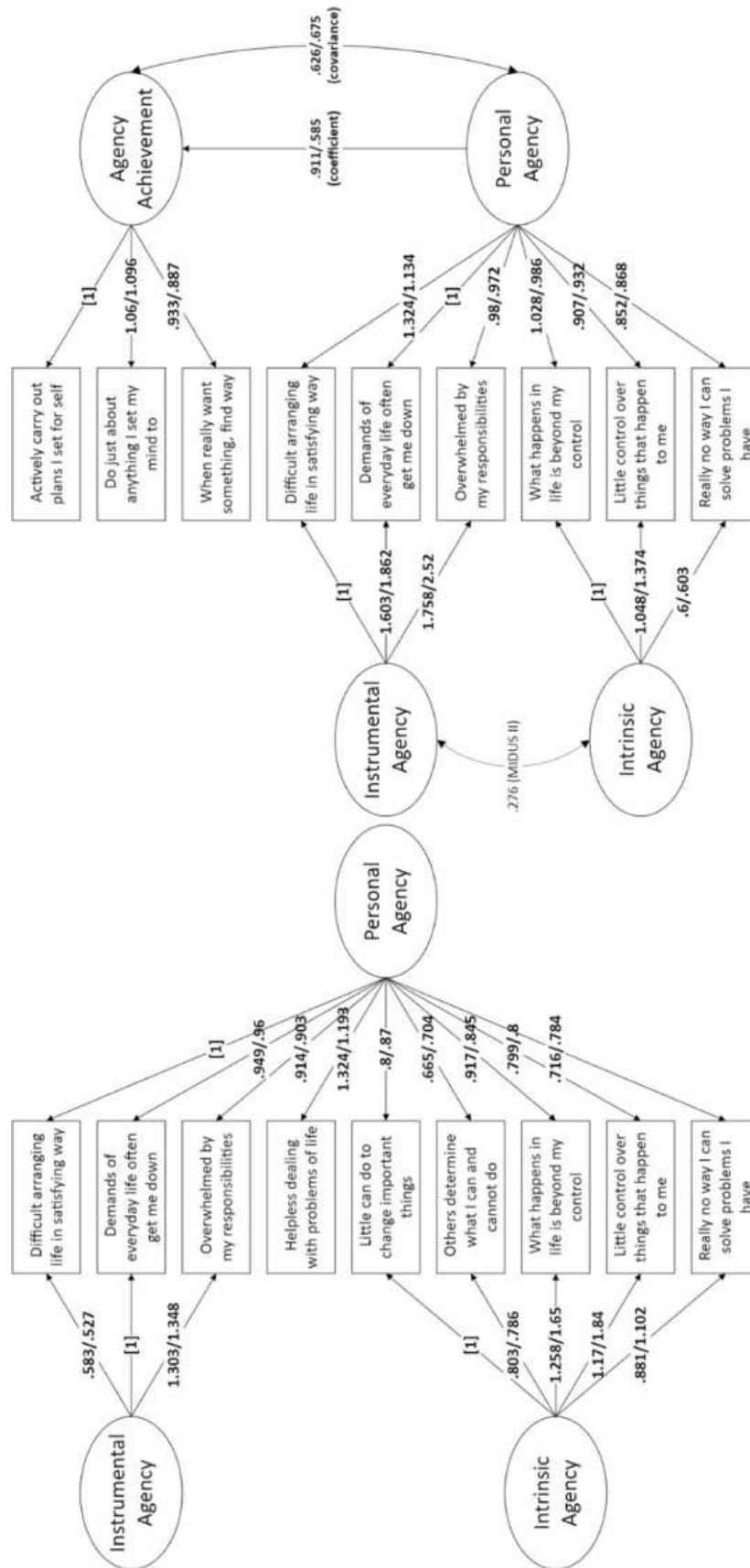


Fig. 2 Path diagrams for GPAS and 9-item GHAI

The weakest correlations were frequently encountered between agency measures and the “Big Five” agency trait, buttressing the argument that the personality construct is distinct from human agency as it is defined in other areas of social science research and offering some indication that the GPAS and GHAI possess discriminant validity. Correlations between MIDUS II GHAI scores and MIDUS III agency correlates were consistent with those in individual waves, albeit with reduced coefficients. The stability of the relationships between measures over time provides preliminary evidence of the GPAS and GHAI’s predictive validity and further supports the criterion validity of the proposed tools. Correlations between the GPAS and 9-item GHAI and agency correlates are found in Tables 7 and 8, respectively, and those between the 13- and 11-item GHAI and agency correlates are included in Appendix E.

## 5 Discussion

Through a rigorous process of item selection and data analysis, this study produced robust initial evidence of construct validity for both a nine-item General Personal Agency Scale (GPAS) and a nine-item General Human Agency Indicators (GHAI) tool measuring personal agency and its subdimensions alongside agency achievement. Face and content validity for the tools were established through a multi-stage item selection process, aggregating items that matched or were similar to those previously used in agency research and ensuring they were representative of agency characteristics, met subjectivity criteria, and fit within existing categories of agency measures. Factor loadings from EFA and fit indices from CFA indicate the content, substantive, and structural validity of the GPAS and 9-item GHAI, demonstrating good-fitting bifactorial structures that align with our proposed conceptual framework for human agency. Although less consistent across MIDUS II and III, some evidence was also produced indicating the factorial and construct validity of expanded 13- and 11-item GHAI, potentially enabling exploration of how one’s ability to solve problems and build a lifestyle to one’s liking may influence relationships between agency subdimensions. Only single-factor GHAI models did not consistently

**Table 7** Pairwise correlations between 9-item personal agency (GPAS) factors and agency correlates

| Construct                             | MIDUS II |      |      | MIDUS III |      |      | TWO-WAVE |      |      |
|---------------------------------------|----------|------|------|-----------|------|------|----------|------|------|
|                                       | (1)      | (2)  | (3)  | (1)       | (2)  | (3)  | (1)      | (2)  | (3)  |
| (1) Personal Agency                   | 1.00     |      |      | 1.00      |      |      | 1.00     |      |      |
| (2) Intrinsic Agency                  | .98*     | 1.00 |      | .98*      | 1.00 |      | .98*     | 1.00 |      |
| (3) Instrumental Agency               | .90*     | .79* | 1.00 | .92*      | .81* | 1.00 | .90*     | .79* | 1.00 |
| (4) Life Satisfaction                 | .49*     | .44* | .50* | .50*      | .46* | .49* | .37*     | .34* | .37* |
| (5) Self-Acceptance (PWB)             | .68*     | .63* | .68* | .69*      | .63* | .70* | .53*     | .49* | .52* |
| (6) Self-Esteem                       | .68*     | .63* | .68* | .67*      | .62* | .66* | .51*     | .47* | .50* |
| (7) Purpose in Life (PWB)             | .65*     | .62* | .60* | .66*      | .63* | .63* | .49*     | .48* | .44* |
| (8) Positive Relations w/Others (PWB) | .53*     | .50* | .51* | .52*      | .48* | .51* | .41*     | .39* | .40* |
| (9) Social Integration                | .35*     | .33* | .33* | .36*      | .34* | .34* | .31*     | .30* | .28* |
| (10) Agency (Big 5)                   | .30*     | .29* | .28* | .28*      | .26* | .26* | .24*     | .23* | .23* |
| (11) Autonomy (PWB)                   | .47*     | .43* | .48* | .47*      | .43* | .47* | .38*     | .35* | .36* |

\* $p < 0.05$

**Table 8** Pairwise correlations between 9-item personal agency and agency achievement (GHAI) factors and agency correlates

| Construct                         | MIDUS II |      |       |      |      |      | MIDUS III |      |       |      |      |      | TWO-WAVE |      |       |      |      |      |
|-----------------------------------|----------|------|-------|------|------|------|-----------|------|-------|------|------|------|----------|------|-------|------|------|------|
|                                   | (1)      | (2)  | (3)   | (4)  | (5)  | (6)  | (1)       | (2)  | (3)   | (4)  | (5)  | (6)  | (1)      | (2)  | (3)   | (4)  | (5)  | (6)  |
| (1) Agency                        | 1.00     |      |       |      |      |      | 1.00      |      |       |      |      |      | 1.00     |      |       |      |      |      |
| (2) Personal Agency               | .98*     | 1.00 |       |      |      |      | .99*      | 1.00 |       |      |      |      | .98*     | 1.00 |       |      |      |      |
| (3) Agency Achievement (2-factor) | .79*     | .67* | 1.00  |      |      |      | .71*      | .60* | 1.00  |      |      |      | .79*     | .67* | 1.00  |      |      |      |
| (4) Intrinsic Agency              | .90*     | .91* | .61*  | 1.00 |      |      | .93*      | .94* | .57*  | 1.00 |      |      | .90*     | .91* | .61*  | 1.00 |      |      |
| (5) Instrumental Agency           | .90*     | .94* | .57*  | .73* | 1.00 |      | .92*      | .95* | .51*  | .78* | 1.00 |      | .90*     | .94* | .57*  | .73* | 1.00 |      |
| (6) Agency Achievement (3-factor) | .80*     | .67* | 1.00* | .61* | .58* | 1.00 | .72*      | .61* | 1.00* | .57* | .53* | 1.00 | .80*     | .67* | 1.00* | .61* | .58* | 1.00 |
| (7) Life Satisfaction             | .52*     | .50* | .43*  | .41* | .50* | .43* | .51*      | .51* | .35*  | .44* | .49* | .36* | .38*     | .38* | .29*  | .31* | .36* | .30* |
| (8) Self-Acceptance (PWB)         | .72*     | .69* | .61*  | .58* | .68* | .62* | .72*      | .70* | .55*  | .60* | .70* | .56* | .56*     | .53* | .47*  | .44* | .51* | .48* |
| (9) Self-Esteem                   | .71*     | .69* | .58*  | .58* | .66* | .58* | .69*      | .67* | .53*  | .59* | .65* | .54* | .53*     | .51* | .43*  | .43* | .49* | .44* |
| (10) Purpose in Life (PWB)        | .66*     | .63* | .53*  | .57* | .59* | .54* | .67*      | .65* | .49*  | .59* | .62* | .50* | .50*     | .48* | .41*  | .43* | .43* | .41* |
| (11) Positive Relations (PWB)     | .55*     | .52* | .46*  | .45* | .51* | .46* | .54*      | .52* | .40*  | .45* | .51* | .41* | .42*     | .40* | .35*  | .34* | .38* | .35* |
| (12) Social Integration           | .36*     | .34* | .29*  | .30* | .32* | .29* | .37*      | .36* | .27*  | .32* | .34* | .27* | .31*     | .30* | .25*  | .27* | .27* | .25* |
| (13) Agency (Big 5)               | .36*     | .32* | .40*  | .28* | .29* | .40* | .33*      | .29* | .36*  | .26* | .27* | .36* | .30*     | .26* | .35*  | .23* | .24* | .35* |
| (14) Autonomy (PWB)               | .50*     | .48* | .44*  | .40* | .47* | .44* | .49*      | .47* | .39*  | .41* | .47* | .40* | .39*     | .38* | .34*  | .32* | .36* | .34* |

\* $p < 0.05$

meet standards for acceptable absolute fit, further supporting conceptualizations of agency as multidimensional. While not all models consistently demonstrated good relative fit, this can be at least partially explained by a combination of convergence issues and challenges related to both intercorrelations among agency items and non-normality of data.

Calculation of Cronbach's alpha coefficients and pairwise Pearson's  $r$  correlations between both individual scale items and latent factors indicated that both the GPAS and GHAI were internally consistent, providing further evidence of their content, substantive, and structural validity, and the significance, direction, and effect sizes of relationships between agency indicators and correlates of agency present preliminary proof that both the GPAS and GHAI possess convergent, discriminant, and criterion validity. Conducting correlational analyses across two waves of data demonstrated the stability of results over time, and correlations between MIDUS II agency measures and MIDUS III agency correlates indicated the predictive validity of the GPAS and GHAI. Altogether, our findings make a strong case for the construct validity of the GPAS and 9-item GPAIs and provide moderate evidence for the expanded 13- and 11-item GHAI.

## 5.1 Contribution

Because human agency is so frequently a misunderstood or misoperationalized construct, the first substantive contributions of this study were to identify common themes and characteristics of agency and organize prior conceptualizations of the construct into a cohesive framework that been implied, but never specified, in previous research. This framework builds on Sen's (1999) dichotomy of agency freedom and agency achievement, and further reifies agency's multidimensionality by clarifying its personal and interpersonal components, under which are situated the subconstructs of intrinsic, instrumental, proxy, and collective agency. The proposed human agency framework also identifies how these components of personal agency align with the essential human needs for autonomy, competence, and relatedness described by Deci and Ryan (1985) and other proponents of SDT, positioning agency as both a critical expression of and mechanism for human thriving and navigation of the social world.

Our item selection process also makes a novel methodological contribution to agency research and scale design. Constructing a composite tool from items identical to or like those used in previous research and validating them through rigorous examination using characteristics and other criteria related to the construct in question reduces the inherent bias of subjective item selection and establishes initial face and content validity for the tool produced. It also for synthesis of new measures from existing tools which, in the present context, was helpful given the lack of consensus regarding the agency construct. This approach should be considered in similar situations where there is disagreement over how a concept should be defined and operationalized.

This study successfully produced global, multidimensional indicators for personal agency and agency achievement that conform to Sen's (1985, 2006) definitions of agency freedom and achievement and align with characteristics of agency established by Alkire (2005, 2008), Burger and Walk (2016), and Kotan (2010). Previous research has explored the effects of agency using proxy measures (Alkire, 2008; Bhattacharyya, 1995), single-item measures (e.g., Graham & Nikolova, 2013; Hojman & Miranda, 2018; Okulicz-Kozaryn, 2015) and unidimensional scales (Kesavayuth et al., 2022; Pleeging et al., 2021; Serdiuk et al., 2018). Rather than relying on purportedly objective proxies or unidimensional indicators, the GPAS and GHAI utilize items that capture respondents' subjective

experience of perceptions and outcomes and capture multiple interrelated dimensions of agency. Although research on SDT (Reis et al., 2018) and psychological well-being (Margolis et al., 2021) has explored agency by combining measures approximating autonomy and competence, it has not integrated these concepts into a broader conceptual framework for human agency like the one articulated in the present study.

To our knowledge, the GPAS and GHAI are the first tools synthesized from existing conceptualizations and measures of human agency. The development of and validation of these multidimensional agency measures has critical implications for the social sciences. It addresses a critical gap in the literature, as there is presently no established measure for operationalizing personal agency (Cavazzoni et al., 2022; Ibrahim & Alkire, 2007). The GPAS and GHAI have utility and value for a variety of disciplines, as personal agency is a fundamental mechanism through which the social world is engaged. They can be used to explore how agency may influence diverse outcomes like prosocial behaviors (Christoph et al., 2014), group membership and volunteerism (Cicognani et al., 2015), community and civic engagement, democratic governance, and the development of other resources and capabilities that support well-being (Alkire, 2005; Ibrahim & Alkire, 2007; Peterson et al., 2008). The GPAS and GHAI may also contribute to the advancement of empowerment theory and the processes by which individuals leverage assets and capabilities to assert control over their lives and participate in change processes (Chan & Mak, 2020; Kieffer, 1984; Rappaport, 1987; Zimmerman, 2000).

The GPAS and GHAI have particular value for the study of community development (Bhattacharyya, 1995), the Capabilities Approach to global development (Sen, 1985, 1999), and subjective well-being (Comim, 2005; Kotan, 2010), as each field understands agency to be essential for human well-being and thriving. If the expansion of freedom, the primary end of development activities, is enacted through agency (Sen, 1999) and the “development” in community development refers to the production of agency among group members (Bhattacharyya, 1995) then, up to this point, these fields have lacked robust indicators for one of their most critical outcomes. This has severely inhibited our ability to probe, among other things, how personal agency might predict desired outcomes like well-being, solidarity, and community and civic engagement, as well as how personal agency might be predicted by environmental factors, policies, and interventions intended to contribute to development initiatives. Now that such indicators have been created and validated, the GPAS and GHAI allow us to assess what is most essential to development—that is, whether the work of the field is advancing freedom by empowering individuals and communities to enact control over their lives and pursue self-determined goals that contribute to outcomes that they value (Alkire, 2008; Bhattacharyya, 1995; Burger & Walk, 2016; Cavazzoni et al., 2022; Kabeer, 1999; Kotan, 2010; Sen, 1999).

The GPAS and GHAI also offer several practical benefits. Because they rely on subjectively determined, global indicators, the GPAS and GHAI are useful for analysis of a general population and can be complemented by supplementary measures or adapted for sub-populations of interest like women (Yount et al., 2020) and children (Poteat et al., 2018; Veronese et al. 2019a, b, 2020a, b). As recommended by Alkire (2008), the tools include items that measure autonomy, ability, direct control, and effective power, and address outcomes phrased so that they can refer to both well-being and other-regarding goals. The latent structures of the GPAS and GHAI also allow for researchers to select from a variety of scales to study agency subconstructs of interest. The GPAS and best-fitting GHAI are only nine items in length, making them easier to deploy alongside other measures without contributing to response attrition, which may be more likely to be experienced when using longer tools like the BIF (Vallacher & Wegner, 1989), ATPA-22 (Lautamo, et al., 2021),

or WAS-61 (Yount et al., 2020). While tools like Hitlin and Elder's (2006) agency model, Snyder's (1991) Agentic Pathways subscale, and Smith and et al.'s (2000) Personal Agency Scale are comparably brief, they conceptualize agency in ways that deviate from how it is commonly discussed in literature, either conflating it with or subsuming it under other constructs like hope, optimism, or planfulness. The GPAS and GHAI do not incorporate proxy measures for agency or indicators for resources that may contribute to agency; however, the brevity of the proposed tools allows them to be deployed alongside measures, potentially enabling analysis of how those resources influence personal agency and agency achievement.

Different versions of the tools also provide distinct benefits: the GPAS provides the most nuanced representation of personal agency and its subfactors, which is typically the construct engaged in agency research. nine-item GHAI provides the best-fitting and most elegant measures of personal agency, its subfactors, and agency achievement, and the 13- and 11-item GHAI enable exploration of the possible influence of one's ability to solve problems and achieve a lifestyle they value on the relationships between intrinsic and instrumental agency and between both personal and instrumental agency and agency achievement, respectively. Finally, because of the nature of their construction and validation we assert that, with further testing, the GPAS and GHAI may emerge as exemplar tools against which other agency indicators might be compared to assess their construct validity. This would be a substantive contribution to the field, as it could serve as a starting point for the development and validation of more bespoke and effective agency indicators.

## 5.2 Limitations

Despite their potential for informing social science research, the GPAS and GHAI have several limitations related to the use of the MIDUS dataset. Our study was not able to identify items previously used to assess interpersonal, proxy, or collective agency. This contributed to a significant gap in our findings as, based on our conceptual framework for human agency, these constructs comprise one half of agency freedom and represent the relatedness component of self-determination theory. While the GHAI are a promising start, without appropriate measures for interpersonal, proxy, and collective agency, our proposed measures are a conspicuously incomplete attempt at developing a full general human agency index.

Due to issues related to sample size and the number of fitted parameters, it was not possible to converge several of the full models recommended by EFA. To test the fit of these models, a larger sample size is required. As this study used a national dataset, the GPAS and GHAI have not been tested for generalizability beyond the United States population, nor have they been shown to be generalizable to subgroups within that population; this is particularly true of children and youth, who were not included in sampling. Because MIDUS does not include established human agency indicators it was not possible to test criterion validity for the proposed tools through direct comparison with exemplars, and further testing of concurrent and predictive validity using regression analysis is needed before the criterion validity of the GPAS and GHAI can be confirmed.

Finally, the GPAS and GHAI are only useful for interpreting agency through a quantitative lens. While helpful for identify general patterns among a given population, such analyses should be complemented by qualitative research to develop a richer and more nuanced understanding of how personal agency is experienced phenomenologically across socioeconomic and cultural contexts. This need for accompanying qualitative research is

compounded by the subjectively determined nature of human agency—as only respondents can discern whether and how they are experiencing agency, qualitative input is necessary to ensure that research on human agency accurately reflects their articulated lived experience.

### 5.3 Future Directions

These limitations offer several directions for future research on human agency. An immediate next step would be to leverage correlational and regression analyses to continue to probe relationships between the GPAS and GHAI and concepts previously associated with personal agency like well-being indicators and participatory behaviors. Here, the identification of significant relationships would reinforce the convergent, discriminant, and predictive validity of the tools and help us understand the effects of agency on outcomes of interest. To test their generalizability, fixed-effects models should be employed to assess whether there are differences in how agency is experienced across subpopulations categorized by socioeconomic indicators like age, race, biological sex, income, and education, and time series models would allow further testing of scales' predictive validity.

Analyses could be performed using the MIDUS dataset; however, collection of primary data may offer additional benefits. International data could be collected to examine agency in a global context, compare how it is experienced across cultures, and better assess the generalizability of the GPAS and GHAI. Primary data incorporating other validated agency measures like the BIF (Vallacher & Wegner, 1989), ATPA-22 (Lautamo, et al. 2020), or Hitlin and Elder's (2006) agency model would enable confirmatory analysis of the GPAS' and GHAI's criterion validity. Further, data collection with a tool comprised of multiple agency measures could expand on the present study by generating GHAI selected from a wider variety of items representing all agency dimensions articulated in our conceptual framework. A larger sample would likely resolve the convergence issues encountered during CFA, potentially clarifying the roles of cross-loaded variables in the 13- and 11-item GHAI. CFA could also test for predictive relationships between agency dimensions and exogenous variables, contributing to our understanding of how agency is experienced, nurtured, and enacted.

Future research should also address the opportunity to use aspects of our item selection process to create measures for interpersonal, proxy, and collective agency to complement and complete the GHAI, either by selecting from existing or developing novel indicators based on established criteria for and characteristics of agency. The GPAS and GHAI are important steps forward; however, on their own they are insufficient for fully operationalizing human agency because they do not capture its social dimensions. Finally, findings that emerge from quantitative analyses utilizing the GPAS and GHAI must be informed by a rich, phenomenological understanding of agency that can only be gained through qualitative research. Should suitable interpersonal agency measures prove difficult to identify, qualitative methods may also be useful for developing novel indicators.

### 5.4 Conclusion

Human agency is an essential mechanism through which the social world is constructed and navigated, but underoperationalization has limited our ability to assess agency and its relationships with social interaction, well-being, and the thriving of individuals and communities. This study advances agency research by engaging past scholarship to identify critical characteristics of agency and constructing a cogent conceptual framework

representing its latent structure. Components of that framework were reified and through the creation and initial validation of the GPAS and GPAIs, contributing a set of tools that assess subjectively determined, multidimensional aspects of personal agency and agency achievement. At a minimum, we hope that these tools will generate further discussion on how to best measure human agency and expand ongoing efforts to come to a unified operationalization of the construct.

## Appendix A: Definitions of agency (adapted from Cavazzoni et al., 2022)

| Article                           | Agency definition                                                                                                                                                                                                                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ahearn (2001), p. 112             | “Agency refers to the socioculturally mediated capacity to act.”                                                                                                                                                                                                                              |
| Alkire (2008), p. 6               | “i) Agency is exercised with respect to goals the person values; ii) agency includes effective power as well as direct control; iii) agency may advance wellbeing or may address other-regarding goals; iv) to identify agency also entails an assessment of the value of the agent’s goals.” |
| Alsop et al., (2006), p. 11       | “Agency is defined as an actor’s or group’s ability to make purposeful choices—that is, the actor is able to envisage and purposively choose options.”                                                                                                                                        |
| Bandura (2001), p. 8              | “Agency thus involves not only the deliberative ability to make choices and action plans, but the ability to give shape to appropriate courses of action and to motivate and regulate their execution.”                                                                                       |
| Barandiaran et al. (2009), p. 369 | “A system doing something by itself according to certain goals or norms within a specific environment.”                                                                                                                                                                                       |
| Barker (2005), p. 632             | “The socially determined capability to act and make a difference.”                                                                                                                                                                                                                            |
| Beer (1995), p. 173               | “Any embodied system [that pursues] internal or external goals by its own actions while inconinuous long-term interaction with the environment in which it is situated.”                                                                                                                      |
| Bentley-Edwards (2016), p. 78     | “The perception of what one is able to do to control their environment or circumstance.”                                                                                                                                                                                                      |
| Berhane et al. (2019), p. S53     | The “ability to define goals, and act on them”                                                                                                                                                                                                                                                |
| Beyers et al. (2003), P. 360      | “AGENCY reflects the possibility of self-directed behavior.”                                                                                                                                                                                                                                  |
| Bhattacharyya (1995), p. 61       | “The capacity of a people to order their world... to create, reproduce, change, and live according to their own meaning systems, the powers effectively to define themselves as opposed to being defined by others.”                                                                          |
| Black, (2016), p. 296             | “Moral agency refers to the ability of individuals to determine their behavior when it affects others’ well-being.”                                                                                                                                                                           |
| Bryan et al., (2014), p. 242      | “The sense that one is in control of one’s life, and is the initiator of one’s own actions.”                                                                                                                                                                                                  |
| Burger and Walk (2016)            | “Individuals’ capacity to gain control over their lives largely independently of social structure (Chin and Phillips 2004),”                                                                                                                                                                  |

| Article                               | Agency definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cadenas et al., (2021), pp. 93–94     | “Critical agency can be conceptualized as a component of critical consciousness that combines motivation and beliefs of self-efficacy to address societal injustices or it is identified as ones perceived ability to make a difference for social change.”                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Cavazzoni et al., (2022), p. 1126     | “People’s ability to exert control over one’s life and pursue goals.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cheong et al., (2017), p. 25          | “The ‘ability to define one’s goals and act upon them.’” (citing Kabeer, 1999, p. 438)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Christensen and Hooker (2000), p. 133 | “AGENTS are entities which engage in normatively constrained, goal-directed, interaction with their environment.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Franklin and Graesser, (1996), p. 25  | “An autonomous agent is a system situated within and a part of an environment that senses that environment and acts on it, over time, in pursuit of its own agenda and so as to effect what it senses in the future.”                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Giddens (1984), p. 14                 | “To be able to intervene in the world, or to refrain from such intervention, with the effect of influencing a specific process or state of affairs.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Graham and Nikolova (2013), p. 4      | “The capacity to pursue a purposeful and fulfilling life.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Grower & Ward, (2018), p. 139         | “Sexual agency is a multidimensional construct that reflects the awareness of self as a sexual being; the ability to identify, negotiate, and communicate one’s sexual needs; and the successful initiation of behaviors that allow for the satisfaction of these needs.”                                                                                                                                                                                                                                                                                                                                                                                                       |
| Habashi & Worley, (2009), p. 44       | “The ability of the agent to reinvent the local resources that are produced by global/local discourse while responding to the same global hegemony.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Harvey (2002), p.173                  | “The capacity of persons to transform existing states of affairs,”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Hitlin and Elder, (2006), p. 38       | “An individual capacity for meaningful and sustained action.” the sense of having the capacity for meaningful and successful action, something related, but not equivalent, to the perception of having structural opportunities to exercise such capacities.” (p. 40) “Agency represents a human capacity to influence one’s own life within socially structured opportunities.” (pp. 56–57) “Agency, in this model, represents an individual capacity, one that is both the result of individual differences (planfulness) as well as achieved successes (self-efficacy) and a sense of temporal, self-reflective understanding about one’s life chances (optimism).” (p. 60) |
| Horvath (1998), p. 139                | “A mode of human functioning that involves self-concern, self-protection, self-determination, self-efficacy, and an instrumental approach to the environment”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Kabeer (1999), p. 438                 | “The ability to define one’s goals and act upon them.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Kauffman, (2000), p. 8                | “A system that can act on its own behalf in an environment.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Klein et al., (2018)                  | “Sexual agency is commonly defined as the ability to act according to one’s own wishes and have control of one’s own sexual life.” (quoting Fahs and McClelland, 2016, p. 396)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Kotan (2010), p. 370                  | “The ability to exert power so as to influence the state of the world, do so in a purposeful way and in line with self-established objectives.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Article                                  | Agency definition                                                                                                                                                                                                                                                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Krauss et al., (2014), p. 1552           | "Psychological agency refers to beliefs about one's abilities in nonsocial environments, such as intellectual or artistic skills (Zimmerman and Zahniser 1991), and the ability to set goals and organize one's actions to achieve them (Bandura 2006; Larson and Angus 2011)."                                                  |
| Lautamo et al., (2021)                   | "The capacity of individuals to act independently and to make their own free choices." (citing Barker, 2005)                                                                                                                                                                                                                     |
| Maes, (1994), p. 136                     | "A system that tries to fulfill a set of goals in a complex, dynamic environment."                                                                                                                                                                                                                                               |
| McWhirter & McWhirter, (2016), p. 553    | "Critical agency combines commitment to and efficacy for taking action against racism and discrimination."                                                                                                                                                                                                                       |
| Moore et al., (2016), p. 890             | "Belief in one's ability to affect change," "Agency refers to the ability to intentionally influence one's life circumstances," (p. 891, citing Bandura, 2006)                                                                                                                                                                   |
| Narayan and Petesch, (2007), p. 15       | "People's ability to act individually or collectively to further <i>their own interests</i> ."                                                                                                                                                                                                                                   |
| Nestadt et al. (2022), pp. NP8819-NP8820 | "Agency is the ability to define one's goals and take action to realize them (Kabeer, 1999). Practically, it is the ability to make choices and act in accordance with what one desires to do without impediment (Blanchard et al., 2013; Kabeer, 1999; Mosedale, 2005)."                                                        |
| Onyx and Bullen (2000), p. 29            | "The capacity of the individual to plan and initiate action."                                                                                                                                                                                                                                                                    |
| Pleeging et al., (2021), p. 1025         | "The belief that we are able to achieve our goals."                                                                                                                                                                                                                                                                              |
| Poteat et al., (2018)                    | "A global belief in one's ability to make and attain goals in general; Snyder et al., (1996)"                                                                                                                                                                                                                                    |
| Reeve & Tseng, (2011), p. 258            | "We define agentic engagement as students' constructive contribution into the flow of the instruction they receive."                                                                                                                                                                                                             |
| Richardson et al., (2019), p. 3          | "Agency is the ability to identify one's goals and act upon them (Kabeer, 1999)."                                                                                                                                                                                                                                                |
| Richardson, (2018), p. 541               | "The ability to make choices and act upon those choices" (citing Malhotra & Schuler, 2005 and Kabeer, 1999)                                                                                                                                                                                                                      |
| Russell and Norvig, (1995), p. 33        | "An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors."                                                                                                                                                                                      |
| Salem et al., (2020), p. 653             | "As women's exercise of choice in [decision-making, freedom of movement, and gender attitudes],"                                                                                                                                                                                                                                 |
| Samari, (2017), p. 562                   | "The ability to define life choices in an evolving historic and social context... Agency includes the ability to formulate one's own strategic choices, to control resources, and to make attitudinal changes under evolving constraints (Crandall et al. 2016; Dyson and Moore 1983; Yount et al. 2016)." (citing Kabeer, 1999) |
| Sen (1985), p. 203                       | "What a person is free to do and achieve in pursuit of whatever goals or values he or she regards as important."                                                                                                                                                                                                                 |
| Mortimer & Shanahan, (2007)              | "The ability to exert influence on one's life."                                                                                                                                                                                                                                                                                  |
| Smith et al., (2000), p. 458             | Personal agency involves "achieving desired outcomes on one's own behalf (e.g., through ability, choices, perseverance, or planning),"                                                                                                                                                                                           |
| Smithers, 1995, p. 97                    | "Agent systems are systems that can initiate, sustain, and maintain an ongoing and continuous interaction with their environment as an essential part of their normal functioning."                                                                                                                                              |

| Article                             | Agency definition                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stattin et al., (2017), p. 309      | “Political agency is defined as person’s intentional attempts to affect other peoples’ minds about political and issues.”                                                                                                                                                                                                                                                |
| Steckermeier (2019), p. 31          | “Agency combines two different aspects: The ability to act independently from others—comparable to the process aspect of freedom in the capabilities approach; and the ability to choose from different opportunities—denoted as the opportunity aspect in the capabilities approach (Sen 2007, p. 10; Archard 2015, p. 5).”                                             |
| Thoits (2003), p. 190               | “The ability to initiate self-change (e.g., Kiecolt, 1994; Lazarus & Folkman, 1984).”                                                                                                                                                                                                                                                                                    |
| Veronese et al. (2018), p. 863      | “Agency can be defined as a creative and dynamic act of resistance to oppose the oppressor and/or occupier (Peteet, 1994).”                                                                                                                                                                                                                                              |
| Veronese et al., (2019b), p. 2      | “The capacity to act positively across space and time with respect to oppressive structures in one’s environment (Jeffrey, 2012).”                                                                                                                                                                                                                                       |
| Veronese et al. (2020a), p. 243     | “The transformational and generative operations by which cognitive models are translated into proficient action... as well as the changes that occur in multilevel regulation of skills as they are perfected.” (quoting Bandura, 1991, p. 61)                                                                                                                           |
| Victor et al. (2013), p. 32         | “Kotan (2010, p. 370) defines agency as ‘the ability to exert power so as to influence the state of the world, do so in a purposeful way and in line with self-established objectives.’”                                                                                                                                                                                 |
| Ward et al. (2018), p. 30           | “Although scholars define sexual agency in many ways, in general it includes the acknowledgment of self as a sexual being; the ability to identify, communicate, and negotiate one’s sexual needs; and the successful initiation of behaviors that allow for the satisfaction of these desires (Fetterolf & Sanchez, 2015; Froyum, 2010; Horne & Zimmer-Gembeck, 2005).” |
| Williams and Merten (2014), p. 1565 | “Agency is more than independence or autonomy; the construct refers to a person’s capacity, willingness, and ability to actively construct their life course (Elder and Hitlin 2006).”                                                                                                                                                                                   |
| Yount et al. (2016)                 | “Women’s agency refers to their ability to make strategic life choices under historically evolving constraints (Kabeer, 1999; VanderEnde et al., n.d.).”                                                                                                                                                                                                                 |
| Yount et al., (2020), p. 6          | “Ability to make strategic choices under constraints.” (citing Kabeer, 1999)                                                                                                                                                                                                                                                                                             |
| Zimmerman et al., (2019), p. 1      | “‘The capacity to make purposeful choices’ (Kabeer, 1999)”                                                                                                                                                                                                                                                                                                               |

## Appendix B: Descriptive Statistics and Tests of Multivariate Normality

| Descriptive statistics—MIDUS II             |     |     |             |      |     |       |      |            |      |     |       |            |      |      |     |       |      |
|---------------------------------------------|-----|-----|-------------|------|-----|-------|------|------------|------|-----|-------|------------|------|------|-----|-------|------|
| Agency items                                | Min | Max | Full sample |      |     |       |      | EFA sample |      |     |       | CFA sample |      |      |     |       |      |
|                                             |     |     | N           | Mean | SD  | Skew  | Kurt | N          | Mean | SD  | Skew  | Kurt       | N    | Mean | SD  | Skew  | Kurt |
| Control over life in general at present*    | 1   | 4   | 4955        | 3.69 | 0.6 | −1.96 | 6.8  | 2478       | 3.68 | 0.6 | −1.9  | 6.5        | 2477 | 3.7  | 0.6 | −2.01 | 7.2  |
| Where there's a will there's a way*         | 1   | 4   | 4014        | 3.21 | 0.8 | −0.71 | 2.9  | 2011       | 3.2  | 0.8 | −0.68 | 2.9        | 2003 | 3.22 | 0.8 | −0.73 | 3    |
| Do what can to change for better*           | 1   | 4   | 4017        | 3.44 | 0.6 | −0.79 | 3    | 2014       | 3.44 | 0.7 | −0.8  | 3          | 2003 | 3.45 | 0.6 | −0.78 | 2.9  |
| Even when feel too much, get it all done*   | 1   | 4   | 4005        | 3.22 | 0.8 | −0.73 | 2.9  | 2007       | 3.21 | 0.8 | −0.69 | 2.7        | 1998 | 3.24 | 0.8 | −0.78 | 3    |
| Know what I want out of life*               | 1   | 4   | 4010        | 3.14 | 0.8 | −0.68 | 2.8  | 2009       | 3.14 | 0.8 | −0.68 | 2.9        | 2001 | 3.14 | 0.8 | −0.69 | 2.8  |
| Not afraid to voice opinions in opposition* | 1   | 7   | 4020        | 5.56 | 1.6 | −1.28 | 3.8  | 2015       | 5.58 | 1.5 | −1.28 | 3.8        | 2005 | 5.53 | 1.6 | −1.27 | 3.8  |
| In charge of situation in which I live*     | 1   | 7   | 4021        | 5.96 | 1.3 | −1.73 | 6    | 2016       | 5.95 | 1.2 | −1.69 | 5.8        | 2005 | 5.97 | 1.3 | −1.77 | 6.2  |
| Difficult arranging life in satisfying way  | 1   | 7   | 4018        | 5    | 1.9 | −0.48 | 1.9  | 2018       | 4.97 | 1.8 | −0.44 | 1.9        | 2000 | 5.03 | 1.9 | −0.52 | 1.9  |

Descriptive statistics—MIDUS II

| Agency items                                      | Min      | Max      | Full sample |             |            |              |            | EFA sample  |             |            |              |            | CFA sample  |             |            |              |            |
|---------------------------------------------------|----------|----------|-------------|-------------|------------|--------------|------------|-------------|-------------|------------|--------------|------------|-------------|-------------|------------|--------------|------------|
|                                                   |          |          | N           | Mean        | SD         | Skew         | Kurt       | N           | Mean        | SD         | Skew         | Kurt       | N           | Mean        | SD         | Skew         | Kurt       |
| Decisions not influenced by others doing*         | 1        | 7        | 4021        | 5.49        | 1.6        | -1.08        | 3.3        | 2018        | 5.39        | 1.6        | -0.99        | 3.1        | 2003        | 5.59        | 1.5        | -1.18        | 3.6        |
| <b>Demands of everyday life often get me down</b> | <b>1</b> | <b>7</b> | <b>4020</b> | <b>4.76</b> | <b>1.9</b> | <b>-0.25</b> | <b>1.7</b> | <b>2016</b> | <b>4.77</b> | <b>1.9</b> | <b>-0.23</b> | <b>1.7</b> | <b>2004</b> | <b>4.74</b> | <b>1.9</b> | <b>-0.28</b> | <b>1.7</b> |
| <b>Actively carry out plans I set for self*</b>   | <b>1</b> | <b>7</b> | <b>4023</b> | <b>5.59</b> | <b>1.5</b> | <b>-1.22</b> | <b>3.9</b> | <b>2018</b> | <b>5.59</b> | <b>1.5</b> | <b>-1.18</b> | <b>3.8</b> | <b>2005</b> | <b>5.6</b>  | <b>1.5</b> | <b>-1.26</b> | <b>4</b>   |
| Judge self by what I think is important*          | 1        | 7        | 4024        | 6.09        | 1.2        | -1.74        | 6.2        | 2019        | 6.08        | 1.2        | -1.74        | 6.1        | 2005        | 6.1         | 1.2        | -1.74        | 6.3        |
| <b>Able to build lifestyle to my liking*</b>      | <b>1</b> | <b>7</b> | <b>4022</b> | <b>5.9</b>  | <b>1.3</b> | <b>-1.71</b> | <b>5.9</b> | <b>2019</b> | <b>5.86</b> | <b>1.4</b> | <b>-1.65</b> | <b>5.6</b> | <b>2003</b> | <b>5.95</b> | <b>1.3</b> | <b>-1.78</b> | <b>6.3</b> |
| Influenced by people with strong opinions         | 1        | 7        | 4022        | 4.74        | 1.8        | -0.16        | 1.7        | 2017        | 4.72        | 1.8        | -0.15        | 1.7        | 2005        | 4.77        | 1.8        | -0.17        | 1.7        |
| Good managing daily responsibilities*             | 1        | 7        | 4020        | 6.09        | 1.3        | -1.97        | 7.1        | 2016        | 6.08        | 1.3        | -1.89        | 6.8        | 2004        | 6.09        | 1.2        | -2.06        | 7.5        |
| Difficult voice opinion on controversial          | 1        | 7        | 4017        | 4.97        | 2          | -0.51        | 1.8        | 2016        | 4.96        | 2          | -0.51        | 1.8        | 2001        | 4.98        | 2          | -0.51        | 1.8        |
| <b>Overwhelmed by my responsibilities</b>         | <b>1</b> | <b>7</b> | <b>4019</b> | <b>4.85</b> | <b>1.9</b> | <b>-0.36</b> | <b>1.7</b> | <b>2016</b> | <b>4.83</b> | <b>1.9</b> | <b>-0.34</b> | <b>1.7</b> | <b>2003</b> | <b>4.87</b> | <b>1.9</b> | <b>-0.38</b> | <b>1.7</b> |

Descriptive statistics—MIDUS II

| Agency items                              | Min | Max | Full sample |      |     |       |      | EFA sample |      |     |       |      | CFA sample |      |     |       |      |
|-------------------------------------------|-----|-----|-------------|------|-----|-------|------|------------|------|-----|-------|------|------------|------|-----|-------|------|
|                                           |     |     | N           | Mean | SD  | Skew  | Kurt | N          | Mean | SD  | Skew  | Kurt | N          | Mean | SD  | Skew  | Kurt |
| Little can do to change important things  | 1   | 7   | 4007        | 5.61 | 1.7 | -1.23 | 3.4  | 2009       | 5.58 | 1.7 | -1.18 | 3.3  | 1998       | 5.65 | 1.7 | -1.29 | 3.6  |
| Helpless dealing with problems of life    | 1   | 7   | 4008        | 5.42 | 1.8 | -0.88 | 2.5  | 2009       | 5.38 | 1.8 | -0.86 | 2.5  | 1999       | 5.47 | 1.7 | -0.91 | 2.5  |
| Do just about anything I set my mind to*  | 1   | 7   | 4014        | 5.8  | 1.4 | -1.55 | 5.1  | 2011       | 5.78 | 1.4 | -1.53 | 5    | 2003       | 5.82 | 1.4 | -1.57 | 5.1  |
| Others determine what I can and cannot do | 1   | 7   | 4012        | 5.88 | 1.5 | -1.37 | 4    | 2012       | 5.83 | 1.5 | -1.31 | 3.7  | 2000       | 5.92 | 1.5 | -1.44 | 4.2  |
| What happens in life is beyond my control | 1   | 7   | 4009        | 5.19 | 1.8 | -0.72 | 2.2  | 2009       | 5.2  | 1.8 | -0.73 | 2.3  | 2000       | 5.17 | 1.8 | -0.7  | 2.2  |
| When really want something, find way*     | 1   | 7   | 4007        | 6    | 1.2 | -1.75 | 6.7  | 2013       | 5.97 | 1.2 | -1.76 | 6.6  | 1994       | 6.03 | 1.1 | -1.72 | 6.8  |
| Many things interfere with what I want do | 1   | 7   | 4009        | 4.3  | 1.8 | -0.02 | 1.7  | 2011       | 4.27 | 1.8 | 0.01  | 1.7  | 1998       | 4.33 | 1.8 | -0.05 | 1.7  |
| Whether I get what want is in own hands*  | 1   | 7   | 4009        | 5.25 | 1.5 | -0.95 | 3.1  | 2012       | 5.24 | 1.5 | -0.91 | 3    | 1997       | 5.26 | 1.6 | -0.98 | 3.2  |

| Descriptive statistics—MIDUS II           |          |     |             |                    |     |            |      |      |            |        |       |      |      |      |      |       |      |
|-------------------------------------------|----------|-----|-------------|--------------------|-----|------------|------|------|------------|--------|-------|------|------|------|------|-------|------|
| Agency items                              | Min      | Max | Full sample |                    |     | EFA sample |      |      | CFA sample |        |       | Kurt |      |      |      |       |      |
|                                           |          |     | N           | Mean               | SD  | Skew       | Kurt | N    | Mean       | SD     | Skew  |      | Kurt | N    | Mean | SD    | Skew |
| Little control over things happen to me   | 1        | 7   | 4008        | 5.42               | 1.6 | −1         | 3    | 2011 | 5.41       | 1.6    | −0.97 | 2.9  | 1997 | 5.44 | 1.6  | −1.04 | 3.1  |
| Really no way I can solve problems I have | 1        | 7   | 4008        | 5.98               | 1.4 | −1.66      | 5.3  | 2012 | 5.97       | 1.4    | −1.63 | 5.2  | 1996 | 6    | 1.4  | −1.68 | 5.4  |
| Feel pushed around in life                | 1        | 7   | 4012        | 5.54               | 1.7 | −0.9       | 2.5  | 2012 | 5.49       | 1.8    | −0.85 | 2.4  | 2000 | 5.59 | 1.7  | −0.95 | 2.6  |
| Happens to me in future depends on me*    | 1        | 7   | 4013        | 5.89               | 1.4 | −1.67      | 5.5  | 2014 | 5.89       | 1.4    | −1.68 | 5.6  | 1999 | 5.89 | 1.4  | −1.66 | 5.5  |
| Rate control over life                    | 0        | 10  | 4002        | 7.91               | 1.8 | −1.24      | 4.8  | 2004 | 7.86       | 1.9    | −1.23 | 4.7  | 1998 | 7.96 | 1.8  | −1.25 | 4.9  |
| *Item reversed                            |          |     |             |                    |     |            |      |      |            |        |       |      |      |      |      |       |      |
| Tests of multivariate normality—MIDUS II  |          |     |             |                    |     |            |      |      |            |        |       |      |      |      |      |       |      |
| Test                                      | Value    |     |             | $\chi^2$           |     |            | DF   |      |            | p      |       |      |      |      |      |       |      |
| Mardia—skewness                           | 75.31973 |     |             | 48,119.183         |     |            | 4960 |      |            | 0.0000 |       |      |      |      |      |       |      |
| Mardia—kurtosis                           | 1207.884 |     |             | 30,643.314         |     |            | 1    |      |            | 0.0000 |       |      |      |      |      |       |      |
| Henze-Zirkler                             | 2.113454 |     |             | 4.98e <sup>9</sup> |     |            | 1    |      |            | 0.0000 |       |      |      |      |      |       |      |
| Doornik-Hansen                            |          |     |             |                    |     |            | 60   |      |            | 0.0000 |       |      |      |      |      |       |      |

Descriptive statistics—MIDUS III

| Agency items                                | Min | Max | Full sample |      |     |       |      | EFA sample |      |     |       |      | CFA sample |      |     |       |      |
|---------------------------------------------|-----|-----|-------------|------|-----|-------|------|------------|------|-----|-------|------|------------|------|-----|-------|------|
|                                             |     |     | N           | Mean | SD  | Skew  | Kurt | N          | Mean | SD  | Skew  | Kurt | N          | Mean | SD  | Skew  | Kurt |
| Control over life in general at present*    | 1   | 4   | 3288        | 3.63 | 0.6 | -1.76 | 5.8  | 1640       | 3.62 | 0.7 | -1.77 | 5.9  | 1648       | 3.63 | 0.6 | -1.75 | 5.8  |
| Where there's a will there's a way*         | 1   | 4   | 2888        | 3.22 | 0.8 | -0.65 | 2.9  | 1439       | 3.2  | 0.8 | -0.68 | 3    | 1449       | 3.23 | 0.7 | -0.62 | 2.8  |
| Do what can to change for better*           | 1   | 4   | 2897        | 3.45 | 0.6 | -0.75 | 2.8  | 1441       | 3.45 | 0.6 | -0.75 | 3    | 1456       | 3.46 | 0.6 | -0.75 | 2.6  |
| Even when feel too much, get it all done*   | 1   | 4   | 2887        | 3.18 | 0.8 | -0.66 | 2.8  | 1439       | 3.14 | 0.8 | -0.6  | 2.6  | 1448       | 3.23 | 0.8 | -0.71 | 3    |
| Know what I want out of life*               | 1   | 4   | 2863        | 3.16 | 0.8 | -0.68 | 2.9  | 1427       | 3.16 | 0.8 | -0.68 | 2.9  | 1436       | 3.16 | 0.8 | -0.67 | 2.8  |
| Not afraid to voice opinions in opposition* | 1   | 7   | 2907        | 5.61 | 1.5 | -1.31 | 4.1  | 1447       | 5.57 | 1.5 | -1.26 | 3.9  | 1460       | 5.64 | 1.5 | -1.35 | 4.3  |
| In charge of situation in which I live*     | 1   | 7   | 2914        | 5.96 | 1.3 | -1.77 | 6.3  | 1452       | 5.93 | 1.3 | -1.74 | 6.2  | 1462       | 5.98 | 1.3 | -1.79 | 6.3  |
| Difficult arranging life in satisfying way  | 1   | 7   | 2906        | 5.11 | 1.8 | -0.6  | 2.1  | 1451       | 5.08 | 1.8 | -0.58 | 2.1  | 1455       | 5.14 | 1.8 | -0.61 | 2.1  |
| Decisions not influenced by others doing*   | 1   | 7   | 2911        | 5.56 | 1.5 | -1.14 | 3.6  | 1450       | 5.56 | 1.5 | -1.19 | 3.7  | 1461       | 5.56 | 1.5 | -1.1  | 3.4  |

Descriptive statistics—MIDUS III

| Agency items                               | Full sample |     |      |      |     | EFA sample |      |      |      |     | CFA sample |      |      |      |     |       |      |
|--------------------------------------------|-------------|-----|------|------|-----|------------|------|------|------|-----|------------|------|------|------|-----|-------|------|
|                                            | Min         | Max | N    | Mean | SD  | Skew       | Kurt | N    | Mean | SD  | Skew       | Kurt | N    | Mean | SD  | Skew  | Kurt |
| Demands of everyday life often get me down | 1           | 7   | 2913 | 4.88 | 1.8 | -0.4       | 1.9  | 1451 | 4.91 | 1.8 | -0.43      | 1.9  | 1462 | 4.85 | 1.9 | -0.37 | 1.8  |
| Actively carry out plans I set for self*   | 1           | 7   | 2910 | 5.59 | 1.5 | -1.22      | 3.9  | 1450 | 5.56 | 1.5 | -1.14      | 3.7  | 1460 | 5.62 | 1.5 | -1.29 | 4.2  |
| Judge self by what I think is important*   | 1           | 7   | 2911 | 5.89 | 1.3 | -1.66      | 5.8  | 1451 | 5.9  | 1.4 | -1.67      | 5.8  | 1460 | 5.89 | 1.3 | -1.66 | 5.9  |
| Able to build lifestyle to my liking*      | 1           | 7   | 2913 | 5.84 | 1.4 | -1.62      | 5.3  | 1451 | 5.81 | 1.4 | -1.58      | 5.2  | 1462 | 5.88 | 1.4 | -1.65 | 5.5  |
| Influenced by people with strong opinions  | 1           | 7   | 2912 | 4.77 | 1.7 | -0.18      | 1.8  | 1452 | 4.73 | 1.7 | -0.14      | 1.8  | 1460 | 4.81 | 1.7 | -0.23 | 1.9  |
| Good managing daily responsibilities*      | 1           | 7   | 2910 | 6.08 | 1.2 | -1.86      | 6.8  | 1451 | 6.07 | 1.2 | -1.81      | 6.7  | 1459 | 6.1  | 1.2 | -1.92 | 7    |
| Difficult voice opinion on controversial   | 1           | 7   | 2907 | 5.06 | 1.9 | -0.61      | 2    | 1450 | 5.06 | 1.9 | -0.61      | 2    | 1457 | 5.06 | 1.9 | -0.61 | 2    |
| Overwhelmed by my responsibilities         | 1           | 7   | 2907 | 4.99 | 1.9 | -0.5       | 1.9  | 1450 | 4.95 | 1.9 | -0.46      | 1.9  | 1457 | 5.02 | 1.9 | -0.55 | 2    |
| Little can do to change important things   | 1           | 7   | 2897 | 5.42 | 1.7 | -1.03      | 2.9  | 1444 | 5.4  | 1.7 | -1.04      | 2.9  | 1453 | 5.43 | 1.7 | -1.02 | 2.9  |

Descriptive statistics—MIDUS III

| Agency items                              | Full sample |     |      |      | EFA sample |       |      |      | CFA sample |     |       |      |      |      |     |       |      |
|-------------------------------------------|-------------|-----|------|------|------------|-------|------|------|------------|-----|-------|------|------|------|-----|-------|------|
|                                           | Min         | Max | N    | Mean | SD         | Skew  | Kurt | N    | Mean       | SD  | Skew  | Kurt | N    | Mean | SD  | Skew  | Kurt |
| Helpless dealing with problems of life    | 1           | 7   | 2898 | 5.44 | 1.7        | -0.91 | 2.6  | 1443 | 5.45       | 1.7 | -0.93 | 2.7  | 1455 | 5.43 | 1.7 | -0.89 | 2.6  |
| Do just about anything I set my mind to*  | 1           | 7   | 2897 | 5.64 | 1.5        | -1.37 | 4.4  | 1442 | 5.6        | 1.5 | -1.34 | 4.2  | 1455 | 5.68 | 1.4 | -1.4  | 4.6  |
| Others determine what I can and cannot do | 1           | 7   | 2899 | 5.84 | 1.5        | -1.36 | 4.1  | 1443 | 5.82       | 1.5 | -1.33 | 4    | 1456 | 5.86 | 1.5 | -1.39 | 4.2  |
| What happens in life is beyond my control | 1           | 7   | 2893 | 5.14 | 1.8        | -0.67 | 2.2  | 1441 | 5.15       | 1.7 | -0.69 | 2.3  | 1452 | 5.13 | 1.8 | -0.66 | 2.2  |
| When really want something, find way*     | 1           | 7   | 2905 | 5.91 | 1.2        | -1.59 | 6.3  | 1447 | 5.87       | 1.2 | -1.54 | 5.9  | 1458 | 5.96 | 1.2 | -1.65 | 6.7  |
| Many things interfere with what I want do | 1           | 7   | 2902 | 4.34 | 1.8        | -0.06 | 1.8  | 1445 | 4.35       | 1.8 | -0.08 | 1.8  | 1457 | 4.32 | 1.8 | -0.03 | 1.8  |
| Whether I get what want is in own hands*  | 1           | 7   | 2900 | 5.17 | 1.5        | -0.9  | 3.1  | 1445 | 5.11       | 1.5 | -0.9  | 3.1  | 1455 | 5.22 | 1.5 | -0.91 | 3.2  |
| Little control over things happen to me   | 1           | 7   | 2904 | 5.31 | 1.6        | -0.89 | 2.8  | 1447 | 5.32       | 1.6 | -0.94 | 2.9  | 1457 | 5.29 | 1.6 | -0.84 | 2.7  |

| Descriptive statistics—MIDUS III                 |          |          |             |                    |            |              |            |             |             |            |             |             |            |
|--------------------------------------------------|----------|----------|-------------|--------------------|------------|--------------|------------|-------------|-------------|------------|-------------|-------------|------------|
| Agency items                                     | Min      | Max      | Full sample |                    |            | EFA sample   |            |             | CFA sample  |            |             | Skew        | Kurt       |
|                                                  |          |          | N           | Mean               | SD         | Skew         | Kurt       | N           | Mean        | SD         | N           |             |            |
| <b>Really no way I can solve problems I have</b> | <b>1</b> | <b>7</b> | <b>2895</b> | <b>5.86</b>        | <b>1.4</b> | <b>−1.44</b> | <b>4.4</b> | <b>1443</b> | <b>5.83</b> | <b>1.4</b> | <b>1452</b> | <b>5.88</b> | <b>4.5</b> |
| Feel pushed around in life                       | 1        | 7        | 2893        | 5.61               | 1.7        | −0.98        | 2.7        | 1442        | 5.61        | 1.7        | 1451        | 5.61        | 2.7        |
| Happens to me in future depends on me*           | 1        | 7        | 2889        | 5.69               | 1.4        | −1.43        | 4.8        | 1441        | 5.67        | 1.4        | 1448        | 5.71        | 4.7        |
| Rate control over life                           | 0        | 10       | 2851        | 7.71               | 2          | −1.17        | 4.5        | 1422        | 7.65        | 2          | 1429        | 7.76        | 4.5        |
| *Item reversed                                   |          |          |             |                    |            |              |            |             |             |            |             |             |            |
| Tests of multivariate normality—MIDUS III        |          |          |             |                    |            |              |            |             |             |            |             |             |            |
| Test                                             | Value    |          |             | $\chi^2$           |            |              | DF         |             |             | p          |             |             |            |
| Mardia—skewness                                  | 78.0815  |          |             | 34,579.611         |            |              | 4960       |             |             | 0.0000     |             |             |            |
| Mardia—kurtosis                                  | 1222.27  |          |             | 23,770.41          |            |              | 1          |             |             | 0.0000     |             |             |            |
| Henze-Zirkler                                    | 1.730598 |          |             | 2.08e <sup>9</sup> |            |              | 1          |             |             | 0.0000     |             |             |            |
| Doornik-Hansen                                   |          |          |             | 30,822.486         |            |              | 60         |             |             | 0.0000     |             |             |            |

### Appendix C: Factor Loadings for Initial 3-, 2-, and Single-Factor IPF Extractions w/Direct Oblimin Rotation

| Item                                       | Three-factor |    |          |    |          |       | Two-factor |    |              |              |            |       | Single factor |    |            |       |              |              |
|--------------------------------------------|--------------|----|----------|----|----------|-------|------------|----|--------------|--------------|------------|-------|---------------|----|------------|-------|--------------|--------------|
|                                            | Factor 1     |    | Factor 2 |    | Factor 3 |       | Factor 1   |    | Factor 2     |              | Uniqueness |       | Factor 1      |    | Uniqueness |       | Factor 1     |              |
|                                            | M2           | M3 | M2       | M3 | M2       | M3    | M2         | M3 | M2           | M3           | M2         | M3    | M2            | M3 | M2         | M3    | M2           | M3           |
| Control over life in general at present    |              |    |          |    |          |       |            |    |              |              | 0.825      | 0.828 |               |    | 0.824      | 0.829 | 0.420        | 0.414        |
| Where there's a will there's a way         | <b>0.700</b> |    | 0.545    |    |          |       |            |    | <b>0.697</b> | <b>0.573</b> | 0.610      | 0.744 |               |    | 0.625      | 0.742 | 0.444        | 0.356        |
| Do what can to change for better           | <b>0.618</b> |    | 0.512    |    |          |       |            |    | <b>0.631</b> | <b>0.574</b> | 0.640      | 0.721 |               |    | 0.639      | 0.719 | 0.492        | 0.402        |
| Even when feel too much, get it all done   | 0.512        |    | 0.433    |    |          |       |            |    | 0.505        | 0.486        | 0.756      | 0.803 |               |    | 0.767      | 0.801 | 0.399        | 0.336        |
| Know what I want out of life               | 0.443        |    | 0.364    |    |          |       |            |    | 0.475        | 0.450        | 0.685      | 0.690 |               |    | 0.697      | 0.723 | 0.514        | 0.484        |
| Not afraid to voice opinions in opposition | 0.359        |    |          |    | 0.334    | 0.358 |            |    | 0.411        | 0.385        | 0.733      | 0.777 |               |    | 0.823      | 0.864 | 0.370        | 0.863        |
| In charge of situation in which I live     | 0.423        |    | 0.419    |    |          |       |            |    | 0.449        | 0.443        | 0.592      | 0.603 |               |    | 0.606      | 0.602 | <b>0.615</b> | <b>0.614</b> |
|                                            |              |    |          |    |          |       |            |    |              |              |            |       |               |    |            |       | 0.622        | 0.623        |

| Item                                       | Three-factor |              |          |       |          |    | Two-factor |       |              |              |              |              | Single factor |       |              |              |            |       |
|--------------------------------------------|--------------|--------------|----------|-------|----------|----|------------|-------|--------------|--------------|--------------|--------------|---------------|-------|--------------|--------------|------------|-------|
|                                            | Factor 1     |              | Factor 2 |       | Factor 3 |    | Uniqueness |       | Factor 1     |              | Factor 2     |              | Uniqueness    |       | Factor 1     |              | Uniqueness |       |
|                                            | M2           | M3           | M2       | M3    | M2       | M3 | M2         | M3    | M2           | M3           | M2           | M3           | M2            | M3    | M2           | M3           | M2         | M3    |
| Difficult arranging life in satisfying way | <b>0.619</b> | <b>0.617</b> |          |       |          |    | 0.497      | 0.483 | <b>0.611</b> | <b>0.602</b> |              |              | 0.537         | 0.516 | <b>0.655</b> | <b>0.682</b> | 0.571      | 0.535 |
| Decisions not influenced by others doing   |              |              | 0.343    |       | 0.340    |    | 0.793      | 0.783 |              |              | 0.386        |              | 0.839         | 0.859 | 0.355        |              | 0.874      | 0.901 |
| Demands of everyday life often get me down | <b>0.651</b> | <b>0.689</b> |          |       |          |    | 0.593      | 0.549 | <b>0.651</b> | <b>0.681</b> |              |              | 0.616         | 0.585 | 0.550        | <b>0.576</b> | 0.698      | 0.668 |
| Actively carry out plans I set for self    |              |              | 0.538    | 0.513 |          |    | 0.638      | 0.649 |              |              | <b>0.557</b> | <b>0.573</b> | 0.636         | 0.650 | 0.546        | 0.507        | 0.702      | 0.744 |
| Judge self by what I think is important    |              |              | 0.430    | 0.342 |          |    | 0.793      | 0.838 |              |              | 0.472        | 0.417        | 0.817         | 0.862 | 0.331        |              | 0.890      | 0.929 |
| Able to build lifestyle to my liking       |              |              | 0.449    | 0.393 |          |    | 0.635      | 0.636 |              |              | 0.474        | 0.439        | 0.641         | 0.638 | <b>0.575</b> | <b>0.581</b> | 0.670      | 0.662 |
| Influenced by people with strong opinions  |              |              |          |       | 0.364    |    | 0.825      | 0.795 |              |              |              |              | 0.914         | 0.923 |              |              | 0.913      | 0.924 |

| Item                                      | Three-factor |              |              |       |          |    | Two-factor |       |              |              |          |       | Single factor |       |              |              |            |       |
|-------------------------------------------|--------------|--------------|--------------|-------|----------|----|------------|-------|--------------|--------------|----------|-------|---------------|-------|--------------|--------------|------------|-------|
|                                           | Factor 1     |              | Factor 2     |       | Factor 3 |    | Uniqueness |       | Factor 1     |              | Factor 2 |       | Uniqueness    |       | Factor 1     |              | Uniqueness |       |
|                                           | M2           | M3           | M2           | M3    | M2       | M3 | M2         | M3    | M2           | M3           | M2       | M3    | M2            | M3    | M2           | M3           | M2         | M3    |
| Good managing daily responsibilities      |              |              | 0.452        | 0.405 |          |    | 0.700      | 0.719 |              |              | 0.479    | 0.467 | 0.704         | 0.727 | 0.504        | 0.471        | 0.746      | 0.778 |
| Difficult voice opinion on controversial  |              |              |              |       |          |    | 0.738      | 0.767 |              |              |          |       | 0.812         | 0.857 | 0.435        | 0.377        | 0.811      | 0.858 |
| Overwhelmed by my responsibilities        | <b>0.642</b> | <b>0.615</b> |              |       |          |    | 0.570      | 0.598 | <b>0.633</b> | <b>0.604</b> |          |       | 0.619         | 0.636 | <b>0.558</b> | <b>0.561</b> | 0.688      | 0.685 |
| Little can do to change important things  | <b>0.569</b> | <b>0.626</b> |              |       |          |    | 0.625      | 0.568 | <b>0.583</b> | <b>0.642</b> |          |       | 0.644         | 0.602 | <b>0.555</b> | <b>0.580</b> | 0.692      | 0.663 |
| Helpless dealing with problems of life    | <b>0.698</b> | <b>0.767</b> |              |       |          |    | 0.445      | 0.378 | <b>0.710</b> | <b>0.774</b> |          |       | 0.442         | 0.377 | <b>0.702</b> | <b>0.739</b> | 0.507      | 0.453 |
| Do just about anything I set my mind to   |              |              | <b>0.554</b> | 0.545 |          |    | 0.629      | 0.602 |              |              | 0.522    | 0.472 | 0.666         | 0.676 | 0.529        | 0.530        | 0.720      | 0.719 |
| Others determine what I can and cannot do | 0.538        | <b>0.596</b> |              |       |          |    | 0.603      | 0.594 | 0.545        | <b>0.605</b> |          |       | 0.601         | 0.592 | <b>0.615</b> | <b>0.609</b> | 0.622      | 0.629 |

| Item                                      | Three-factor |              |              |              |          |    | Two-factor |       |              |              |              |              | Single factor |       |              |              |            |       |
|-------------------------------------------|--------------|--------------|--------------|--------------|----------|----|------------|-------|--------------|--------------|--------------|--------------|---------------|-------|--------------|--------------|------------|-------|
|                                           | Factor 1     |              | Factor 2     |              | Factor 3 |    | Uniqueness |       | Factor 1     |              | Factor 2     |              | Uniqueness    |       | Factor 1     |              | Uniqueness |       |
|                                           | M2           | M3           | M2           | M3           | M2       | M3 | M2         | M3    | M2           | M3           | M2           | M3           | M2            | M3    | M2           | M3           | M2         | M3    |
| What happens in life is beyond my control | <b>0.626</b> | <b>0.654</b> |              |              |          |    | 0.561      | 0.521 | <b>0.638</b> | <b>0.670</b> |              |              | 0.615         | 0.554 | <b>0.560</b> | <b>0.619</b> | 0.687      | 0.617 |
| When really want something, find way      |              |              | <b>0.646</b> | <b>0.722</b> |          |    | 0.518      | 0.514 |              |              | <b>0.615</b> | <b>0.687</b> | 0.556         | 0.563 | <b>0.601</b> | 0.526        | 0.639      | 0.723 |
| Many things interfere with what I want do | <b>0.668</b> | <b>0.656</b> |              |              |          |    | 0.614      | 0.574 | <b>0.684</b> | <b>0.666</b> |              |              | 0.611         | 0.572 | 0.531        | <b>0.602</b> | 0.719      | 0.638 |
| Whether I get what want is in own hands   |              |              | 0.387        | 0.544        |          |    | 0.682      | 0.624 |              |              | 0.336        | 0.442        | 0.740         | 0.738 | 0.505        | 0.470        | 0.745      | 0.779 |
| Little control over things happen to me   | <b>0.649</b> | <b>0.726</b> |              |              |          |    | 0.508      | 0.465 | <b>0.660</b> | <b>0.743</b> |              |              | 0.567         | 0.503 | <b>0.603</b> | <b>0.631</b> | 0.636      | 0.602 |
| Really no way I can solve problems I have | <b>0.639</b> | <b>0.702</b> |              |              |          |    | 0.519      | 0.450 | <b>0.653</b> | <b>0.716</b> |              |              | 0.550         | 0.461 | <b>0.625</b> | <b>0.691</b> | 0.609      | 0.523 |
| Feel pushed around in life                | <b>0.636</b> | <b>0.617</b> |              |              |          |    | 0.555      | 0.581 | <b>0.637</b> | <b>0.618</b> |              |              | 0.575         | 0.587 | <b>0.607</b> | <b>0.610</b> | 0.632      | 0.628 |

| Item                                  | Three-factor |       |          |       |          |       | Two-factor |       |          |       |          |       | Single factor |              |              |       |            |    |
|---------------------------------------|--------------|-------|----------|-------|----------|-------|------------|-------|----------|-------|----------|-------|---------------|--------------|--------------|-------|------------|----|
|                                       | Factor 1     |       | Factor 2 |       | Factor 3 |       | Uniqueness |       | Factor 1 |       | Factor 2 |       | Uniqueness    |              | Factor 1     |       | Uniqueness |    |
|                                       | M2           | M3    | M2       | M3    | M2       | M3    | M2         | M3    | M2       | M3    | M2       | M3    | M2            | M3           | M2           | M3    | M2         | M3 |
| Happens to me in future depends on me |              |       | 0.354    | 0.497 |          |       | 0.730      | 0.651 |          |       | 0.412    | 0.822 | 0.737         | 0.416        | 0.483        | 0.827 | 0.767      |    |
| Rate control over life                | 0.371        |       |          | 0.371 |          |       | 0.668      | 0.637 | 0.379    | 0.320 |          |       | 0.667         | <b>0.558</b> | <b>0.576</b> | 0.688 | 0.669      |    |
| Eigenvalue                            | 8.348        | 8.268 | 1.507    | 1.632 | 0.866    | 0.962 |            |       | 8.322    | 8.236 | 1.483    | 1.605 |               |              | 8.269        | 8.180 |            |    |
| Proportion of variance                | 0.779        | 0.761 | 0.141    | 0.150 | 0.081    | 0.089 |            |       | 0.849    | 0.837 | 0.151    | 0.163 |               |              | 1.000        | 1.000 |            |    |

Bold numbers indicate a loading of good or better ( $\lambda \geq 0.55$ ); table does not include loadings  $< 0.32$

## Appendix D: Factor Loadings and Path Diagrams for 13- and 11-Item Agency Scales

### Factor Loadings for 13-Item Personal Agency And Agency Achievement Scales (GHAs)

| Full Scale Factor Loadings                |               |              |            |                 |              |                    |          |            |       |                  |              |                     |    |                    |       |            |  |
|-------------------------------------------|---------------|--------------|------------|-----------------|--------------|--------------------|----------|------------|-------|------------------|--------------|---------------------|----|--------------------|-------|------------|--|
| Item                                      | SINGLE FACTOR |              |            | 2-FACTOR        |              |                    | 3-FACTOR |            |       |                  |              |                     |    |                    |       |            |  |
|                                           | Agency        |              | Uniqueness | Personal agency |              | Agency achievement |          | Uniqueness |       | Intrinsic agency |              | Instrumental agency |    | Agency achievement |       | Uniqueness |  |
|                                           | M2            | M3           | M2         | M3              | M2           | M3                 | M2       | M3         | M2    | M3               | M2           | M3                  | M2 | M3                 | M2    | M3         |  |
| Little can do to change important things  | <b>0.620</b>  | <b>0.650</b> | 0.616      | 0.578           | <b>0.623</b> | <b>0.703</b>       |          | 0.599      | 0.545 | <b>0.570</b>     | <b>0.686</b> |                     |    | 0.580              | 0.508 |            |  |
| Others determine what I can and cannot do | <b>0.637</b>  | <b>0.634</b> | 0.594      | 0.598           | <b>0.572</b> | <b>0.624</b>       |          | 0.593      | 0.589 | 0.495            | <b>0.579</b> |                     |    | 0.583              | 0.569 |            |  |
| What happens in life is beyond my control | <b>0.605</b>  | <b>0.660</b> | 0.634      | 0.564           | <b>0.701</b> | <b>0.714</b>       |          | 0.581      | 0.530 | <b>0.723</b>     | <b>0.759</b> |                     |    | 0.523              | 0.461 |            |  |
| Little control over things happen to me   | <b>0.648</b>  | <b>0.684</b> | 0.580      | 0.532           | <b>0.739</b> | <b>0.754</b>       |          | 0.524      | 0.488 | <b>0.757</b>     | <b>0.772</b> |                     |    | 0.460              | 0.426 |            |  |
| Really no way I can solve problems I have | <b>0.680</b>  | <b>0.735</b> | 0.538      | 0.460           | <b>0.711</b> | <b>0.745</b>       |          | 0.506      | 0.440 | <b>0.726</b>     | <b>0.655</b> |                     |    | 0.446              | 0.423 |            |  |

| Full Scale Factor Loadings                 |               |              |            |                 |              |                    |                  |       |                     |       |                    |              |                   |
|--------------------------------------------|---------------|--------------|------------|-----------------|--------------|--------------------|------------------|-------|---------------------|-------|--------------------|--------------|-------------------|
| Item                                       | SINGLE FACTOR |              |            | 2-FACTOR        |              |                    | 3-FACTOR         |       |                     |       |                    |              |                   |
|                                            | Agency        |              | Uniqueness | Personal agency |              | Agency achievement | Intrinsic agency |       | Instrumental agency |       | Agency achievement | Uniqueness   |                   |
|                                            | M2            | M3           |            | M2              | M3           |                    | M2               | M3    | M2                  | M3    | M2                 | M3           |                   |
| Helpless dealing with problems of life     | <b>0.752</b>  | <b>0.798</b> | 0.435      | 0.363           | <b>0.692</b> | <b>0.764</b>       | 0.430            | 0.356 | 0.374               | 0.404 | 0.383              | 0.453        | 0.434 0.353       |
| Difficult arranging life in satisfying way | <b>0.663</b>  | <b>0.693</b> | 0.560      | 0.520           | 0.524        | <b>0.558</b>       | 0.567            | 0.525 |                     |       | <b>0.595</b>       | <b>0.686</b> | 0.480 0.415       |
| Demands of everyday life often get me down | <b>0.585</b>  | <b>0.621</b> | 0.658      | 0.615           | <b>0.574</b> | <b>0.604</b>       | 0.648            | 0.609 |                     |       | <b>0.730</b>       | <b>0.709</b> | 0.477 0.470       |
| Overwhelmed by my responsibilities         | <b>0.578</b>  | <b>0.571</b> | 0.666      | 0.674           | <b>0.576</b> | <b>0.571</b>       | 0.654            | 0.666 |                     |       | <b>0.718</b>       | <b>0.664</b> | 0.490 0.544       |
| Able to build lifestyle to my liking       | 0.514         | 0.510        | 0.736      | 0.740           |              |                    | 0.657            | 0.653 |                     |       | 0.316              | 0.389        | 0.453 0.635 0.637 |
| Actively carry out plans I set for self    | 0.486         | 0.435        | 0.764      | 0.811           |              |                    | 0.665            | 0.632 |                     |       | 0.494              | 0.498        | 0.670 0.646       |

Full Scale Factor Loadings

| Item                                    | SINGLE FACTOR |       |            | 2-FACTOR        |              |                    | 3-FACTOR   |       |                  |                     |       |                    |              |              |       |       |
|-----------------------------------------|---------------|-------|------------|-----------------|--------------|--------------------|------------|-------|------------------|---------------------|-------|--------------------|--------------|--------------|-------|-------|
|                                         | Agency        |       | Uniqueness | Personal agency |              | Agency achievement | Uniqueness |       | Intrinsic agency | Instrumental agency |       | Agency achievement |              | Uniqueness   |       |       |
|                                         | M2            | M3    | M2         | M3              | M2           | M3                 | M2         | M3    | M2               | M3                  | M2    | M3                 | M2           | M3           |       |       |
| Do just about anything I set my mind to | 0.481         | 0.458 | 0.769      | 0.790           | <b>0.604</b> | <b>0.530</b>       | 0.617      | 0.655 |                  |                     |       |                    | <b>0.612</b> | 0.515        | 0.595 | 0.638 |
| When really want something, find way    | 0.541         | 0.442 | 0.707      | 0.805           | <b>0.777</b> | <b>0.734</b>       | 0.430      | 0.506 |                  |                     |       |                    | <b>0.782</b> | <b>0.781</b> | 0.401 | 0.392 |
| Eigenvalue                              | 4.742         | 4.952 |            |                 | 4.792        | 4.996              | 0.737      | 0.811 |                  | 4.846               | 5.050 | 0.772              | 0.867        | 0.609        | 0.602 |       |
| Proportion of variance                  | 1.000         | 1.000 |            |                 | 0.867        | 0.860              | 0.133      | 0.140 |                  | 0.778               | 0.775 | 0.124              | 0.133        | 0.098        | 0.092 |       |

Bold numbers indicate a loading of good or better ( $\lambda \geq 0.55$ ); table does not include loadings  $< 0.2$

Loadings on Individual Factors

| Item                                      | Personal agency |              | Uniqueness |       | Intrinsic agency |              | Instrumental agency |    | Agency achievement |    | Uniqueness |    |
|-------------------------------------------|-----------------|--------------|------------|-------|------------------|--------------|---------------------|----|--------------------|----|------------|----|
|                                           | M2              | M3           | M2         | M3    | M2               | M3           | M2                  | M3 | M2                 | M3 | M2         | M3 |
| Little can do to change important things  | <b>0.634</b>    | <b>0.671</b> | 0.599      | 0.549 | <b>0.659</b>     | <b>0.710</b> |                     |    | 0.566              |    | 0.496      |    |
| Others determine what I can and cannot do | <b>0.631</b>    | <b>0.637</b> | 0.602      | 0.594 | <b>0.634</b>     | <b>0.660</b> |                     |    | 0.598              |    | 0.564      |    |
| What happens in life is beyond my control | <b>0.636</b>    | <b>0.680</b> | 0.596      | 0.538 | <b>0.683</b>     | <b>0.719</b> |                     |    | 0.533              |    | 0.483      |    |

| Loadings on Individual Factors             |                 |              |            |       |                  |              |                     |              |                    |              |            |       |
|--------------------------------------------|-----------------|--------------|------------|-------|------------------|--------------|---------------------|--------------|--------------------|--------------|------------|-------|
| Item                                       | Personal agency |              | Uniqueness |       | Intrinsic agency |              | Instrumental agency |              | Agency achievement |              | Uniqueness |       |
|                                            | M2              | M3           | M2         | M3    | M2               | M3           | M2                  | M3           | M2                 | M3           | M2         | M3    |
| Little control over things happen to me    | <b>0.676</b>    | <b>0.708</b> | 0.543      | 0.499 | <b>0.718</b>     | <b>0.744</b> |                     |              |                    |              | 0.485      | 0.446 |
| Really no way I can solve problems I have  | <b>0.698</b>    | <b>0.748</b> | 0.513      | 0.441 | <b>0.743</b>     | <b>0.766</b> |                     |              |                    |              | 0.447      | 0.413 |
| Helpless dealing with problems of life     | <b>0.756</b>    | <b>0.801</b> | 0.429      | 0.359 | <b>0.694</b>     | <b>0.742</b> | <b>0.703</b>        | <b>0.755</b> |                    |              | 0.506      | 0.430 |
| Difficult arranging life in satisfying way | <b>0.643</b>    | <b>0.668</b> | 0.586      | 0.553 |                  |              | <b>0.735</b>        | <b>0.768</b> |                    |              | 0.460      | 0.411 |
| Demands of everyday life often get me down | <b>0.599</b>    | <b>0.629</b> | 0.641      | 0.604 |                  |              | <b>0.699</b>        | <b>0.720</b> |                    |              | 0.511      | 0.482 |
| Overwhelmed by my responsibilities         | <b>0.594</b>    | <b>0.585</b> | 0.647      | 0.658 |                  |              | <b>0.686</b>        | <b>0.665</b> |                    |              | 0.529      | 0.557 |
| Able to build lifestyle to my liking       |                 |              |            |       |                  |              | 0.491               | 0.499        | <b>0.554</b>       | 0.535        | 0.693      | 0.713 |
| Actively carry out plans I set for self    |                 |              |            |       |                  |              |                     |              | <b>0.580</b>       | <b>0.591</b> | 0.664      | 0.651 |
| Do just about anything I set my mind to    |                 |              |            |       |                  |              |                     |              | <b>0.618</b>       | <b>0.605</b> | 0.618      | 0.634 |
| When really want something, find way       |                 |              |            |       |                  |              |                     |              | <b>0.755</b>       | <b>0.726</b> | 0.431      | 0.473 |
| Eigenvalue                                 | 3.845           | 4.206        |            |       | 2.852            | 3.148        | 2.235               | 2.368        | 1.595              | 1.529        |            |       |

**Factor loadings for 11-item personal agency and agency achievement scales (GHAI)**

| Full Scale Factor Loadings                 |               |              |       |                 |                    |              |                     |                  |       |                    |              |            |       |       |
|--------------------------------------------|---------------|--------------|-------|-----------------|--------------------|--------------|---------------------|------------------|-------|--------------------|--------------|------------|-------|-------|
| Item                                       | SINGLE FACTOR |              |       | 2-FACTOR        |                    |              | 3-FACTOR            |                  |       |                    |              |            |       |       |
|                                            | Agency        | Uniqueness   |       | Personal agency | Agency achievement |              | Instrumental agency | Intrinsic agency |       | Agency achievement |              | Uniqueness |       |       |
|                                            | M2            | M3           | M2    | M3              | M2                 | M3           | M2                  | M3               | M2    | M3                 | M2           | M3         | M2    | M3    |
|                                            |               |              |       |                 |                    |              |                     |                  |       |                    |              |            |       |       |
| What happens in life is beyond my control  | <b>0.577</b>  | <b>0.627</b> | 0.667 | 0.607           | <b>0.610</b>       | <b>0.625</b> |                     | 0.640            | 0.595 | <b>0.685</b>       | <b>0.753</b> |            | 0.528 | 0.439 |
| Little control over things happen to me    | <b>0.632</b>  | <b>0.651</b> | 0.601 | 0.576           | <b>0.663</b>       | <b>0.661</b> |                     | 0.569            | 0.558 | <b>0.803</b>       | <b>0.747</b> |            | 0.387 | 0.415 |
| Really no way I can solve problems I have  | <b>0.647</b>  | <b>0.709</b> | 0.582 | 0.497           | <b>0.617</b>       | <b>0.690</b> |                     | 0.572            | 0.488 | <b>0.625</b>       | <b>0.588</b> |            | 0.494 | 0.441 |
| Helpless dealing with problems of life     | <b>0.750</b>  | <b>0.784</b> | 0.438 | 0.386           | <b>0.713</b>       | <b>0.784</b> |                     | 0.424            | 0.364 | 0.418              | 0.522        | 0.309      | 0.444 | 0.379 |
| Difficult arranging life in satisfying way | <b>0.674</b>  | <b>0.713</b> | 0.546 | 0.492           | <b>0.603</b>       | <b>0.662</b> |                     | 0.545            | 0.492 | <b>0.614</b>       | <b>0.711</b> |            | 0.483 | 0.415 |

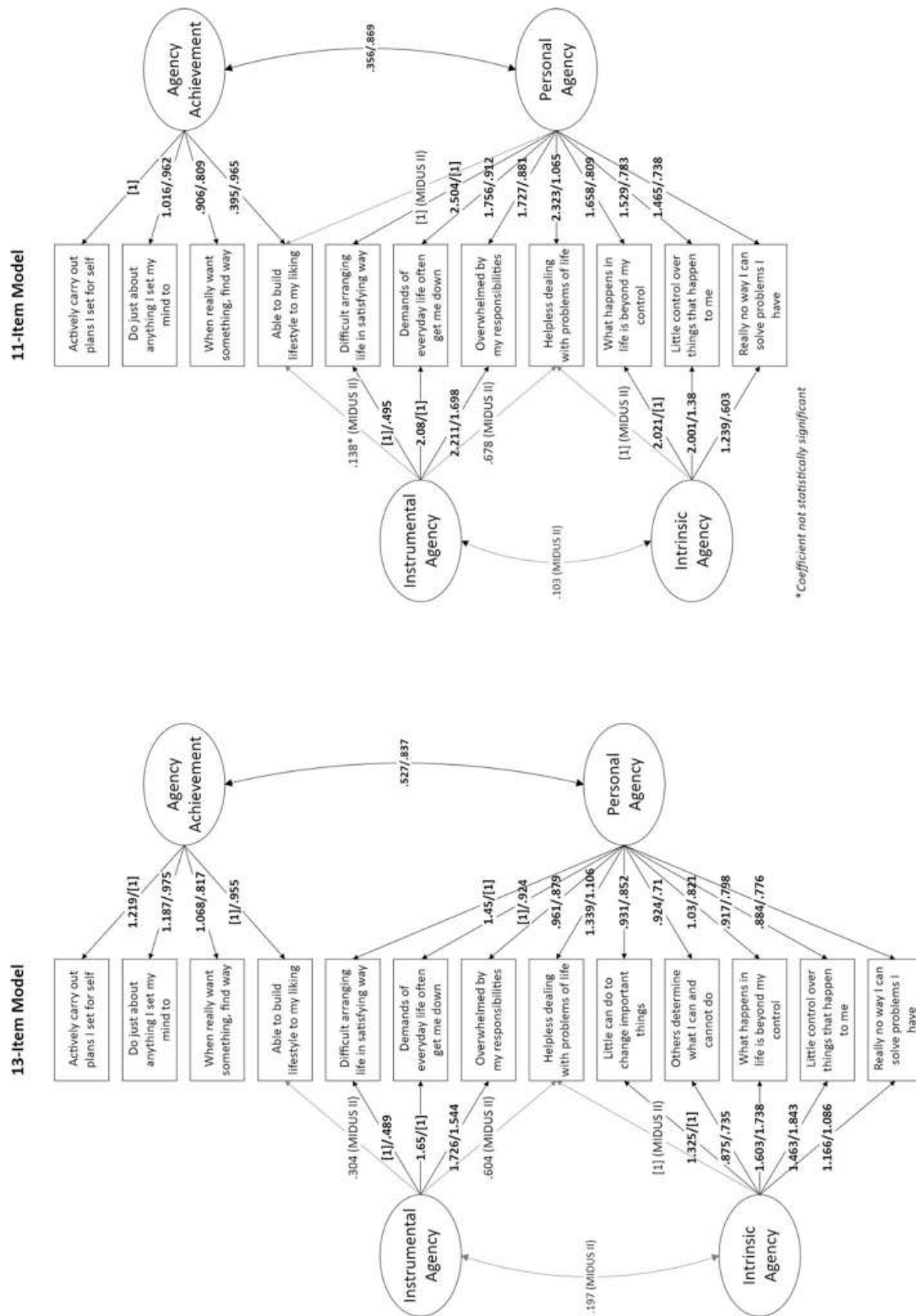
| Full Scale Factor Loadings                 |               |              |       |                 |                    |              |            |       |                     |                  |              |                    |              |              |       |
|--------------------------------------------|---------------|--------------|-------|-----------------|--------------------|--------------|------------|-------|---------------------|------------------|--------------|--------------------|--------------|--------------|-------|
| Item                                       | SINGLE FACTOR |              |       | 2-FACTOR        |                    |              | 3-FACTOR   |       |                     |                  |              |                    |              |              |       |
|                                            | Agency        | Uniqueness   |       | Personal agency | Agency achievement |              | Uniqueness |       | Instrumental agency | Intrinsic agency |              | Agency achievement |              | Uniqueness   |       |
|                                            | M2            | M3           | M2    |                 | M3                 | M2           | M3         | M2    |                     | M3               | M2           | M3                 | M2           | M3           | M2    |
| Demands of everyday life often get me down | <b>0.601</b>  | <b>0.639</b> | 0.639 | 0.591           | <b>0.682</b>       | <b>0.713</b> |            | 0.586 | 0.541               | <b>0.734</b>     | <b>0.721</b> |                    |              | 0.477        | 0.469 |
| Overwhelmed by my responsibilities         | <b>0.593</b>  | <b>0.590</b> | 0.649 | 0.652           | <b>0.683</b>       | <b>0.678</b> |            | 0.592 | 0.598               | <b>0.722</b>     | <b>0.661</b> |                    |              | 0.491        | 0.545 |
| Able to build lifestyle to my liking       | 0.534         | 0.534        | 0.714 | 0.715           | 0.228              | 0.301        |            | 0.674 | 0.675               | 0.315            | 0.402        |                    | 0.443        | 0.370        | 0.637 |
| Actively carry out plans I set for self    | 0.498         | 0.459        | 0.752 | 0.790           |                    |              |            | 0.677 | 0.677               |                  |              |                    | 0.491        | 0.492        | 0.644 |
| Do just about anything I set my mind to    | 0.484         | 0.470        | 0.766 | 0.779           |                    |              |            | 0.598 | 0.636               |                  |              |                    | <b>0.617</b> | 0.521        | 0.632 |
| When really want something, find way       | 0.548         | 0.454        | 0.699 | 0.794           |                    |              |            | 0.386 | 0.390               |                  |              |                    | <b>0.784</b> | <b>0.778</b> | 0.400 |
| Eigenvalue                                 | 3.947         | 4.122        |       |                 | 4.009              | 4.177        |            |       |                     | 4.070            | 4.235        |                    | 0.757        | 0.831        | 0.521 |

| Full Scale Factor Loadings                                                                                         |              |              |                 |       |              |                     |              |                  |       |                    |       |            |
|--------------------------------------------------------------------------------------------------------------------|--------------|--------------|-----------------|-------|--------------|---------------------|--------------|------------------|-------|--------------------|-------|------------|
| SINGLE FACTOR                                                                                                      |              |              | 2-FACTOR        |       |              | 3-FACTOR            |              |                  |       |                    |       |            |
| Agency                                                                                                             |              |              | Personal agency |       |              | Instrumental agency |              | Intrinsic agency |       | Agency achievement |       | Uniqueness |
| M2                                                                                                                 | M3           |              | M2              | M3    |              | M2                  | M3           | M2               | M3    | M2                 | M3    |            |
| Item                                                                                                               | M2           | M3           | M2              | M3    |              | M2                  | M3           | M2               | M3    | M2                 | M3    | M3         |
| Proportion of variance                                                                                             | 1.000        |              | 0.846           | 0.838 | 0.154        | 0.162               |              | 0.755            | 0.758 | 0.141              | 0.149 | 0.093      |
| Bold numbers indicate a loading of good or better ( $\lambda \geq 0.55$ ); table does not include loadings $< 0.2$ |              |              |                 |       |              |                     |              |                  |       |                    |       |            |
| Loadings on Individual Factors                                                                                     |              |              |                 |       |              |                     |              |                  |       |                    |       |            |
| Personal agency                                                                                                    |              |              | Uniqueness      |       |              | Instrumental agency |              | Intrinsic agency |       | Agency achievement |       | Uniqueness |
| M2                                                                                                                 | M3           |              | M2              | M3    |              | M2                  | M3           | M2               | M3    | M2                 | M3    | M3         |
| What happens in life is beyond my control                                                                          | <b>0.597</b> | <b>0.639</b> | 0.643           | 0.591 | <b>0.690</b> | <b>0.726</b>        |              |                  |       | 0.524              | 0.473 |            |
| Little control over things happen to me                                                                            | <b>0.651</b> | <b>0.666</b> | 0.576           | 0.557 | <b>0.759</b> | <b>0.748</b>        |              |                  |       | 0.425              | 0.440 |            |
| Really no way I can solve problems I have                                                                          | <b>0.650</b> | <b>0.719</b> | 0.578           | 0.483 | <b>0.723</b> | <b>0.775</b>        |              |                  |       | 0.478              | 0.400 |            |
| Helpless dealing with problems of life                                                                             | <b>0.763</b> | <b>0.797</b> | 0.417           | 0.365 | <b>0.661</b> | <b>0.705</b>        | <b>0.703</b> | <b>0.755</b>     |       | 0.506              | 0.430 |            |
| Difficult arranging life in satisfying way                                                                         | <b>0.676</b> | <b>0.713</b> | 0.543           | 0.492 |              |                     | <b>0.735</b> | <b>0.768</b>     |       | 0.460              | 0.411 |            |
| Demands of everyday life often get me down                                                                         | <b>0.635</b> | <b>0.666</b> | 0.596           | 0.557 |              |                     | <b>0.699</b> | <b>0.720</b>     |       | 0.511              | 0.482 |            |
| Overwhelmed by my responsibilities                                                                                 | <b>0.628</b> | <b>0.622</b> | 0.606           | 0.613 |              |                     | <b>0.686</b> | <b>0.665</b>     |       | 0.529              | 0.557 |            |

| Loadings on Individual Factors          |                 |       |            |       |                  |       |                     |       |                    |              |            |       |
|-----------------------------------------|-----------------|-------|------------|-------|------------------|-------|---------------------|-------|--------------------|--------------|------------|-------|
| Item                                    | Personal agency |       | Uniqueness |       | Intrinsic agency |       | Instrumental agency |       | Agency achievement |              | Uniqueness |       |
|                                         | M2              | M3    | M2         | M3    | M2               | M3    | M2                  | M3    | M2                 | M3           | M2         | M3    |
| Able to build lifestyle to my liking    | 0.469           | 0.478 | 0.780      | 0.772 |                  |       | 0.491               | 0.499 | <b>0.554</b>       | 0.535        | 0.693      | 0.713 |
| Actively carry out plans I set for self |                 |       |            |       |                  |       |                     |       | <b>0.580</b>       | <b>0.591</b> | 0.664      | 0.651 |
| Do just about anything I set my mind to |                 |       |            |       |                  |       |                     |       | <b>0.618</b>       | <b>0.605</b> | 0.618      | 0.634 |
| When really want something, find way    |                 |       |            |       |                  |       |                     |       | <b>0.755</b>       | <b>0.726</b> | 0.431      | 0.473 |
| Eigenvalue                              | 3.261           | 3.570 |            |       | 2.010            | 2.185 | 2.235               | 2.368 | 1.595              | 1.529        |            |       |

For cross-loaded items in 3-factor solutions, uniqueness reported is for factor on which items had highest loading

Path Diagrams for 13- and 11-Item GHAls



## Appendix E: Internal Consistency Statistics and Construct Validity

### Internal Consistency Statistics for Selected Agency Items and Final GPAS and GHAI Scales

| Original agency items      |         |       | Intrinsic agency scales |       |                           |       |        |       |        |       |
|----------------------------|---------|-------|-------------------------|-------|---------------------------|-------|--------|-------|--------|-------|
|                            | 9-ITEM  |       | 5-ITEM                  |       | 3-ITEM                    |       |        |       |        |       |
|                            | M2      | M3    | M2                      | M3    | M2                        | M3    |        |       |        |       |
| Alpha                      | 0.910   | 0.911 | 0.809                   | 0.833 | 0.753                     | 0.782 |        |       |        |       |
| Std. Alpha                 | 0.914   | 0.913 | 0.812                   | 0.835 | 0.760                     | 0.786 |        |       |        |       |
| Avg. Interitem covariance  | 0.556   | 0.561 | 1.177                   | 1.294 | 1.312                     | 1.411 |        |       |        |       |
| Avg. interitem correlation | 0.261   | 0.260 | 0.464                   | 0.503 | 0.513                     | 0.550 |        |       |        |       |
| Final agency scales        |         |       |                         |       | Instrumental agency scale |       |        |       |        |       |
|                            | 13-ITEM |       | 11-ITEM                 |       | 9-ITEM                    |       | 3-ITEM |       |        |       |
|                            | M2      | M3    | M2                      | M3    | M2                        | M3    | M2     | M3    |        |       |
| Alpha                      | 0.876   | 0.884 | 0.855                   | 0.864 | 0.817                     | 0.829 | 0.724  | 0.738 |        |       |
| Std. Alpha                 | 0.877   | 0.884 | 0.857                   | 0.864 | 0.820                     | 0.830 | 0.717  | 0.731 |        |       |
| Avg. Interitem covariance  | 0.912   | 0.966 | 0.913                   | 0.956 | 0.882                     | 0.924 | 1.265  | 1.283 |        |       |
| Avg. interitem correlation | 0.354   | 0.370 | 0.353                   | 0.366 | 0.337                     | 0.351 | 0.388  | 0.405 |        |       |
| Personal agency scales     |         |       |                         |       | Agency achievement scales |       |        |       |        |       |
|                            | 9-ITEM  |       | 7-ITEM                  |       | 6-ITEM                    |       | 4-ITEM |       | 3-ITEM |       |
|                            | M2      | M3    | M2                      | M3    | M2                        | M3    | M2     | M3    | M2     | M3    |
| Alpha                      | 0.864   | 0.881 | 0.839                   | 0.858 | 0.801                     | 0.826 | 0.715  | 0.703 | 0.672  | 0.666 |
| Std. Alpha                 | 0.867   | 0.884 | 0.842                   | 0.861 | 0.805                     | 0.830 | 0.722  | 0.709 | 0.685  | 0.678 |
| Avg. Interitem covariance  | 1.216   | 1.316 | 1.307                   | 1.392 | 1.228                     | 1.328 | 0.706  | 0.715 | 0.754  | 0.760 |
| Avg. interitem correlation | 0.420   | 0.458 | 0.433                   | 0.470 | 0.408                     | 0.448 | 0.393  | 0.379 | 0.420  | 0.412 |

### Pairwise Pearson's $r$ correlations between final agency items

| MIDUS II                                       |       |       |       |       |       |       |       |     |     |      |      |      |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-----|-----|------|------|------|
| Item                                           | (1)   | (2)   | (3)   | (4)   | (5)   | (6)   | (7)   | (8) | (9) | (10) | (11) | (12) |
|                                                |       |       |       |       |       |       |       |     |     |      |      | (13) |
| (1) Little can do to change important things   | 1     |       |       |       |       |       |       |     |     |      |      |      |
| (2) Others determine what I can and cannot do  | 0.41* | 1     |       |       |       |       |       |     |     |      |      |      |
| (3) What happens in life is beyond my control  | 0.43* | 0.45* | 1     |       |       |       |       |     |     |      |      |      |
| (4) Little control over things happen to me    | 0.44* | 0.40* | 0.55* | 1     |       |       |       |     |     |      |      |      |
| (5) Really no way I can solve problems I have  | 0.50* | 0.46* | 0.45* | 0.54* | 1     |       |       |     |     |      |      |      |
| (6) Helpless dealing with problems of life     | 0.51* | 0.44* | 0.48* | 0.47* | 0.52* | 1     |       |     |     |      |      |      |
| (7) Difficult arranging life in satisfying way | 0.34* | 0.39* | 0.35* | 0.35* | 0.37* | 0.53* | 1     |     |     |      |      |      |
| (8) Demands of everyday life often get me down | 0.31* | 0.31* | 0.32* | 0.35* | 0.34* | 0.49* | 0.48* | 1   |     |      |      |      |

| MIDUS II                                      |       |       |       |       |       |       |       |       |       |       |       |       |      |
|-----------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
| Item                                          | (1)   | (2)   | (3)   | (4)   | (5)   | (6)   | (7)   | (8)   | (9)   | (10)  | (11)  | (12)  | (13) |
| (9) Overwhelmed by my responsibilities        | 0.30* | 0.32* | 0.32* | 0.34* | 0.34* | 0.47* | 0.48* | 0.54* | 1     |       |       |       |      |
| (10) Able to build lifestyle to my liking     | 0.23* | 0.28* | 0.22* | 0.25* | 0.27* | 0.39* | 0.44* | 0.31* | 0.30* | 1     |       |       |      |
| (11) Actively carry out plans I set for self  | 0.23* | 0.26* | 0.22* | 0.25* | 0.28* | 0.34* | 0.33* | 0.26* | 0.23* | 0.41* | 1     |       |      |
| (12) Do just about anything I set my mind to  | 0.27* | 0.29* | 0.26* | 0.25* | 0.29* | 0.32* | 0.29* | 0.21* | 0.20* | 0.31* | 0.34* | 1     |      |
| (13) When really want something, find way     | 0.29* | 0.30* | 0.26* | 0.28* | 0.31* | 0.35* | 0.32* | 0.24* | 0.24* | 0.39* | 0.39* | 0.54* | 1    |
| MIDUS III                                     |       |       |       |       |       |       |       |       |       |       |       |       |      |
| (1) Little can do to change important things  | 1     |       |       |       |       |       |       |       |       |       |       |       |      |
| (2) Others determine what I can and cannot do | 0.42* | 1     |       |       |       |       |       |       |       |       |       |       |      |
| (3) What happens in life is beyond my control | 0.46* | 0.48* | 1     |       |       |       |       |       |       |       |       |       |      |
| (4) Little control over things happen to me   | 0.51* | 0.48* | 0.58* | 1     |       |       |       |       |       |       |       |       |      |

| MIDUS II                                       |       |       |       |       |       |       |       |       |       |       |       |       |      |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
| Item                                           | (1)   | (2)   | (3)   | (4)   | (5)   | (6)   | (7)   | (8)   | (9)   | (10)  | (11)  | (12)  | (13) |
| (5) Really no way I can solve problems I have  | 0.54* | 0.49* | 0.51* | 0.57* | 1     |       |       |       |       |       |       |       |      |
| (6) Helpless dealing with problems of life     | 0.56* | 0.49* | 0.49* | 0.50* | 0.57* | 1     |       |       |       |       |       |       |      |
| (7) Difficult arranging life in satisfying way | 0.39* | 0.42* | 0.39* | 0.42* | 0.45* | 0.56* | 1     |       |       |       |       |       |      |
| (8) Demands of everyday life often get me down | 0.36* | 0.36* | 0.35* | 0.38* | 0.41* | 0.54* | 0.52* | 1     |       |       |       |       |      |
| (9) Overwhelmed by my responsibilities         | 0.31* | 0.35* | 0.36* | 0.35* | 0.37* | 0.49* | 0.51* | 0.55* | 1     |       |       |       |      |
| (10) Able to build lifestyle to my liking      | 0.25* | 0.29* | 0.26* | 0.28* | 0.32* | 0.40* | 0.47* | 0.32* | 0.30* | 1     |       |       |      |
| (11) Actively carry out plans I set for self   | 0.23* | 0.27* | 0.23* | 0.25* | 0.29* | 0.32* | 0.36* | 0.26* | 0.23* | 0.41* | 1     |       |      |
| (12) Do just about anything I set my mind to   | 0.27* | 0.27* | 0.28* | 0.27* | 0.31* | 0.33* | 0.28* | 0.25* | 0.21* | 0.29* | 0.33* | 1     |      |
| (13) When really want something, find way      | 0.26* | 0.25* | 0.25* | 0.24* | 0.28* | 0.29* | 0.29* | 0.21* | 0.20* | 0.34* | 0.37* | 0.53* | 1    |

\* $p < 0.05$

All correlations significant ( $p \leq 0.05$ )

| 2-Factor Personal Agency/Achievement Models |          |           |          |           |          |           |
|---------------------------------------------|----------|-----------|----------|-----------|----------|-----------|
|                                             | 13-Item  |           | 11-Item  |           | 9-Item   |           |
|                                             | MIDUS II | MIDUS III | MIDUS II | MIDUS III | MIDUS II | MIDUS III |
| MIDUS II                                    | (1)      | (2)       | (1)      | (2)       | (1)      | (2)       |
| (1) Personal agency                         | 1        | 1         | 1        | 1         | 1        | 1         |
| (2) Agency achievement                      | 0.61     | 1         | 0.59     | 1         | 0.54     | 1         |
|                                             |          |           |          |           |          | 0.45      |
|                                             |          |           |          |           |          | 1         |

All correlations significant ( $p \leq 0.05$ )

3-Factor Personal Agency/Agency Achievement Models

|                         | 13-Item  |      |     |           |      |     | 11-Item  |      |     |           |      |     | 9-item   |      |     |           |      |     |
|-------------------------|----------|------|-----|-----------|------|-----|----------|------|-----|-----------|------|-----|----------|------|-----|-----------|------|-----|
|                         | MIDUS II |      |     | MIDUS III |      |     | MIDUS II |      |     | MIDUS III |      |     | MIDUS II |      |     | MIDUS III |      |     |
|                         | (1)      | (2)  | (3) | (1)       | (2)  | (3) | (1)      | (2)  | (3) | (1)       | (2)  | (3) | (1)      | (2)  | (3) | (1)       | (2)  | (3) |
| MIDUS II                |          |      |     |           |      |     |          |      |     |           |      |     |          |      |     |           |      |     |
| (1) Intrinsic agency    | 1        |      |     | 1         |      |     | 1        |      |     | 1         |      |     | 1        |      |     | 1         |      |     |
| (2) Instrumental agency | 0.64     | 1    |     | 0.67      | 1    |     | 0.62     | 1    |     | 0.65      | 1    |     | 0.62     | 1    |     | 0.67      | 1    |     |
| (3) Agency achievement  | 0.54     | 0.48 | 1   | 0.43      | 0.39 | 1   | 0.50     | 0.48 | 1   | 0.40      | 0.40 | 1   | 0.50     | 0.45 | 1   | 0.44      | 0.38 | 1   |

All correlations significant ( $p \leq 0.05$ )

### Pairwise correlations between 13-item human agency factors and correlates of agency

| Construct              | MIDUS II |       |       |           |     |     | MIDUS III |       |       |           |     |     | TWO-WAVE |       |       |          |     |     |
|------------------------|----------|-------|-------|-----------|-----|-----|-----------|-------|-------|-----------|-----|-----|----------|-------|-------|----------|-----|-----|
|                        | MIDUS II |       |       | MIDUS III |     |     | MIDUS III |       |       | MIDUS III |     |     | TWO-WAVE |       |       | TWO-WAVE |     |     |
|                        | (1)      | (2)   | (3)   | (4)       | (5) | (6) | (1)       | (2)   | (3)   | (4)       | (5) | (6) | (1)      | (2)   | (3)   | (4)      | (5) | (6) |
| (1) Agency             | 1.00     |       |       |           |     |     | 1.00      |       |       |           |     |     | 1.00     |       |       |          |     |     |
| (2) Personal Agency    | 0.98*    | 1.00  |       |           |     |     | 0.99*     | 1.00  |       |           |     |     | 0.98*    | 1.00  |       |          |     |     |
| (3) Agency Achievement | 0.86*    | 0.74* | 1.00  |           |     |     | 0.77*     | 0.68* | 1.00  |           |     |     | 0.86*    | 0.74* | 1.00  |          |     |     |
| (4) Intrinsic Agency   | 0.94*    | 0.98* | 0.68* | 1.00      |     |     | 0.95*     | 0.97* | 0.62* | 1.00      |     |     | 0.94*    | 0.98* | 0.68* | 1.00     |     |     |

| Construct                     | MIDUS II |       |       |       |       |       | MIDUS III |       |       |       |       |       | TWO-WAVE |       |       |       |       |       |
|-------------------------------|----------|-------|-------|-------|-------|-------|-----------|-------|-------|-------|-------|-------|----------|-------|-------|-------|-------|-------|
|                               |          |       |       |       |       |       |           |       |       |       |       |       |          |       |       |       |       |       |
|                               | (1)      | (2)   | (3)   | (4)   | (5)   | (6)   | (1)       | (2)   | (3)   | (4)   | (5)   | (6)   | (1)      | (2)   | (3)   | (4)   | (5)   | (6)   |
| (5) Instrumental Agency       | 0.90*    | 0.87* | 0.76* | 0.76* | 1.00  |       | 0.92*     | 0.91* | 0.67* | 0.80* | 1.00  |       | 0.90*    | 0.87* | 0.76* | 0.76* | 1.00  |       |
| (6) Agency Achievement        | 0.79*    | 0.66* | 0.97* | 0.64* | 0.60* | 1.00  | 0.71*     | 0.61* | 0.99* | 0.58* | 0.56* | 1.00  | 0.79*    | 0.66* | 0.97* | 0.64* | 0.60* | 1.00  |
| (3-factor)                    |          |       |       |       |       |       |           |       |       |       |       |       |          |       |       |       |       |       |
| (7) Life Satisfaction         | 0.53*    | 0.49* | 0.53* | 0.44* | 0.53* | 0.47* | 0.53*     | 0.51* | 0.46* | 0.46* | 0.53* | 0.41* | 0.39*    | 0.37* | 0.38* | 0.33* | 0.39* | 0.33* |
| (8) Self-Acceptance (PWB)     | 0.75*    | 0.69* | 0.74* | 0.62* | 0.72* | 0.67* | 0.74*     | 0.71* | 0.68* | 0.62* | 0.75* | 0.61* | 0.57*    | 0.53* | 0.57* | 0.48* | 0.54* | 0.51* |
| (9) Self-Esteem               | 0.73*    | 0.68* | 0.69* | 0.63* | 0.70* | 0.62* | 0.71*     | 0.69* | 0.63* | 0.62* | 0.70* | 0.58* | 0.54*    | 0.51* | 0.52* | 0.46* | 0.52* | 0.46* |
| (10) Purpose in Life (PWB)    | 0.69*    | 0.65* | 0.63* | 0.62* | 0.62* | 0.58* | 0.70*     | 0.68* | 0.59* | 0.63* | 0.66* | 0.54* | 0.52*    | 0.50* | 0.48* | 0.47* | 0.46* | 0.44* |
| (11) Positive Relations (PWB) | 0.58*    | 0.53* | 0.57* | 0.49* | 0.55* | 0.51* | 0.56*     | 0.53* | 0.50* | 0.48* | 0.55* | 0.45* | 0.45*    | 0.41* | 0.43* | 0.38* | 0.42* | 0.39* |
| (12) Social Integration       | 0.38*    | 0.35* | 0.36* | 0.33* | 0.35* | 0.33* | 0.39*     | 0.37* | 0.34* | 0.34* | 0.37* | 0.31* | 0.33*    | 0.31* | 0.31* | 0.29* | 0.30* | 0.28* |
| (13) Agency (Big 5)           | 0.36*    | 0.31* | 0.42* | 0.29* | 0.30* | 0.41* | 0.32*     | 0.29* | 0.39* | 0.26* | 0.29* | 0.38* | 0.30*    | 0.25* | 0.35* | 0.24* | 0.24* | 0.35* |
| (14) Autonomy (PWB)           | 0.52*    | 0.47* | 0.51* | 0.43* | 0.49* | 0.47* | 0.50*     | 0.48* | 0.45* | 0.43* | 0.49* | 0.41* | 0.40*    | 0.38* | 0.39* | 0.35* | 0.37* | 0.36* |

\* $p < 0.05$

**Pairwise correlations between 11-item human agency factors and correlates of agency**

| Construct                         | MIDUS II |       |       |       |       |       | MIDUS III |       |       |       |       |       | TWO-WAVE |       |       |       |       |       |
|-----------------------------------|----------|-------|-------|-------|-------|-------|-----------|-------|-------|-------|-------|-------|----------|-------|-------|-------|-------|-------|
|                                   | (1)      |       |       | (2)   |       |       | (3)       |       |       | (4)   |       |       | (5)      |       |       | (6)   |       |       |
|                                   | (1)      | (2)   | (3)   | (4)   | (5)   | (6)   | (1)       | (2)   | (3)   | (4)   | (5)   | (6)   | (1)      | (2)   | (3)   | (4)   | (5)   | (6)   |
| (1) Agency                        | 1.00     |       |       |       |       |       | 1.00      |       |       |       |       |       | 1.00     |       |       |       |       |       |
| (2) Personal Agency               | 0.98*    | 1.00  |       |       |       |       | 0.99*     | 1.00  |       |       |       |       | 0.98*    | 1.00  |       |       |       |       |
| (3) Agency Achievement (2-factor) | 0.82*    | 0.71* | 1.00  |       |       |       | 0.75*     | 0.64* | 1.00  |       |       |       | 0.82*    | 0.71* | 1.00  |       |       |       |
| (4) Instrumental Agency           | 0.93*    | 0.94* | 0.65* | 1.00  |       |       | 0.95*     | 0.96* | 0.59* | 1.00  |       |       | 0.93*    | 0.94* | 0.65* | 1.00  |       |       |
| (5) Intrinsic Agency              | 0.89*    | 0.92* | 0.61* | 0.75* | 1.00  |       | 0.91*     | 0.92* | 0.57* | 0.79* | 1.00  |       | 0.89*    | 0.92* | 0.61* | 0.75* | 1.00  |       |
| (6) Agency Achievement (3-factor) | 0.80*    | 0.68* | 1.00* | 0.62* | 0.60* | 1.00  | 0.73*     | 0.62* | 1.00* | 0.57* | 0.56* | 1.00  | 0.80*    | 0.68* | 1.00* | 0.62* | 0.60* | 1.00  |
| (7) Life Satisfaction             | 0.55*    | 0.52* | 0.48* | 0.53* | 0.42* | 0.47* | 0.54*     | 0.53* | 0.41* | 0.53* | 0.44* | 0.41* | 0.40*    | 0.39* | 0.34* | 0.39* | 0.32* | 0.33* |
| (8) Self-Acceptance (PWB)         | 0.76*    | 0.72* | 0.68* | 0.73* | 0.59* | 0.66* | 0.76*     | 0.73* | 0.62* | 0.75* | 0.60* | 0.61* | 0.58*    | 0.55* | 0.52* | 0.55* | 0.46* | 0.51* |
| (9) Self-Esteem                   | 0.74*    | 0.71* | 0.63* | 0.70* | 0.60* | 0.61* | 0.72*     | 0.70* | 0.58* | 0.69* | 0.60* | 0.57* | 0.55*    | 0.53* | 0.47* | 0.52* | 0.44* | 0.46* |
| (10) Purpose in Life (PWB)        | 0.68*    | 0.66* | 0.58* | 0.63* | 0.58* | 0.57* | 0.69*     | 0.67* | 0.54* | 0.66* | 0.59* | 0.54* | 0.51*    | 0.49* | 0.44* | 0.46* | 0.44* | 0.44* |

| Construct                     | MIDUS II |       |       |       |       |       | MIDUS III |       |       |       |       |       | TWO-WAVE |       |       |       |       |       |
|-------------------------------|----------|-------|-------|-------|-------|-------|-----------|-------|-------|-------|-------|-------|----------|-------|-------|-------|-------|-------|
|                               | (1)      | (2)   | (3)   | (4)   | (5)   | (6)   | (1)       | (2)   | (3)   | (4)   | (5)   | (6)   | (1)      | (2)   | (3)   | (4)   | (5)   | (6)   |
| (11) Positive Relations (PWB) | 0.58*    | 0.55* | 0.52* | 0.55* | 0.45* | 0.50* | 0.56*     | 0.54* | 0.45* | 0.55* | 0.45* | 0.45* | 0.44*    | 0.42* | 0.39* | 0.42* | 0.35* | 0.39* |
| (12) Social Integration       | 0.38*    | 0.36* | 0.33* | 0.35* | 0.31* | 0.32* | 0.39*     | 0.38* | 0.31* | 0.37* | 0.32* | 0.31* | 0.33*    | 0.31* | 0.29* | 0.30* | 0.27* | 0.28* |
| (13) Agency (Big 5)           | 0.37*    | 0.32* | 0.42* | 0.30* | 0.28* | 0.41* | 0.33*     | 0.30* | 0.38* | 0.29* | 0.26* | 0.38* | 0.30*    | 0.26* | 0.36* | 0.24* | 0.23* | 0.35* |
| (14) Autonomy (PWB)           | 0.51*    | 0.49* | 0.47* | 0.49* | 0.40* | 0.46* | 0.50*     | 0.48* | 0.42* | 0.49* | 0.41* | 0.41* | 0.40*    | 0.38* | 0.36* | 0.37* | 0.33* | 0.35* |

\* $p < 0.05$

## Declarations

**Conflict of interest** The authors have no competing interests to declare that are relevant to the content of this article. No funding has been received to support the development of this study, nor is the author affiliated with or involved in any organization or entity with any financial or non-financial interest in the subject matter or materials discussed in this manuscript. As of January 1, 2024, one of the authors began serving as a co-editor of Social Indicators Research; to avoid any conflict of interest, an alternate editor was involved in the review and editing of this manuscript.

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- 3.4 The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals (Okulicz-Kozaryn et al. 2014)

## RESEARCH REPORT

# The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals

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Political scientists traditionally have analyzed the effect of politics on subjective well-being (SWB) at the collective level, finding that more liberal countries report greater SWB. Conversely, psychologists have focused primarily on SWB at the individual level and shown that being more conservative corresponds in greater SWB. We integrate the theoretical foundations of these 2 literatures (e.g., livability and system justification theories) to compare and contrast the effects of country- and individual-level political orientation on SWB simultaneously. Using a panel of 16 West European countries representative of 1,134,384 individuals from 1970 to 2002, we demonstrated this SWB political paradox: More liberal countries and more conservative individuals had higher levels of SWB. More important, we explored measurement as a moderator of the political orientation–SWB relationship to shed some light on why this paradox exists. When orientation is measured in terms of enacted values (i.e., what the government actually does), liberalism corresponds in higher SWB, but when politics is measured in terms of espoused values (i.e., what individuals believe), greater conservatism coincided in higher SWB.

**Keywords:** subjective well-being, politics, conservatism, liberalism

Happiness or subjective well-being (SWB), defined as “people’s cognitive and affective evaluations of their lives” (Diener, Diener, & Diener, 1995, p. 851) has long been of interest to scholars and philosophers across various disciplines. Henceforth, we use the term *SWB* to encompass the related constructs of happiness and life satisfaction but sometimes use the terms *happiness* and *SWB* interchangeably to be consistent with the literature we review. Within the social psychology literature, for instance, scholars have examined pertinent antecedents of SWB, such as national culture and personality (Diener, Suh, Lucas, & Smith, 1999; Steel & Ones, 2002). Within the political science literature, the idea that governments should promote well-being dates back centuries to Adam Smith, Jeremy Bentham, and even ancient philosophers (Kahneman, Wakker, & Sarin, 1997; Nussbaum, 2005; Rasmussen, 2006). Furthermore, economists have focused on how income impacts

citizens’ well-being (Alesina, Di Tella, & MacCulloch, 2004; Clark, Frijters, & Shields, 2008; Easterlin, 1974; Frey & Stutzer, 2000), and organizational scholars’ interest in SWB dates back prior to the Hawthorne studies and extends to topics such as job satisfaction, job engagement, and organizational commitment (Ashkanasy, 2011; Fisher, 2010; Hackman & Oldham, 1980; Hassard, 2012; Judge, Locke, Durham, & Kluger, 1998; Maier & Brunstein, 2001). Despite such widespread attention across disciplines, SWB is usually approached at different levels of analysis without considering how variables may impact SWB differently at each level. As Ashkanasy (2011) pointed out, there can be surprisingly different relationships between SWB and its predictors, depending on the level of analysis.

As an example, the effects of politics on SWB have been studied at both the country and the individual levels. Political scientists have found that societies led by leftist or liberal governments (also referred to as welfare states) report the highest SWB (Bok, 2010; Radcliff, 2001), yet they often overlook the possible connection between individual-level political orientation and SWB. A *welfare state* is a term used to describe a government on the liberal end of the political continuum wherein the state protects and promotes equity and public responsibility and makes broader investments (e.g., health care, education, social services) to provide its citizens with more protections from social, economic, and political hardships (Green-Pedersen, 2004). In contrast, psychologists have found that more conservative individuals report higher SWB than more liberal folks (Napier & Jost, 2008) but largely fail to consider that politics at the country level also could affect SWB. As such, there is little consensus across the literatures on how politics affect

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SWB at both levels of analyses simultaneously. Generating consensus on how politics affect SWB at both levels is important not only in terms of SWB itself but also because variables that lead to lower SWB are related to a number of organizational issues, such as greater absenteeism and turnover (McKee-Ryan & Harvey, 2011; Mullainathan & Shafir, 2013), increased health problems (Dooley et al., 1996; Wilkinson & Pickett, 2006, 2010), IQ decrements (Mullainathan & Shafir, 2013), and lower job satisfaction and organizational commitment (Ensher et al., 2001; McKee-Ryan & Harvey, 2011; Simon et al., 2010).

The purpose of our study is to reconcile the seemingly disparate findings between the literatures on how politics affect SWB at the country and individual level. This research not only responds to the call for multidisciplinary and multilevel examination of SWB (Ashkanasy, 2011; Judge & Kammeyer-Mueller, 2011) but also further illustrates how research implications can be misguided when taken from only one level of analysis (Earley & Mosakowski, 2002; Klein & Kozlowski, 2000). Drawing on livability (Veenhoven, 2008; Veenhoven & Ehrhardt, 1995) and system justification theories (Jost & Banaji, 1994), we use a panel of West European market democracies to shed light on why the relationship between politics and SWB ostensibly has competing relationships at the country and individual levels.

## Background and Theoretical Development

### Livability Theory and Collective Political Orientation

Governments impact the interests of their citizenry. Livability theory posits (Veenhoven & Ehrhardt, 1995; Veenhoven, 2008) and empirical research supports (Appleton & Song, 2008; Bjørnskov, Dreher, & Fischer, 2008; Judge, Piccolo, Podsakoff, Shaw, & Rich, 2010; Okulicz-Kozaryn, 2012) the idea that improvements in one's living conditions can lead to greater well-being. As such, SWB is expected to increase if governmental policies produce collective provisions that create livable conditions satisfying citizen needs and, to some extent, their desires (Radcliff, 2001; Veenhoven & Ehrhardt, 1995). The conundrum for governments is that most consist of differing political parties whose divergent perspectives often lead them to disagree with what the collective provisions should be and how they should be allocated to citizens (Powell & Whitten, 1993; Radcliff, 2001; van Oorschot, 2006). Whereas the collective political orientation of countries falls on a continuum, countries can be and often are populated by a combination of citizens on both extremes of the spectrum. Collectively, liberal or left-wing governments occupy one end and conservative or right-wing governments occupy the other end of the political continuum (Hibbs, 1977).

On the basis of livability theory, political scientists tend to equate liberal or left-wing governments with increased well-being (Álvarez-Díaz, Gonzalez, & Radcliff, 2010; Bok, 2010; Pacek & Radcliff, 2008; Radcliff, 2001), believing that the lack of welfare and over reliance on the free market will result in lower SWB. For example, Esping-Andersen (1990) argued that "the market becomes to the worker a prison within which it is imperative to behave as a commodity in order to survive" (p. 36). In this regard, lower SWB can be expected when workers perceive themselves to be commodities (Easterlin, 2009). Additionally, Lane (2000) contended that because markets are indifferent to the fate of individ-

uals, SWB should be less when there is no safety net protecting citizens from economic hardships. More liberal governments tend to take greater efforts to ensure better health care, public education, and pension systems and do more to protect their citizens from the externalities (e.g., unemployment) of the market economy (Bok, 2010; Scruggs & Allan, 2006). They also actively seek to reduce income inequality that is commonly linked to poor health, crime, and overall misery among citizens (Wilkinson & Pickett, 2006). It is important to note that no government can fully eliminate adverse conditions, but more liberal governments tend to do more to shield citizens by providing a buffer against these hardships and their impact on SWB (Helliwell & Huang, 2008).

Considering the far-reaching negative factors that contribute to low levels of SWB, it is no wonder why many governments have been interested in alleviating these negative factors through the collective provisions and policies that they enact. In fact, the happiest countries in the world, the Scandinavian countries, have lower income inequality and, politically, are considered more liberal countries. On the basis of evidence in support of livability theory, we hypothesize that SWB will be higher in more liberal countries.

*Hypothesis 1: Citizens of more liberal countries will report levels of SWB that are higher than those reported in more conservative countries.*

### System Justification Theory and Individual Political Orientation

Although livability theory predicts that country-level liberalism enhances SWB, system justification theory predicts a different pattern altogether for individuals. System justification theory posits that people generally tend to (a) support and rationalize the status quo and (b) believe that the current political, social, and economic state of affairs has been realized from relatively fair and legitimate processes (Jost & Banaji, 1994; Jost & Hunyady, 2003; Jost, Pelham, & Carvallo, 2002). The theory also contends that people are hesitant to call attention to and work to disrupt inequities because of their desire to see themselves and the world as generally fair and meritocratic (Benabou & Tirole, 2006; Jost & Banaji, 1994). System-justifying ideology results partly because it provides individuals with the cognitive structure to cope with uncertainty and threats (Cheung, Noel, & Hardin, 2011; Laurin, Kay, & Shepherd, 2011). As such, individuals are motivated to believe in the fairness, legitimacy, and meritocracy of the surrounding sociopolitical system (i.e., maintain structure) because exposing its flaws and contradictions calls into question their own outcomes, reduces their ability to predict cause-and-effect relationships, and detracts from their understanding of the world, all of which introduce instability and erode structure (Jost & Hunyady, 2003; Jost, Kivetz, Rubini, Guermandi, & Mosso, 2005).

In their recent examination of why "conservatives are happier than liberals," Napier and Jost (2008, p. 565) concluded that an important factor related to system justification endorsement is one's individual political orientation. They found that more conservative individuals were more likely to endorse system-justifying beliefs and, more surprisingly, to report higher SWB (Napier & Jost, 2008). This is the case because those who are more politically conservative more readily accept and endorse the legit-

imacy of their social, political, and economic reality in spite of any adverse conditions within the nation in which they live. This acceptance and endorsement heuristic leads to the belief that individuals are responsible for their outcomes and, ultimately, get what they deserve (i.e., rationalize the status quo). This desire to rationalize the status quo serves as a “*palliative function* of system-justifying ideology” (Jost & Hunyady, 2003; Napier & Jost, 2008, p. 565; italics in original). Thus, Napier and Jost argued that more conservative people report higher SWB because they ignore or rationalize away many of the factors that otherwise might negatively influence their SWB (e.g., unemployment, inequality, disparate health care and education). In support of this notion, evidence (e.g., DiTomaso, 2013) indicates that those with more conservative beliefs were much more likely to attribute unemployment to laziness than to structural issues within the sociopolitical system. Conversely, those holding more liberal views report lower SWB because they benefit less (if at all) from this palliative function because, at least in part, to greater empathy for those experiencing adverse conditions (Graham, Haidt, & Nosek, 2009; Napier & Jost, 2008; Savani & Rattan, 2012). On the basis of system justification theory, we hypothesize the following:

*Hypothesis 2:* Those with more politically conservative views will report higher SWB.

### Juxtaposing the Two Perspectives: Measurement as a Moderator

So how can it be that political liberalism is tied to greater well-being at the country level (Hypothesis 1) but lesser well-being at the individual level (Hypothesis 2)? We contend that this apparent paradox between the literatures stems from differences in the constructions of political orientation across disciplines and levels of analysis. On the one hand, more macro perspectives on political liberalism (e.g., political science, sociology, public policy) tend to focus on what governments provide or do not provide for their citizens. Countries offering greater state-supported housing, health care, education, and remediation of income inequality are deemed more liberal, whereas those that do not are labeled more conservative (Álvarez-Díaz et al., 2010). Likewise, scholars rate governmental political orientation largely on the basis of interest group ratings of legislators’ behavioral support (i.e., voting, advocacy) for liberal and conservative policies (Berry, Ford, Ringquist, Hanson, & Klarner, 2010). On the other hand, more micro approaches (e.g., psychology, behavioral economics) center on individual beliefs about what a government should or should not do for the people it serves (Jost, Federico, & Napier, 2009). Those believing governments should be more involved in the provision of social services are tabbed as more liberal, whereas those who prefer a more hands-off approach to government assistance are considered more conservative.

Although these two methods of defining political orientation might appear similar at first glance, a closer look illustrates a key difference between the two. Namely, the macro approach focuses more on what governments actually do while the micro perspective centers on beliefs about what governments should do. This difference resembles that of the distinction between espoused and enacted values in the organizational literature (Argyris & Schon, 1978). The former pertains to things that an organization (or its

membership) purports to endorse whereas the latter describes the ideals embodied by its actions. Applied to the current context, it appears that while scholars with a micro SWB perspective have explored the impact of espoused political orientation, scholars with a macro perspective have devoted their attention to enacted political orientation.

Recognizing this division in the theoretical and empirical construction of political orientation across disciplines provides a potential reconciliation for the seemingly paradoxical findings produced by micro and macro SWB research. Inconsistent results often call for greater attention to prospective moderators that determine the nature of a relationship between two variables. In this case, it appears that measurement type may be such a moderator. When political orientation is determined by what countries actually do (enacted), the tenets of livability theory should apply and the nature of the liberalism–SWB linkage should be positive because the measured government actions (or lack thereof) are influencing individuals’ lives. Conversely, when espoused political orientation is the focus, system justification theory’s prediction of a negative relationship between liberalism and SWB should hold true because what is being measured is an aggregation of individual worldviews as opposed to actual governmental action.

*Hypothesis 3:* The country-level political orientation effect on citizen’s SWB is moderated by the type of measurement. The effect is positive when enacted political orientation was measured but negative when espoused political orientation was measured.

## Method

### Sample and Data Collection

Archival data served as the sample for this study. All individual-level (Level 1) variables were obtained from the Mannheim Eurobarometer data set (ICPSR 4357), which is a collection of Eurobarometers conducted each year in Europe and commonly used in political science research. This data set contains 1,134,384 individuals nested within Western European countries between the years of 1970 to 2002. About 2,000 people per country were surveyed in earlier years and up to 4,000 people per country were surveyed in more recent years. The 16 European countries included are France, Belgium, the Netherlands, Germany, Italy, Luxembourg, Denmark, Ireland, the United Kingdom, Greece, Spain, Portugal, Finland, Sweden, Austria, and Norway. Country-level (Level 2) variables were obtained from the World Bank and the Organisation for Economic Co-operation and Development (OECD). Two (i.e., decommodification and equality) were produced by political scientists (Scruggs & Allan, 2006) and economists (Deininger & Squire, 1996). Great Britain in the Eurobarometer data is matched with the United Kingdom in the OECD and Scruggs and Allan’s data sets.

### Individual-level (Level 1) measures

**SWB.** We measured SWB using the following item from the Mannheim Eurobarometer: “On the whole, are you very satisfied, fairly satisfied, not very satisfied, or not at all satisfied with the life you lead?” The scale ranges from 1 = *not at all satisfied* to 4 =

very satisfied. Previous research has found this measure to be reliable and valid (Di Tella & MacCulloch, 2006; Myers, 2000) and closely correlates with objective measures of well-being, such as brain activity measures (Layard, 2005).

**Political orientation.** Political orientation, the degree to which one is more conservative (right) or liberal (left), was measured using the following item: "In political matters people talk of 'the left' and 'the right,' how would you place your views on this scale?" The scale ranges from 1 = *left* to 10 = *right*. Research has found that self-placement on the right-left self-placement scale has "good test-retest reliability and strong predictive validity" (Napier & Jost, 2008, p. 568; see also Jost, 2006; Jost et al., 2009). Scores on this measure were (a) reversed so that higher values indicate greater liberalism and (b) aggregated to the country level and also included as a Level 2 measure.

### Country-Level (Level 2) Measures

To enhance the robustness of our results, we triangulate the measurement for country-level political orientation using three measures: decommodification, welfare spending, and equality, where the higher the value, the more liberal, or, alternatively speaking, the closer the country to the "welfare state ideal."

**Decommodification.** The decommodification index was retrieved from Scruggs and Allan's (2006) research and captures three dimensions: (a) the ease of access to welfare, (b) welfare value, and (c) breadth of coverage. The key welfare programs captured in this index are pensions, sickness benefits, and unemployment compensation (Cronbach's  $\alpha = .55$ ).

**Welfare spending.** Welfare spending is measured as the total public social expenditure as a percentage of gross domestic product (GDP). This variable, which can range from 0 to 100, was retrieved from the Welfare Expenditure Report produced by the OECD.

**Equality.** The Gini index, which ranges from 0 (*perfect equality*) to 1 (*perfect inequality*), is used commonly in economics and sociology research to capture societal inequity. To make this measure more consistent with the other Level 2 measures, we

reverse coded it and multiplied it by 100 so that equality scores ranged from 0 to 100. The Gini index was retrieved from Deininger and Squire's (1996) research.

### Data Analyses

Because the participants in our study were nested within countries and years, their SWB scores are likely not independent and ordinary least squares is inappropriate. We have data about countries over time, but for reasons of parsimony and computational convenience, we model it using a country-year random intercept, as opposed to a three-level model. A one-way analysis of variance (ANOVA) indicated that the variation in SWB attributable to country-year was statistically significant ( $ICC = .14$ ,  $F(333, 631834) = 290.18$ ,  $p < .001$ ). Accordingly, we used multilevel modeling (MLM) with the "xtmixed" command in Stata statistical software (Rabe-Hesketh & Skrondal, 2005). Moreover, because we incorporated variables at both the country (i.e., welfare spending, decommodification, equality) and the individual levels (i.e., SWB and political orientation), we used random intercept models, a specific type of MLM, to test all hypotheses (Kreft & de Leeuw, 1998). Like prior researchers (e.g., Park & DeShon, 2010; Porter, Webb, & Gogus, 2010), we present Rosenthal and Rubin's (2003)  $r_{\text{equivalent}}$  as an indicator of effect size for our MLM analyses.

### Results

Descriptive statistics and correlations are presented in Table 1. When modeling SWB, results tend to be similar in ordinal logistic and linear models (Ferrer-i-Carbonell & Frijters, 2004); hence, we used a linear model for ease of interpretation. To avoid multicollinearity, each model includes only one measure of political orientation at the country level (see Table 2 for a summary). Hypothesis 1 proposed that citizens of more liberal countries (i.e., welfare states) would report higher SWB than those in more conservative countries. In support of this hypothesis, we observed moderate to large effects of decommodification (.03,  $p < .01$ ,  $r = .60$ ), welfare spending (.03,  $p < .01$ ,  $r = .50$ ) and equality (.02,  $p < .01$ ,  $r = .35$ ) on SWB.

Table 1  
Individual- and Country-Level Descriptive Statistics and Correlations

| Variable                   | M     | SD   | 1       | 2      | 3      | 4       | 5 |
|----------------------------|-------|------|---------|--------|--------|---------|---|
| Individual level (Level 1) |       |      |         |        |        |         |   |
| 1. Political orientation   | 5.69  | 2.09 | —       |        |        |         |   |
| 2. Subjective well-being   | 3.06  | 0.75 | −0.10** | —      |        |         |   |
| Country level (Level 2)    |       |      |         |        |        |         |   |
| 1. Decommodification       | 28.51 | 4.29 | —       |        |        |         |   |
| 2. Welfare spending        | 23.71 | 4.61 | 0.72**  | —      |        |         |   |
| 3. Equality                | 69.37 | 4.77 | 0.10    | 0.18   | —      |         |   |
| 4. Political orientation   | 5.64  | 0.42 | −0.08   | −0.00  | −0.26* | —       |   |
| 5. Subjective well-being   | 3.07  | 0.28 | 0.54**  | 0.48** | 0.45** | −0.38** | — |

*Note.* Political orientation is on a right-to-left continuum ranging from more conservative (1) to more liberal (10). At country level a unit of analysis is a country-year, that is, a given country observed in a given year. In country level (Level 2) panel, subjective well-being scores are aggregated to the country-year. Reported correlations are pairwise with about 300 country-year observations and only 77 in cases of equality because of missing data. At individual level, there are at least 400,000 observations and usually more than 700,000 observations.

\*  $p < .05$ . \*\*  $p < .01$ .

Table 2  
Summary of Multilevel Models Predicting Subjective Well-Being

| Variable                  | Model 1          | Model 2          | Model 3           |
|---------------------------|------------------|------------------|-------------------|
| Person level (Level 1):   |                  |                  |                   |
| Intercept                 | 3.114** (0.011)  | 3.082** (0.014)  | 3.045** (0.025)   |
| Political orientation     | −0.029** (0.001) | −0.026** (0.001) | −0.033*** (0.001) |
| $r_{\text{equivalent}}$   | .09              | .08              | .10               |
| Country level (Level 2)   |                  |                  |                   |
| Decommodification         | 0.030** (0.003)  |                  |                   |
| $r_{\text{equivalent}}$   | .60              |                  |                   |
| Welfare spending          |                  | 0.030** (0.003)  |                   |
| $r_{\text{equivalent}}$   |                  | .50              |                   |
| Equality                  |                  |                  | 0.020** (0.006)   |
| $r_{\text{equivalent}}$   |                  |                  | .35               |
| Political orientation     | −0.301** (0.026) | −0.229** (0.036) | −0.283*** (0.050) |
| $r_{\text{equivalent}}$   | .60              | .36              | .31               |
| <i>N</i>                  | 407,333          | 430,091          | 114,517           |
| Level 1 residual variance | .46              | .48              | .50               |
| Level 2 residual variance | .03              | .05              | .04               |
| Deviance                  | 843,007          | 908,862          | 246,296           |

Note. Numbers in parentheses are standard errors. Political orientation is on a right-to-left continuum ranging from more conservative (1) to more liberal (10). At the individual level, political orientation is centered within cluster. At the country level, all variables are grand-mean centered. We also calculate country-year mean of political orientation at Level 2. Intraclass correlation (1) for the null model is .14.

\*\*  $p < .01$ . \*\*\*  $p < .001$ .

Hypothesis 2 predicted that more conservative participants would report higher SWB than their more liberal counterparts. Because we centered political orientation within cluster at Level 1, it is noteworthy that its effects at Levels 1 and 2 are independent and the Level 1 variable indicates the participant's relative (to the mean) political orientation within their country. Regressing SWB onto political orientation revealed that both country-level ( $b$  ranged from  $-.30$  to  $-.23$ ,  $ps < .01$ ,  $r$  ranged from  $.31$  to  $.60$ ) and individual-level ( $b$  ranged from  $-.03$  to  $-.03$ ,  $ps < .01$ ,  $r$  ranged from  $.08$  to  $.10$ ) political orientation predicted SWB as expected in all three models (i.e., across the different enacted measures). Thus, we find support for Hypothesis 2.

Hypothesis 3 posited that measurement type moderates the effect of political orientation such that the effect is positive when enacted political orientation was measured but negative when espoused political orientation was measured. Because the proposed moderation involves testing differences in the relationships between the various indicators of political orientation and SWB, it cannot be tested by computing a product term between an independent variable and a moderator variable. Rather, the task is to determine whether the coefficients reported for country-level espoused and enacted political orientation differ significantly from

one another. Toward this end, we used two tactics. First, Stata provides a post hoc function that uses a Wald test to compare the equivalence of any specified coefficients derived from the multilevel analysis. Second, we (a) computed standardized coefficients with confidence intervals for the models reported in Table 2 and (b) conducted a Monte Carlo simulation of 50,000 estimates of the differences between country-level espoused political orientation coefficient and each enacted political orientation coefficient. All of these results led to identical conclusions: Each of the enacted measures (decommodification, welfare spending, and equality) differed from the espoused measure at the country level (see Table 3 for a summary of these results). As Figure 1 illustrates, the effect of the espoused measure was negative, whereas the effects of the three enacted measures were positive, thereby supporting Hypothesis 3.

### Supplemental Analyses

Although not hypothesized, the multilevel nature of our data presents an opportunity to test for possible interactions among the different measures of political orientation. First, we analyzed cross-level interactions of political orientation at the individual and

Table 3  
Summary of Tests of Moderation by Measurement Type

| Variable                                 | Decommodification      | Welfare spending       | Equality               |
|------------------------------------------|------------------------|------------------------|------------------------|
| Enacted political orientation            | 0.46** [0.36, 0.56]    | 0.49** [0.36, 0.63]    | 0.34** [0.06, 0.63]    |
| Espoused political orientation (Level 2) | −0.45** [−0.55, −0.35] | −0.34** [−0.48, −0.20] | −0.42** [−0.62, −0.23] |
| Difference                               | 0.91** [0.75, 1.07]    | 0.83** [0.65, 1.03]    | 0.76** [0.50, 1.03]    |
| $\chi^2$                                 | 164.23**               | 50.48**                | 38.02**                |

Note. Coefficients are standardized. Espoused political orientation is on a right-to-left continuum ranging from more conservative (1) to more liberal (10). Numbers in parentheses are 99% confidence intervals. Confidence intervals for differences were derived via Monte Carlo simulation.

\*\*  $p < .01$ .

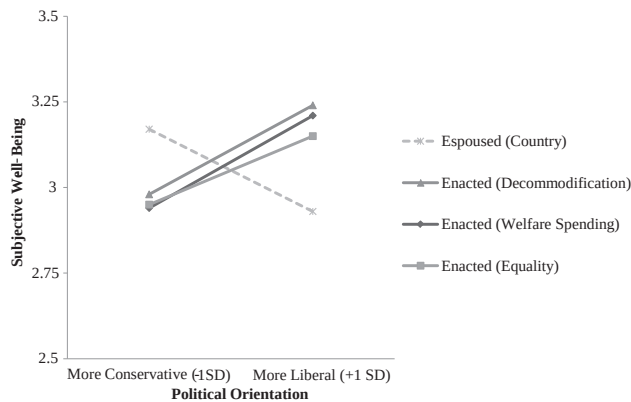


Figure 1. The moderating effect of measurement type on the political orientation–subjective well-being relationship.

country levels but found that none were significant with the exception of the interaction of political orientation (right–left) and welfare spending ( $\gamma = .001$ ,  $p = .02$ ). This indicated that the negative association between political orientation and SWB was slightly larger when welfare spending was lower ( $b = -.03$ ,  $p < .01$ ) than when it was higher ( $b = -.02$ ,  $p < .01$ ). Second, we tested interactions among the country-level (Level 2) variables and many were significant. For instance, the interaction between welfare spending and decommodification was positive ( $\gamma = .01$ ,  $p < .01$ ) and simple slopes indicated welfare spending related positively to SWB when decommodification was higher ( $\gamma = .02$ ,  $p < .01$ ) but negatively when decommodification was lower ( $\gamma = -.04$ ,  $p < .01$ ). We also performed a robustness check by including a host of controls commonly used in past research on SWB (age, income, employment status, marital status, per capita gross domestic product, unemployment, and inflation), which did not alter the pattern of results reported.

## Discussion

Integrating livability and system justification theories, we investigated how politics may lead to ostensibly inverse relationships with SWB at the country and individual levels. Consistent with livability theory, more liberal countries had higher SWB. Conversely, more conservative people reported greater SWB in line with system justification theory. To reconcile these paradoxical findings, we proposed and found that measurement moderated the relationship between political orientation and SWB at the country level. When measures capture what a country does (enacted political orientation), greater liberalism corresponds with higher SWB, but when measures tap what citizens believe (espoused political orientation), the pattern is the opposite. We now turn our attention to the implication of these findings.

## Implications for Research and Practice

Perhaps the most significant implication of this work is that it calls attention to a previously unacknowledged paradox within the existent multidisciplinary research on the association between politics and SWB. On the one hand, political scientists measure collective political orientation via enacted values or how countries

govern their citizens. In this regard, countries providing greater safety nets are designated more liberal countries (i.e., welfare states). On the other hand, psychologists measure individual political orientation by measuring espoused values, or what individuals believe. In this regard, more conservative values help people rationalize the status quo, thereby buffering their SWB from the negative impact of any existing adverse societal conditions. We found that when individual political orientation is aggregated to the country level, more conservative countries also reported higher SWB. Thus, it is not just the level at which political orientation is measured but rather what the measures reflect that determines the nature of the politics–SWB linkage. As such, our results reconcile the disparate findings and generate consensus regarding how politics affect SWB across the two literatures. It appears that both political scientists and psychologists are correct regarding the relationship between politics and SWB—the culprit of the paradox is measurement differences across the two literatures.

It also appears that the effects of espoused and enacted political orientation on SWB may be largely independent. Individuals living in countries with greater enacted liberalism reported higher SWB than those in countries with less enacted liberalism, irrespective of their own political views or those of their fellow residents. Moreover, more conservative individuals reported higher SWB than their more liberal counterparts irrespective of how their governments actually behaved (i.e., enacted political orientation). This is important because scholars with macro and micro perspectives frequently debate the relative contributions of the person and environment in predicting attitudes and behavior and often settle on a compromise involving interactions between the two (Kristof-Brown & Guay, 2011). In the present study, the effects of political person and environment characteristics on SWB appeared to operate in relative isolation from one another.

To place our effects in perspective, it helps to take a closer look at what the coefficients mean. In our data, decommodification ranged from 18 (Ireland in 1974) to 36 (Sweden in 1994), and such a discrepancy translates into a more than half point difference in SWB [(36 – 18).03 = .54] on a 4-point scale. Similarly, the observed ranges in welfare spending (from 11 in Portugal in 1985 to 35 in Sweden in 1994) and equality (from 56 in France in 1975 to 77 in the United Kingdom in 1977) corresponded with SWB differences of .72 and .42, respectively. All else being equal, two individuals on opposite ends of the espoused political orientation spectrum should differ in SWB by 0.3, whereas two countries at the poles would differ by the entire range of the scale. This clearly indicates that one's political environment and perspective contribute to SWB (albeit in different ways), which is important because SWB influences workplace productivity (Judge, Ilies, & Dimotakis, 2010; Judge & Kammeyer-Mueller, 2011; Judge & Locke, 1993; Judge et al., 1998; Judge, Piccolo, et al., 2010), team cohesion, organizational citizenship behavior, counterproductive work behavior, and turnover (Chiaburu, Peng, Oh, Banks, & Lomeli, 2013; Eatough, Chang, Miloslavic, & Johnson, 2011; Simon et al., 2010).

Governments and organizations depend on each other to provide for the citizenry. Jobs, education, and security, which contribute considerably to individual well-being, are just a few issues requiring the two to work together effectively (Diener et al., 1995; Diener & Fujita, 1995; Diener, Sandvik, Seidlitz, & Diener, 1993; Judge & Kammeyer-Mueller, 2011). This should be especially

important to policy makers and managers because a growing body of literature is uncovering how factors that occur outside of organizations can affect employee attitudes and behaviors within them (Brief et al., 2005; Martins, Eddleston, & Veiga, 2002; Pugh, Dietz, Brief, & Wiley, 2008). Because politics is omnipresent, managers and policy makers should be aware of how these factors contribute to SWB so that they can make better decisions. For instance, our findings contribute to the current debate about the effect of welfare (or its components, e.g., health care or education) on SWB. Sometimes this debate is implicit—current health care reform in the United States is assumed to increase people's well-being (Blanchflower & Oswald, 2011). In other cases, the debate is more explicit. In fact, economics Nobel Prize laureates Joseph Stiglitz and Amartya Sen explicitly argued, in a recent report by the Commission on the Measurement of Economic Performance and Social Progress, that welfare increases SWB (Stiglitz, Sen, & Fitoussi, 2009). Our findings provide a measure of empirical support for their argument.

### Study Limitations, Future Research Directions, and Conclusions

Our results should be interpreted in light of the study's limitations. First, although our measures for SWB and political orientation have seen use in previous research and have been shown to be reliable and valid (Di Tella & MacCulloch, 2006; Myers, 2000; Napier & Jost, 2008), using a single item indicator does not allow us to easily estimate the internal consistency of scores on these measures. Future researchers should use multiple-item measures when possible. Second, the low internal consistency of scores on the decommodification composite suggests a fairly high degree of measurement error, which typically attenuates relationships between constructs. Although the Spearman–Brown prophecy formula indicates that doubling the number of indicators (from three to six) would achieve a conventionally acceptable level of internal consistency (i.e., .70), we urge researchers to continue developing measures of this construct. Third, despite the large individual-level sample size, our study was limited to 16 Western European countries. Although this relatively small number of countries may make our results conservative, it is also possible that there may be a different relationship between politics and SWB in other parts of the world (e.g., Asia) because of different values or culture. Scholars generalizing our results beyond a Western European context should do so with appropriate caution and may wish to consider country-level moderators.

As with any new findings, we suggest that future researchers should attempt to replicate our findings using additional measures and data sets such as the International Social Survey Programme, newer Eurobarometers that have more recently become available, the World Values Survey, and the Database of Political Institutions from the World Bank. Future researchers also might further explore potential interactions involving political variables. For instance, we observed several significant interactions between most country-level (Level 2) variables as briefly mentioned in the results. Moreover, subsequent inquiry could focus on refining the concept of political ideology (right–left) by perhaps creating subdimensions and exploring how they relate to SWB. Notably, views on fiscal and social issues are often misaligned. Accordingly, political views (e.g., welfare spending, taxation, preferences for

size of government, preferences for redistribution, public good provision, health care, and education) could be used to define conservatism and liberalism. The goal of this exercise would be to find out what it is specifically about the broad concept of conservatism that contributes to SWB. Finally, our study found an inverse relationship between enacted and espoused values. Previous research (House, Hanges, Javidan, Dorfman, & Gupta, 2004; Taras, Steel, & Kirkman, 2010) also uncovered several examples where values and practices have been found to relate negatively without identifying the specific reason for each case. We suggest future researchers investigate the specific reasons these inverse relationships occur to contribute to the theoretical understanding of the values-to-practice phenomenon.

Politics are everywhere—individuals have political views and countries have political cultures. In this study, we showed how politics at the individual and collective levels simultaneously and differentially influence SWB. Our results explained the SWB political paradox by uncovering that how scholars measure politics matters. It is interesting that both enacted left-wing politics (liberalism) and espoused right-wing politics (conservatism) relate positively to SWB.

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#### 4 papers: urban-rural happiness gradient

##### 4.1 Urbanism and happiness: A test of Wirths theory of urban life (Okulicz-Kozaryn and Mazelis 2018)

# Urbanism and happiness: A test of Wirth's theory of urban life

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## Abstract

Social scientists have long studied the effects of cities on human wellbeing and happiness. This article demonstrates that people in cities are less happy, confirming a long-standing argument in the literature. But it had not yet been tested whether it is urbanism that negatively affects happiness, or if urban problems such as crime and poverty are to blame. Wirth posited that urbanism itself led to negative effects, but Fischer noted the necessity of empirical tests of Wirth's ideas. This study uses a happiness measure to provide a new look at the old question of urban unhappiness. Using the Behavioral Risk Factor Surveillance System from the US Centers for Disease Control and Prevention, we aim to untangle the effects of the city itself and urban problems on happiness in the USA. We find that the core characteristics of urban life (in particular size and density) contribute to urban unhappiness, controlling for urban problems. Urban unhappiness persists regardless of urban characteristics.

## Keywords

city, happiness, life satisfaction, subjective wellbeing, urbanism

## 摘要

社会科学研究者一直在研究城市对人类福祉和幸福感的影响。本文证明城市里的人们确实不那么幸福，印证了研究文献中长期存在的一个论点。但到底是都市生活对幸福感产生了负面影响，还是应归咎于犯罪和贫困等贫困问题，这一点尚未得到验证。沃斯 (Wirth) 认为都市生活本身导致了负面影响，但费歇尔 (Fischer) 指出有必要对沃斯思想进行实证检验。本研究采用了一项幸福度指标来重新审视都市生活之不幸福的旧问题。我们使用美国疾病控制和预防中心的行为风险因素监测系统，旨在厘清城市本身和城市问题对美国幸福感的影响。我们发现，在假定城市问题不产生影响的情况下，城市生活的核心特征（特别是规模和密度）促成了城市中的不幸福感。不论城市特征如何，城市中的不幸福感都会持续下去。

## 关键词

城市、幸福、生活满意度、主观幸福感、都市生活

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This article addresses a fundamental social science question: how does the modern world affect those who live in it? Modernisation and industrialisation brought urbanisation, rife with complex problems. Tönnies ([1887] 2002) and Durkheim ([1893] 1997) challenged us to investigate what living in cities does to us. The first American sociologists, Louis Wirth among them, were urban sociologists, similarly concerned with city characteristics, urban problems and happiness (Wirth, 1938). They did not use the term happiness, but other terms to denote the negative side of the urban way of life, such as ‘malaise’.<sup>1</sup> Wirth’s theory of urban life focused on how urbanism led to various negative consequences, including (1) cognitively, in terms of alienation, (2) behaviourally, in terms of deviance, and (3) structurally, in terms of anomie (normlessness), all of which would lead to unhappiness.

There is a long-standing debate about whether people in cities are unhappy. The academic debate started in sociology (Durkheim, [1893] 1997; Park, 1915; Weber et al., [1921] 1962; Wirth, 1938), but was abandoned by sociologists in the 1970s after Fischer’s major works were published. Recently, the topic was taken up by other disciplines, yet it has not addressed sociological research, and has used other approaches and foci. Regardless of the fact that urban life has much to offer residents, there is a consensus emerging among scholars that city living makes us, on average, unhappy and unhealthy (Adams, 1992; Adams and Serpe, 2000; Amato and Zuo, 1992; Balducci and Checchi, 2009; Berry and Okulicz-Kozaryn, 2009, 2011; Evans, 2009; Lederbogen et al., 2011). Those authors, however, used small data sets with few variables, typically operationalised city simply as the size of settlement, or used a dichotomous urban/rural variable. No study has controlled for city-level variables such as per capita income; no study so far has used a

model that accounts for both personal and city-level factors that affect happiness. A fundamental question as framed by Fischer (1972, 1973, 1975) remains unanswered: is it cities themselves, in terms of size, density, and heterogeneity, that lead to unhappiness, or is it problems people typically associate with cities – notably poverty, crime and lack of support – that make people unhappy?

The first urban sociologists proposed that urbanisation created malaise (unhappiness) because of the core characteristics of cities: increased population size created anonymity and impersonality, density created sensory overload and withdrawal from social life, and heterogeneity led to anomie and deviance (see Park et al., [1925] 1984; Simmel, 1903; Tönnies [1887] 2002; Wirth, 1938). Some might assert that cities are defined by their problems, and argue that measures can be used interchangeably to examine both urbanism and urban problems. However, we follow Park, Wirth, and others to define urbanism by using population size and density. We seek to advance the understanding of urban unhappiness by better measuring urbanism using several size and density variables, and controlling for urban characteristics and problems to test whether urban unhappiness persists regardless of them. Recent large-scale survey data representative of a wide range of cities, as well as counties that are not cities as a comparison group, make this test possible.

## Theoretical background

There are many advantages to cities. Labour specialisation, economies of scale, invention and creativity are all made possible by high-density living (e.g. Florida, 2008; Glaeser, 2011; O’Sullivan, 2009). Human civilisation as we know it was made possible by cities and the labour specialisation that took place in them (O’Sullivan, 2009); this made organic solidarity possible (Durkheim,

[1893] 1997). Average standard of living has improved tremendously, even with high levels of inequality: by one estimate, those in the bottom decile today have a better standard of living than all but the top 1% did 100 years ago (Bok, 2010). For an overview of the progress that Americans have made since colonial times, see Fischer (2010). Progress in standard of living and urbanisation are closely related, leading some scholars to conclude that we are happier in cities than we are elsewhere (Glaeser, 2011); this is not so. In fact, Americans have not become any happier over time (Easterlin, 1974), and we are, on average, the least happy in cities (Berry and Okulicz-Kozaryn, 2011). The most celebrated thinkers in America's intellectual history almost universally expressed ambivalence or animosity toward the city, among them: Jefferson, Emerson and Thoreau (White and White, 1977). Wirth (1938) famously summarised the negatives of city life, building on the worries of earlier European sociological theories (Tönnies, [1887] 2002). Cities are collections of heterogeneous people with different specialisations; there is a lot of variability among city dwellers. Relations between them are anonymous, superficial, impersonal, transitory, unstable and insecure (Wirth, 1938). Early theorists lamented what they saw as a lack of norms, which led to moral deviance such as crime (Wirth, 1938). Cities are fast-paced and crowded, creating a psychological overload because of the intensification of stimuli for urban dwellers and occasioning detachment, and a retreat to social isolation (Simmel, 1903). Simmel argued that people necessarily adapted to urban life by developing a blasé attitude: with the rapid pace and all the people we come across, we would not be able to handle it if we had to deeply engage with everyone and everything, so we become detached. Simmel noted that this reserve provides more personal freedom and privacy to urban dwellers, but it also necessitates

impersonal interactions with those we encounter (Simmel, 1903). Cities generate moral *disorder*: misunderstandings, tension and conflict (Smelser and Alexander, 1999). For more elaboration see Fischer and Merton (1976).

Recent evidence raises other concerns about cities: density and heterogeneity predict lower social capital (Helliwell and Wang, 2010) and low social capital is associated with poor health and higher mortality (Kawachi et al., 1997; Subramanian et al., 2002). Poverty is concentrated in cities, and the more concentrated the poverty, the worse it is for people (Jargowsky, 1997). There is an abundance of crime (Bettencourt et al., 2007, 2010) and segregation (Jargowsky, 1997; Massey and Denton, 1993). There is more crime in cities than there is elsewhere, and high crime predicts poor health (Lynch et al., 2004; Zimmerman and Bell, 2006).

High density predicts low happiness (Fassio et al., 2013; Lawless and Lucas, 2011). One specific reason why there may be less happiness in cities is that 'the pecuniary nexus tends to displace personal relations' (Wirth, 1938: 1). In addition, 'fashion tends to take the place of custom' (Park, 1915: 605), and consumption does not have a lasting impact on life satisfaction because people adapt to material goods (Easterlin, 2003). Recent neurological evidence confirms early sociological theory: great cities actually overstimulate our brains to the point where it is not healthy (Lederbogen et al., 2011). Schwartz (2004) shows that the abundance of choice, exemplified in cities, often leads to depression, loneliness, anxiety and stress. Americans do not merely prefer smaller areas as documented earlier by Fuguitt and Zuiches (1975) and Fuguitt and Brown (1990), they are also happier in smaller areas (Lawless and Lucas, 2011).

There have been many attempts at discerning the effect of city living on happiness. But the limitation of even the most recent

research is that it does not control for city-level variables such as per capita income, and city disadvantages such as crime and poverty, something that Fischer suggested researchers should do over 40 years ago. Fischer (1973: 222) acknowledged that people are less happy in the biggest cities, but he was not sure whether it was urbanism or other confounding variables that make residents unhappy in cities:

(1) Such preferences [for rural living] may be a function of idealized images founded on popular conceptions of urban and rural life. (2) They may indicate utopian hopes of maintaining urban opportunities in small communities. [...] (3) These evaluations may result from the contemporary state of American cities rather than from the nature of cities per se.

The remainder of this article will try to answer item (3) above. We aim to discover whether city residents are less happy, controlling for problems people typically associate with contemporary cities. If so, these problems cannot be the cause of lower happiness in cities. Rather, urbanism's core characteristics – large numbers of people and high density – along with other factors, make city dwellers unhappy:

H1: people in cities are less happy than are those in other areas if cities are defined by:

H1a: population size

H1b: density

H2: urban unhappiness will persist after controlling for problems people typically associate with cities.

In testing the above hypotheses we control for city characteristics that may affect happiness and for known person-level predictors of happiness.

## Data and measurement

We analyse the 2005 Behavioral Risk Factor Surveillance System (BRFSS) from the

United States Centers for Disease Control and Prevention. The BRFSS is representative of the USA, with a total sample size of over 100,000 people per year, well suited for the study of cities. The regular BRFSS data set, however, is not representative of most counties, so we use the SMART (Selected Metropolitan/Micropolitan Area Risk Trends) version of the BRFSS that is representative of counties; for simplicity we will refer to it as the BRFSS. The downside of the SMART version is a smaller number of counties, 232, but it still provides a good sample of the USA with enough variation.<sup>2</sup> See the appendix (available online) for a listing of counties. In addition to the 2005 BRFSS, as a robustness check, we include models using data for 2006, 2007 and 2008 in the appendix (available online), because the more recent waves have bigger samples, and to account for potential lag effects from variables measuring urban problems at the county level. In the main text we report the 2005 BRFSS results, because the county-level data we use are either for 2003, 2004 or 2005. All county-level data come from the Inter-university Consortium for Political and Social Research: County Characteristics, 2000–2007.

We use data at person and county levels. We do not have data at the city level. City attributes are measured at the county level; many metropolitan areas cover the whole county. Even where a metropolitan area covers part of a county, it seldom ends abruptly and changes into a non-urban area or open land. Where a metropolitan area abruptly ends, then it is due to a natural boundary such as an ocean in the case of coastal cities, but in that case the county ends as well. Furthermore, this limitation is ameliorated by another operationalisation of cities: the urban–rural continuum, which differentiates between metropolitan areas, cities, towns of various sizes and smaller areas. Another triangulation of measurement is performed

using a person-level location variable: metropolitan status code.

### *The person-level dependent variable: Happiness*

As noted above, we treat malaise as unhappiness, following research by Fischer (1972, 1973), who pioneered using happiness data to study urban malaise. Yet data and statistical software did not enable a proper test in the early 1970s. We use large-scale data and a model controlling for both personal and ecological characteristics.

Happiness is a multidimensional concept (e.g. Hills and Argyle, 2001), though for simplicity we use the terms happiness, life satisfaction and subjective wellbeing interchangeably (the dimensions overlap). The BRFSS has only one survey item measuring happiness.<sup>3</sup>

The survey item reads ‘In general, how satisfied are you with your life?’. For simplicity, we recoded answers so that a higher numeric value means more happiness: 1 is ‘very dissatisfied’, 2 ‘dissatisfied’, 3 ‘satisfied’ and 4 ‘very satisfied’. Over 90% of respondents were either satisfied or very satisfied with their lives. The same variable has been used in recent happiness studies by Oswald and Wu (2010, 2011), Glaeser et al. (2014) and Winters and Li (2015).

The happiness measure, even though self-reported and subjective, is reliable, valid (Myers, 2000), and closely correlates with similar, more objective measures such as brain waves (Layard, 2005). Unhappiness strongly correlates with suicide incidence and mental health problems (Bray and Gunnell, 2006). Finally, to be clear, we study here general/overall happiness, not a domain-specific happiness such as neighbourhood or community satisfaction.

### *The main person- and county-level independent variables: City size/city population*

In this study, we sacrifice breadth of measurement of urbanism at the cost of precision and use the most precise measures available: population size and density. Population size is used in two ways: as an urban–rural continuum with nine categories defined by the USDA (US Department of Agriculture) and as a metropolitan status code based on a survey item from the BRFSS. We chose continuous/ordinal measures because they convey more information than do binary or nominal ones (see Meyer, 2013; Wirth, 1938). The county-level urban–rural continuum ranges from 1 (‘metro > 1 million’) to 9 (‘non-metro < 2500, not adjacent to metro’).<sup>4</sup>

The metropolitan status code differs within a county; it is a person-level variable and reflects the urban dweller status more precisely than does the county-level urban–rural continuum. The metropolitan status code is broken down into the following dummy variables: ‘not in an MSA’, ‘inside a suburban county of the MSA’ and ‘outside centre OR in MSA without centre’. ‘Centre city of MSA’ is the base case, and MSA stands for Metropolitan Statistical Area, a geographical area associated with an urban area of 50,000 people or more, as defined by the United States Office of Management and Budget.

### *The main county-level independent variable: Population density*

We measure density as population (in thousands) per square mile, and there are staggering differences; density ranges from four people per square mile in Sweetwater County, WY to 70,000 people per square mile in New York County, NY.

The measures of urbanism we use do not capture heterogeneity directly. We do not include heterogeneity for three reasons. First, it is not obvious which type of heterogeneity is to be measured. In modern times we think of racial, ethnic and class diversity, but in Wirth's conception, influenced by Durkheim's focus on the increasing division of labour and occupational specialisation that accompanied urbanisation, heterogeneity may have been more about occupational diversity; Wirth writes of heterogeneity in 'social groups' (1938). It also may have been more about diversity in nation of origin, which while related to contemporary understandings of ethnicity, is not synonymous with the latter. As Wilson (2012) and Finney and Simpson (2005) note, sometimes the focus on race and ethnicity is misplaced. So we might consider measures of ethnicity or national origin, race, religion, income or occupation, for example. Second, there is precedent in recent literature to omit heterogeneity as part of a definition of urbanism, though literature consistently considers size and density (e.g. Meyer, 2013). Third, we looked at various measures of racial diversity, but did not find a consistent and robust relationship.

### *County-level controls: Measures of urban characteristics*

As we have argued above, this paper's key contribution is to disentangle the effects of the city (size, density) and urban problems on the happiness of city residents. To achieve that, we control for the following city characteristics: crime, housing stress, low education, low employment, persistent poverty, population loss, personal income (USD 1000)/cap, and percent Black. This set of controls addresses a fundamental confounding factor and an alternative explanation for city unhappiness: there are many poor people either stuck because they cannot afford to move, or who come to

cities seeking a better life. Poverty and crime are seen as fundamental urban problems (Jargowsky, 1997).<sup>5</sup> Classical theorists posited that lack of support from others is an issue in cities (Park, 1915; Wirth, 1938), though Fischer (1982, 1995) argues that this urban problem is a myth.<sup>6</sup> However, we control for lack of support with a person-level variable: social/emotional support.

We control for eight city characteristics, but there are countless characteristics one could hypothetically include. Therefore we use county fixed effects (we can measure city size at person level, because there is a person-level survey item asking respondents about the location of their residence). Finally, we focus on several of the happiest and least happy counties. We focus on issues thought to be major urban problems, such as poverty and crime, as defined in the literature (e.g. Jargowsky, 1997), as well as other variables that might have an effect, such as personal income and percent Black. 'Problems' such as congestion or high cost of living are arguably very closely related to population density, and hence are characteristics of urbanism as defined here.

### *Person-level controls: Predictors of happiness*

We control for several variables because they predict happiness as shown in the literature. Exploring the interactions of these variables with city size and density may be an interesting topic for future research, but it is beyond the scope of this study.

Increased income boosts happiness and unemployment depresses it (e.g. Di Tella and MacCulloch, 2006; Di Tella et al., 2001a, 2001b). Married people are happier than those who are unmarried (e.g. Diener and Seligman, 2004; Myers, 2000). African Americans are less happy than are Whites in the USA (e.g. Berry and Okulicz-Kozaryn, 2009, 2011), and they are concentrated in

cities, so it is important to control for race in the context of urban unhappiness. Happiness is U-shaped with age, that is, both the young and the old are happier than are people during midlife (e.g. Sanfey and Teksoz, 2005).

Health is consistently an important predictor of happiness, so we control for health status, and we also control for education, which some authors find to predict happiness (see Dolan et al., 2008). There are substantial urban–rural differences in education. In general, urban relationships are impersonal and transitory (e.g. Wirth, 1938) and some posit that there is less support in cities (White and White, 1977) and less trust in cities (Berry and Okulicz-Kozaryn, 2011). We measure lack of support using the following question: ‘How often do you get the social and emotional support you need?’. It ranges from 1 (never) to 5 (always).<sup>7</sup> Together with several county-level variables this accounts for urban problems in testing whether it is such urban problems – or cities themselves – that create urban unhappiness.

The appendix (available online) lists all the variables used in this study along with their definitions, and includes summary statistics as well.

## Model and results

Our modelling approach is similar to Fernandez and Kulik (1981) who also tested the effect of place on happiness and found that people are happier in smaller areas. By comparison with their work, the sample in the present study is over ten times larger, uses additional robustness checks and controls for what are typically characterised as urban problems.

We use a standard OLS regression with clustered standard errors by county. We also use BRFSS sampling weights to account for oversampling. We treat the four-step happiness variable as continuous (see Blanchflower and Oswald, 2011; Ferrer-i-Carbonell and Frijters, 2004).<sup>8</sup>

Results are shown in Table 1. We first estimate a base model (A) that includes known predictors of happiness. Column (B) adds two basic proxies for urban problems: crime and poverty. We also include percent Black, not as an urban problem but because many US central cities include a large African American population, African Americans score lower on happiness measures than do white Americans, and therefore the lack of this variable in a model could bias results because of different levels of happiness between racial groups. Finally, column (C) includes several additional controls for city characteristics and possible urban problems: low education, housing stress, low employment, population loss and personal income per capita.

We start with a person-level location variable. City size dummies are listed first in the table and indicate difference relative to the centre city. The most significant, both statistically and practically, is the ‘not in MSA’ dummy. As hypothesised, people are happiest outside of cities. Estimates are stable across specifications; it does not matter whether the city has stereotypical urban problems. People living outside of metropolitan areas as compared with central cities are happier by 0.05 on a scale from 1 to 4. This difference may appear to be small in practical terms, but it is still important. First, it contradicts claims that people are happier in cities, notably those by Glaeser (2011). Second, happiness is largely due to person-level influences such as unemployment, and ecology is secondary. Across 232 counties in our sample, happiness ranges between 3.18 and 3.56, with the vast majority of counties, 209 of the 232, in the 3.30 and 3.40 ranges. Third, an ecological difference of 0.05 (coefficient on ‘not in an MSA’ from Table 1) on a happiness scale from 1 to 4 can translate to large effects: for example, all else being equal, if one-third of 1% of the US population, about 1 million people, who live in

**Table 1.** Survey weighted regressions of happiness.

|                                          | A          | B          | C          |
|------------------------------------------|------------|------------|------------|
| Outside centre OR in MSA without centre  | −0.0001    | 0.0002     | −0.0017    |
| Inside suburban county of MSA            | −0.0024    | 0.0098     | 0.0079     |
| Not in an MSA                            | 0.0467***  | 0.0445**   | 0.0451**   |
| <i>County-level controls</i>             |            |            |            |
| Crime rate index                         |            | 7.90e−06** | 7.90e−06*  |
| Persistent poverty                       |            | 0.0061     | 0.0175     |
| % Black                                  |            | −0.0012**  | −0.0009*   |
| Low education                            |            |            | 0.017      |
| Housing stress                           |            |            | −0.0019    |
| Low employment                           |            |            | −0.028     |
| Population loss                          |            |            | −0.02      |
| Pers. inc. (USD 1000)/cap                |            |            | 0.0001     |
| <i>Person-level controls</i>             |            |            |            |
| Income                                   | 0.0255***  | 0.0260***  | 0.0259***  |
| Married or member of an unmarried couple | 0.1406***  | 0.1410***  | 0.1409***  |
| Unemployed                               | −0.1687*** | −0.1676*** | −0.1672*** |
| Age                                      | −0.0078*** | −0.0081*** | −0.0081*** |
| Age squared                              | 0.0001***  | 0.0001***  | 0.0001***  |
| White                                    | −0.0558*** | −0.0581*** | −0.0558*** |
| Education level                          | −0.0085+   | −0.0110*   | −0.0110*   |
| Soc/emo support                          | 0.1847***  | 0.1842***  | 0.1844***  |
| General health                           | 0.1342***  | 0.1357***  | 0.1356***  |
| N                                        | 138,453    | 132,677    | 131,657    |

Note: + $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

central cities instead lived in non-urban areas, it would mean about 50,000 (0.05\*1 million) people would be satisfied with their lives rather than dissatisfied.

All person-level variables are significant, while most county-level variables are not significant. It could be that the location variable is significant because of the large sample size at the person level. Yet, in Table 2 we use a county-level size variable, the urban–rural continuum. It ranges from 1 (‘metro > 1 million’) to 9 (‘non-metro < 2500, not adjacent to metro’). This triangulation of measurement serves as a robustness check; the smaller the place, the more happiness, no matter how the city is measured.

The coefficient estimates are 0.01, and there are eight categories on the urban–rural continuum, so a change from 1 (‘metro > 1 million’) to 9 (‘non-metro < 2500, not

adjacent to metro’) is associated with a happiness difference of about 0.1. What is striking about this second set of results is the persistence of statistical significance at the  $p < 0.01$  level on the urban–rural continuum variable; the significance of all county-level control variables is much less stable. Urban unhappiness exists, and hence we find support for hypotheses H1a and H1b. To account for possible lag effects from contextual, county-level variables on happiness, we reran these models using the 2006, 2007 and 2008 BRFSS data (the results are in the appendix, available online).

Finally, Table 3 shows that the denser the county, the less happy people are there. Adding controls attenuates the effect. Crime and poverty explain away some of the city effect. Percent Black also explains some of the city effect. Yet, density remains negative

**Table 2.** Survey weighted regressions of happiness.

|                                             | A          | B           | C          |
|---------------------------------------------|------------|-------------|------------|
| Urban–rural continuum                       | 0.0121***  | 0.0083**    | 0.0094**   |
| <i>County-level controls</i>                |            |             |            |
| Crime rate index                            |            | 7.198e–06** | 7.366e–06* |
| Persistent poverty                          |            | 0.0024      | 0.0144     |
| % Black                                     |            | –0.0011**   | –0.0008+   |
| Low education                               |            |             | 0.0174     |
| Housing stress                              |            |             | –0.0013    |
| Low employment                              |            |             | –0.0282    |
| Population loss                             |            |             | –0.0191    |
| Pers. inc. (USD 1000)/cap                   |            |             | 0.0002     |
| <i>Person-level controls</i>                |            |             |            |
| Income                                      | 0.0256***  | 0.0261***   | 0.0259***  |
| Married or member of an<br>unmarried couple | 0.1401***  | 0.1409***   | 0.1408***  |
| Unemployed                                  | –0.1683*** | –0.1674***  | –0.1670*** |
| Age                                         | –0.0077*** | –0.0080***  | –0.0081*** |
| Age squared                                 | 0.0001***  | 0.0001***   | 0.0001***  |
| White                                       | –0.0573*** | –0.0582***  | –0.0561*** |
| Education level                             | –0.0085+   | –0.0111*    | –0.0111*   |
| Soc/emo support                             | 0.1846***  | 0.1841***   | 0.1843***  |
| General health                              | 0.1342***  | 0.1357***   | 0.1356***  |
| N                                           | 138,453    | 132,677     | 131,657    |

Note: + $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

and significant when controlling for a full set of covariates.

Results from all models are consistent. Defining the city in multiple ways and controlling for commonly considered urban problems and other characteristics changes the results little, which suggests that it is the defining features of cities – size and density – that are responsible for the urban–rural happiness gradient, indicating support for hypothesis H2. Solving urban problems, if it were possible, would not likely make people in cities as happy as people are elsewhere. Urbanism – in terms of large population size and high density – is associated with lower happiness.

We use a county fixed effects model to account for potential omitted variable bias, any unobserved heterogeneity, and anything specific about the area that may affect results. US cities differ from each other

substantially. It is not possible to control for all urban problems (or all confounding variables), but any such factors would be picked up in the fixed effects model. Lower happiness in urban areas persists in the fixed effects model.

Finally, we focus on several specific cases and make some illustrative comparisons. A ranking of happy counties (table 8 in the appendix, available online) shows that the happiest counties are not cities, and the least happy counties are large cities. The three happiest counties (happiness > 3.5) in this sample are rural, yet close to large cities. They therefore have the amenities nearby that cities offer, but they escape potential downfalls of larger size and higher density. Douglas County, CO, borders Denver, but remains largely rural and has a population density of 300/sq. mile. Shelby County, TN, is partially urban; it houses Memphis, yet

**Table 3.** Survey weighted regressions of happiness.

|                                                      | A          | B          | C          |
|------------------------------------------------------|------------|------------|------------|
| Population density per sq mile,<br>05–09 * 1,000,000 | –1.5241 +  | –0.9695    | –1.3860*   |
| <i>County-level controls</i>                         |            |            |            |
| Crime rate index                                     |            | 5.560e–06* | 5.825e–06* |
| Persistent poverty                                   |            | 0.0050     | 0.0171     |
| % Black                                              |            | –0.0009*   | –0.0006    |
| Low education                                        |            |            | 0.0165     |
| Housing stress                                       |            |            | 0.001      |
| Low employment                                       |            |            | –0.0101    |
| Population loss                                      |            |            | –0.0241*   |
| Pers. inc. (USD 1000)/cap                            |            |            | 0.0007     |
| <i>Person-level controls</i>                         |            |            |            |
| Income                                               | 0.0254***  | 0.0258***  | 0.0255***  |
| Married or member of<br>an unmarried couple          | 0.1388***  | 0.1405***  | 0.1406***  |
| Unemployed                                           | –0.1676*** | –0.1671*** | –0.1670*** |
| Age                                                  | –0.0077*** | –0.0080*** | –0.0081*** |
| White                                                | –0.0598*** | –0.0591*** | –0.0559*** |
| Education level                                      | –0.0078    | –0.0107*   | –0.0109*   |
| Soc/emo support                                      | 0.1846***  | 0.1842***  | 0.1845***  |
| General health                                       | 0.1341***  | 0.1356***  | 0.1354***  |
| Age squared                                          | 0.0001***  | 0.0001***  | 0.0001***  |
| N                                                    | 138,453    | 132,677    | 131,657    |

Note: + $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

most of the county is rural and the population density is relatively low at 1200/sq. mile. Johnson County, KS, is mostly rural, yet close to Kansas City and has a population density of 1100/sq. mile. On the other side of the scale, the counties with the lowest happiness scores (happiness < 3.2) in this sample are urban: the first one, St Louis city, MO, is all urban; it covers the central city of St Louis, and has about five times the population density of two of the three happiest counties, at 5700/sq. mile (and 19 times the density of the happiest county). The second and third least happy counties, Bronx, NY and Kings, NY (Brooklyn) are parts of the American city with the largest population, New York City; both counties have a density of over 30,000/sq. mile. The happiness differences between urban and non-urban counties, even though they appear small, are

substantial because most people (>90%) are either satisfied '3' or very satisfied '4' with their lives and the standard deviation across counties in this sample is only 0.06. Hence, the urban–rural happiness gradient is quite substantial relative to the overall variability.

## Discussion and future research

Fischer asked the right question about urban unhappiness: is it the city itself – in terms of its core characteristics of size and density – or is it what are typically thought of as urban problems that generate unhappiness? We have excluded alternative explanations (our controls and fixed effects at the county level), that urban problems such as poverty or crime are solely responsible for urban unhappiness. We have confirmed that people

**Table 4.** Fixed effects survey weighted regressions of happiness.

|                                             | FE         |
|---------------------------------------------|------------|
| Outside centre OR<br>in MSA without centre  | −0.003     |
| Inside suburban county of MSA               | 0.0265     |
| Not in an MSA                               | 0.0927*    |
| Income                                      | 0.0261***  |
| Married or member<br>of an unmarried couple | 0.1361***  |
| Unemployed                                  | −0.1650*** |
| Age                                         | −0.0076*** |
| Age squared                                 | 0.0001***  |
| White                                       | −0.0570*** |
| Education level                             | −0.0082 +  |
| Soc/emo support                             | 0.1843***  |
| General health                              | 0.1337***  |
| Constant                                    | 2.1122***  |
| N                                           | 138,453    |

are less happy in cities, in part because of population size and density, the defining features of the city. Other variables also have an effect, but size and density matter, controlling for urban problems.

There is clearly a ‘city paradox’: cities provide sought-after resources, but we pay a price: the relative unhappiness of city life. High-density living fosters specialisation, information exchange, labour pooling, skills matching, knowledge spillovers, and more. On the other hand it can also foster crime, spread of disease, stress, cognitive and sensory overload, and other problems. Cities act like a magnifying glass, bringing out the best and the worst in us. This synergy has been estimated at the rate of 1.15 relative to the population – doubling population size increases everything (say crime or creativity) that an average person does by 15% (Bettencourt et al., 2007, 2010). Cities do not make people happier as Glaeser (2011) argues. People prefer to live outside of cities (Fuguitt and Brown, 1990; Fuguitt and Zuiches, 1975; Palmer, 2012) and, as we document here, people are happier outside of them.

Yet Americans are moving to metropolitan areas. This may point to an explanation of the Easterlin Paradox (1974) that economic growth does not result in happiness growth; perhaps this is due to people moving to less happy places.<sup>9</sup> Moving to a city will not necessarily make one less happy, nor will moving to a small town or village make one more happy. How moving to an area affects personal happiness is beyond the scope of this cross-sectional study. Yet, there is an urban–rural happiness gradient and it persists, regardless of urban problems such as poverty or crime. By pointing to urban unhappiness, we do not argue that urbanisation should be countered. There are many benefits to cities, and of course, some people are happier in cities than they would be elsewhere.

This study is about the USA and its results should not be generalised to other countries. At best these results can be generalised to other developed countries. However, in developing countries, there is likely to be a negative relationship between happiness and the city (Berry and Okulicz-Kozaryn, 2009).

As with any study, there needs to be more work done in this area; even though we continue a long-standing line of research, this is the first study to use county-level data to investigate the urban unhappiness hypothesis empirically. Future research may study how neighbourhoods (for example, census tracts) affect the wellbeing of residents. Cities are collections of neighbourhoods, and all characteristics, including happiness and urban problems, are distributed unevenly within a city and across its neighbourhoods. There are notable spatial inequalities (Ballas, 2013), so segregation and inequality are key fields for future work. Finer geographic resolution also has disadvantages; there are no surveys representative of neighbourhoods across a wide range of cities, so a neighbourhood study is not an

alternative or an improvement over the present study but a complement to it. While neighbourhoods often seem independent or self-contained ('a mosaic of little worlds which touch but do not interpenetrate' (Park, 1915: 608)), there are spillovers as well; for example, poverty and crime in one neighbourhood does indirectly affect people and their happiness in other neighbourhoods. While people in public space may not engage in focused interaction, effectively being 'alone, together', people do encounter others in public transportation and in public spaces, especially in large cities (Morrill et al., 2005; Putnam, 2001).

Future research would do well to investigate why people choose to live in cities, given that urbanites are less happy, using person-level panel data. Future work might also explore the effect of the city on the happiness of different social groups. Additional research could use case studies of the happiest cities as opposed to the least happy cities and investigate more deeply than is possible with currently available data what makes residents happy. As more data become available, future research can focus on the exploration of multiple advantages and disadvantages of cities. We have striven to include the most important and easily measurable controls, but it is important to extend this research further by including an even larger set of city characteristics in the future. Notably, segregation, inequality, various types of pollution, and natural amenities are topics for future research in this area (Ballas, 2013; Brereton et al., 2008).

It may be possible that 'deviant' people choose to live in cities – and they may be less happy than are others and so that may depress overall city happiness. Indeed, Wirth (1938) argued that deviance would be higher in cities; Fischer (1995) referred to this as unconventionality, a more positive framing, but with the same notion that people with unusual interests or members of small

subcultures would be more common in cities. These qualities may be to some degree picked up by person-level controls: income, education and unemployment, and by the county-level in the crime rate index. But otherwise, there is room for improvement, and future research could help in this area. In general, though, deviance and 'qualities that make people unhappy' are difficult to measure. There is also a possibility that unhappy people are pulled to cities; people who feel they do not belong in smaller areas and people who are unhappy where they live migrate to cities, but it is possible that even after urban migration their happiness remains. LGBTQ people (e.g. Doderer, 2011) and immigrants (e.g. Damm, 2009) prefer cities. But it is equally possible that happy and energetic extroverts move to cities to realise their full potential.

While there may be an element of self-selection, we think that it is possible that cities have the ability to change people in such a way that they become less happy. For instance, cities may intensify the pecuniary and consumerist orientation in people, and make them more stressed and overworked. Indeed:

To say that self-selection explains some of the differences between town and city is not to say that urbanism is irrelevant to those very same differences. It remains an indirect cause by stimulating selective migration. Although, part of urban/nonurban self-selection is coincidental, most of it results from processes originating in the nature of urbanism itself. (Fischer, 1982: 256)

And this is our interpretation and conclusion: urbanism is at least partially responsible for lower happiness in cities, regardless of urban problems.

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(Okulicz-Kozaryn) on this subject inspired this study – to search for the effects of city size per se as opposed to the effects of compositional differences of bigger cities. We thank Paul Jargowsky and Arielle Kuperberg for their advice and comments, and our fellow affiliated scholars of the Center for Urban Research and Education at Rutgers-Camden for pushing us to be clearer in our argument. We are also grateful to the students in our courses, who have helped us think and rethink ideas about cities, but we take full responsibility for weaknesses in our arguments.

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### Notes

1. We use happiness, life satisfaction and (subjective) wellbeing interchangeably, as in much of the literature; happiness is defined in the 'Data and measurement' section. Wirth, who used the term 'malaise' in his influential article, 'Urbanism as a way of life', referred to general dissatisfaction and unhappiness, and hence we use a measure of happiness to study urban malaise, as Fischer (1972, 1973) did. In other words, we interpret the early 20th century focus on 'urban malaise' as 21st century 'urban unhappiness or dissatisfaction'.
2. As a robustness check, we reran our analyses using the regular BRFSS and found similar results, which are available from the corresponding author upon request.
3. The happiness measure used here is quite comprehensive and also popular in the existing literature. This does not preclude the possibility that some aspects of happiness/life satisfaction/wellbeing may be actually improved among city dwellers. This is a topic for future research, but would require a different type of data with a very detailed questionnaire about happiness, or an in-depth qualitative study which would necessarily reduce the size and representativeness of such a data set. For future research, it may be useful to use data sets measuring distinct cognitive and affective aspects of subjective

wellbeing. As of 2015, we are unaware of a data set representative of a large number of US counties that includes such measures.

4. For all categories of both size variables, see Table 7 in the appendix, available online.
5. Only very recently have crime and poverty been suburbanising. This study, however, uses 2005 data.
6. It is better to overcontrol than undercontrol, especially in a large data set as used here (Gujarati, 2002).
7. This survey item does not capture levels of trust or social capital, which may also be important, but it is the best measure available in the BRFSS to measure support, which is related to trust and social capital. Future research should try to use other measures in order to capture trust and social capital.
8. We used the following Stata command: regress <happiness> <person-level variables> <county-level variables> [pw=\_cntywt], robust cluster(<county>). We also tried a multilevel logit model, and the results were similar. The bivariate relationship between size of a place and happiness is negative. Stata code and additional results are available upon request from the corresponding author.
9. The Easterlin Paradox shows that Americans did not become happier over the past several decades despite enormous economic progress over that time. The fact that over the same time America (sub)urbanised substantially, and that people are less happy in metropolitan areas as shown here, may help explain the Easterlin Paradox.

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